
Framing the XQ Competencies in Research & Practice

XQ Competencies —
Literature Review



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Introduction:

Since 2016, XQ Institute has been recognized as the nation's leading organization dedicated to rethinking high school. At the heart of XQ's work lies a set of foundational beliefs about what knowledge, skills, and capacities young people will need to thrive in an increasingly complex world. Those beliefs are articulated by the **XQ Learner Outcomes**, five aspirational but achievable goals for what every high school graduate should be: **Original Thinkers for an Uncertain World, Generous Collaborators for Tough Problems, Learners for Life, Masters of All Fundamental Literacies**, and **Holders of Foundational Knowledge**.

These outcomes were developed by synthesizing evidence and research from the fields of developmental and cognitive science, the science of learning, and adolescent neuroscience; from analyses of the future of work and of civic and democratic participation; and from broad engagement with researchers and other experts, educators, youth, and community members.

Rather than focusing solely on content or narrow academic benchmarks, the XQ Learner Outcomes reflect a first-principles approach: they ask not only "What do students need to know and be able to do?" but also "What kind of people do our students need to become?" This shift repositions high school as a launching point for lifelong learning and development. XQ Learning nurtures students not merely to acquire isolated facts but rather to apply complex knowledge in new and unfamiliar contexts; and to develop the habits of mind, identity, purpose, and capacity for lifelong learning essential for thriving today.

In order to operationalize the XQ Learner Outcomes—that is, to provide detailed guidance for how to teach and measure progress toward them—XQ then developed the **XQ Competencies (XQCs)**. The XQ Competencies are a set of 37 interactive competencies, each broken into component skills that are teachable, observable, and supported by research. The XQCs are designed to enable schools to translate an ambitious, necessary vision for student development into day-to-day high school teaching and learning and school conditions that promote adolescent well-being and growth.

Learning Science and the XQ Design Principles

The XQ Competencies are grounded in research from developmental psychology, cognitive science, neuroscience, and educational theory, all of which affirm that adolescence is a critical period for developing complex competencies. This period is marked by heightened plasticity in brain systems related to executive function, identity formation, emotional regulation, and social reasoning. As such, high school is uniquely positioned to help students cultivate habits of mind, self-understanding, and adaptive skills that will serve them for life.

However, students do not develop these competencies in a vacuum. The science of learning tells us that context matters deeply. The pedagogical practices and environmental conditions of schools are

crucial. To support schools in aligning their practice with the XQCs, XQ developed the **XQ Design Principles**—six evidence-based practices that promote deep learning and meaningful growth:



The interplay between these conditions and the XQ Competencies is critical. For example, developing competencies such as *Self-Motivation* or *Self-Regulation* is closely linked to learning environments that honor student voice and give students real responsibility. Building *Logical Thinking* is significantly enhanced in classrooms that engage students in inquiry, debate, and real-world problem-solving.

These insights are not new, but implementation remains challenging—especially when educators face rigid schedules, accountability pressures, or narrow assessments. The XQ Competencies aim to support, not prescribe, how schools adapt pedagogical practices to better serve adolescent development.

Interconnected and Overlapping Design

While specific skills or constructs may be emphasized differently within academic disciplines or by individual high schools, the structure of the XQ Competencies captures the breadth and depth necessary for whole-student development. The XQ Competencies are not designed to operate as discrete silos. Rather, they reflect the matrixed and integrated nature of real human capacities, in which skills like synthesis, communication, reflection, or questioning emerge in different contexts

with different expressions. For a complete technical analysis of the integrated structure of the XQ Competencies framework, see [Pittman and Irby \(2024\)](#).

For example, the skill of **synthesis** appears both as a component of the competency *Making Meaning* (within *Masters of All Fundamental Literacies*) and as a full competency under *Original Thinkers for an Uncertain World*. In a literacy context, synthesis refers to integrating multiple texts to construct meaning; in a problem-solving context, it underlies creative ideation and innovation. Similarly, a skill like **collaboration** supports multiple outcomes—from solving tough problems in teams, to navigating conflict, to co-creating knowledge across differences.

Rather than eliminating these overlaps, the XQ framework embraces them intentionally. In fact, decades of research teach us that cognitive, emotional, and social phenomena “continually interact in essential ways to create learning” and that, furthermore, learning is unlikely to occur in the absence of such interaction (Farrington et al., 2012; p. 2). Additional research confirms that knowledge, skills, and habits are strengthened through varied application across contexts (Bliss & Collingridge, 1993; Willis, 2007). The more settings in which a student practices a competency, the more durable and transferable it becomes. Therefore, overlap in the framework is not redundancy—it is resilience.

Each competency is purposefully situated to support different aspects of the XQ Learner Outcomes. When apparent overlap occurs, it can often be resolved by examining the different goals or contexts of application. For instance, the competency *Problem Solving* appears within *Original Thinkers* rather than *Generous Collaborators* because its core focus is cognitive and creative rather than interpersonal. These distinctions help maintain clarity while honoring the interconnectedness of the dynamic, mutually influential constructs—that is, the diverse and yet integrative cognitive, social, and emotional processes—that cumulatively comprise real-world learning for students (Lerner et al., 2024).

Intentional Omissions

As with any framework, questions may arise about what’s not explicitly named. Notably, some well-known constructs—such as design thinking, growth mindset, or grit—do not appear as standalone competencies. This is intentional.

For example, design thinking is a method composed of multiple phases (empathize, ideate, prototype, test), each of which draws on a mix of competencies like *Diverse Perspectives*, *Problem Seeking*, and *Creative Process*. Rather than elevating one specific approach, the XQCs provide the underlying cognitive and social tools that can be applied across many such models, including but not limited to design thinking (Razzouk & Schute, 2012).

Similarly, growth mindset (Dweck) and grit (Duckworth) are influential constructs, but each reflects broader domains of self-regulation, persistence, and identity development—all of which are well-represented in competencies such as *Understanding Self*, *Self-Motivation*, and *Pursuing Goals*.

By remaining agnostic to brand-name constructs, the XQCs preserve flexibility and resist being tethered to any one theoretical framework or pedagogical trend. This ensures the competencies remain relevant across evolving educational models and future innovations.

Finally, biological/physical health and wellness are not included here. While XQ believes that physical wellness is a key component of student thriving, our 37 competencies focus on the areas of growth and development that frontline educators are best equipped to cultivate.

Ultimately, these decisions regarding the content and structure of the XQ Competencies framework were made to optimize alignment to both the current scientific consensus on how students learn and the current sociocultural consensus on what young people need in order to thrive in a 21st century economy and society. “Thriving” is not a steady state, but rather a dynamic and lifelong process (Cantor et al., 2021). Therefore, we built the XQ Competencies to encompass the knowledge, behaviors, and contexts that enable not merely content mastery or skill development, but the full complement of academic, cognitive, social, and emotional processes that enable positive growth in adolescents and throughout the lifespan.

Design of the Literature Review

The XQ Competencies reflect a rigorous, research-aligned, and future-oriented synthesis of what high school graduates need to thrive. They respond to the developmental readiness of adolescents and are designed to be teachable and observable across multiple settings and disciplines. Backed by learning science and grounded in real pedagogical environments, the competencies emphasize what’s durable, transferable, and essential.

This review of the relevant scientific and pedagogical literature—organized to mirror the structure of the XQ Competencies framework—provides an annotated bibliography of the core academic research that underlies each XQ Competency and its component skills. We provide only a sampling of the relevant literature for each competency, focusing on research studies, literature reviews and syntheses, and theoretical articles that are most recent, most relevant, and/or most seminal in the fields of research and practice that drive each competency.

In sum, the XQ Competencies provide not only a research-validated framework for 21st-century learning but also a practical toolset for transforming how schools support student growth—intellectually, personally, and socially. This integrative approach is essential if we are to prepare young people for the “increasingly complex, global, and rapidly changing environment” in which they will assume the mantle of adulthood. Furthermore, this approach to adolescent learning is essential to *us all*, as the documented benefits of such an approach impact not only individual students but public health, economic growth, and the well-being of our society as a whole (Jones and Kahn, 2017, p. 11).

We invite you to dive into the research upon which the XQ Competencies are based in order to understand more deeply how the XQ Competencies work, how they might be most effectively

implemented in schools and youth programs, and why they are so critical to the success of students in high school and beyond.

Works Referenced

Allen, K., Kern, M. L., Vella-Brodrick, D., Hattie, J., & Waters, L. (2018). What schools need to know about fostering school belonging: A meta-analysis. *Educational Psychology Review*, 30(1), 1–34. <http://www.jstor.org/stable/45379661>

Ballard, P. J., & Syme, S. L. (2016). Engaging youth in communities: a framework for promoting adolescent and community health. *Journal of Epidemiology and Community Health* (1979–), 70(2), 202–206. <http://www.jstor.org/stable/44017690>

Bliss, T., & Collingridge, G. (1993). A synaptic model of memory: long-term potentiation in the hippocampus. *Nature*, 361, 31–39. <https://doi.org/10.1038/361031a0>

Cantor, P., Lerner, R. M., Pittman, K. J., Chase, P. A., & Gomperts, N. (2021). *Whole-child development, learning, and thriving: A dynamic systems approach*. Cambridge University Press.

Chen, C., & Yang, Y. (2019). Revisiting the effects of project-based learning on students' academic achievement: A meta-analysis investigating moderators. *Educational Research Review*, 26, 71–81. <https://doi.org/10.1016/j.edurev.2018.11.001>

Duckworth, A., & Gross, J. J. (2014). Self-control and grit: Related but separable determinants of success. *Current Directions in Psychological Science*, 23(5), 319–325. <http://www.jstor.org/stable/44318793>

Ellerbrock, C. R., & Kiefer, S. M. (2013). The interplay between adolescent needs and secondary school structures: Fostering developmentally responsive middle and high school environments across the transition. *The High School Journal*, 96(3), 170–194. <http://www.jstor.org/stable/23351973>

Farrington, C.A., Roderick, M., Allensworth, E., Nagaoka, J., Keyes, T.S., Johnson, D.W., & Beechum, N.O. (2012). Teaching adolescents to become learners. The role of noncognitive factors in shaping school performance: A critical literature review. Chicago: University of Chicago Consortium on Chicago School Research. <https://files.eric.ed.gov/fulltext/ED542543.pdf>

Flanagan, C., & Levine, P. (2010). Civic engagement and the transition to adulthood. *The Future of Children*, 20(1), 159–179. <https://doi.org/10.1353/foc.0.0043>

Gardner, H. (2007). The synthesizing mind: Making sense of the deluge of information. In M. Sánchez Sorondo, E. Malinvaud, & P. Léna (Eds.), *Globalization and education: Proceedings of the joint working group, The Pontifical Academy of Sciences, The Pontifical Academy of Social Sciences*, 19 – 17 November 2005, Casino Pio IV (pp. 3–18). De Gruyter. <https://doi.org/10.1515/9783110207019-004>

Giri, V., & Paily, M.U. (2020). Effect of scientific argumentation on the development of critical thinking. *Science & Education*, 29, 673–690. <https://doi.org/10.1007/s11191-020-00120-y>

Goldman, S. R., Britt, M. A., Brown, W., Cribb, G., George, M., Greenleaf, C., Lee, C.D., & Project READI. (2016). Disciplinary literacies and learning to read for understanding: A conceptual framework for disciplinary literacy. *Educational Psychologist*, 51(2), 219–246.
<https://doi.org/10.1080/00461520.2016.1168741>

Gowdy, G., & Spencer, R. (2021). It's who you know that matters: Identifying which type of informal mentor is most likely to promote economic mobility for vulnerable youth. *The Journal of Primary Prevention*, 42(3), 217–238. <https://doi.org/10.1007/s10935-021-00630-7>

Gruber M.J., Gelman B.D., Ranganath C. (2014). States of curiosity modulate hippocampus-dependent learning via the dopaminergic circuit. *Neuron*, 84(2), 486–96.
<https://doi.org/10.1016/j.neuron.2014.08.060>

Hardy, S. A., & Carlo, G. (2005). Identity as a source of moral motivation. *Human Development*, 48(4), 232–256. <https://doi.org/10.1159/000086859>

Hathaway, W. H. (1917). A course in socialized high-school civics. *The School Review*, 25(10), 731–743.
<http://www.jstor.org/stable/1077975>

Holmes, V. L., & Hwang, Y. (2016). Exploring the effects of project-based learning in secondary mathematics education. *The Journal of Educational Research*, 109(5), 449–463.
<https://doi.org/10.1080/00220671.2014.979911>

Immordino-Yang, M. H., & Damasio, A. (2007). We feel, therefore we learn: The relevance of affective and social neuroscience to education. *Mind, Brain, and Education*, 1(1), 3–10.
<https://doi.org/10.1111/j.1751-228X.2007.00004.x>

Jones, B. D., Ruff, C., & Osborne, J. W. (2015). Fostering students' identification with mathematics and science. In K. A. Renninger, M. Nieswandt, & S. Hidi (Eds.), *Interest in mathematics and science learning* (pp. 331–352). American Educational Research Association.
<http://www.jstor.org/stable/j.ctt1s474j0.25>

Jones, S.M., & Kahn, J. (2017). The evidence base for how we learn: Supporting students' social, emotional, and academic development. National Commission on Social, Emotional, and Academic Development, Aspen Institute.
<https://www.aspeninstitute.org/wp-content/uploads/2017/10/SEAD-Research-Brief-11.17.pdf>

Lerner, R. M., Yu, D., Abbasi-Asl, R., Keces, N., Gonçalves, C., Buckingham, M. H., ... & Dunham, Y. (2024). Towards a dynamic, idiographic approach to describing, explaining, and enhancing the development of SEL. *Social and Emotional Learning: Research, Practice, and Policy*, 4, 100050.
https://www.sciencedirect.com/science/article/pii/S277323392400024X?ref=pdf_download&fr=RR-2&rr=94ea12cc3cd18fc6

Loyens, S.M.M., van Meerten, J.E., Schaap, L., & Wijnia, L. (2023). Situating higher-order, critical, and critical-analytic thinking in problem- and project-based learning environments: A systematic review. *Educational Psychology Review*, 35, 39. <https://doi.org/10.1007/s10648-023-09757-x>

- Maxwell, N. L., Mergendoller, J. R., & Bellisimo, Y. (2005). Problem-based learning and high school macroeconomics: A comparative study of instructional methods. *The Journal of Economic Education*, 36(4), 315–329. <https://doi.org/10.3200/JECE.36.4.315-331>
- Monahan, K. C., Steinberg, L., Cauffman, E., & Mulvey, E. P. (2009). Trajectories of antisocial behavior and psychosocial maturity from adolescence to young adulthood. *Developmental Psychology*, 45(6), 1654–1668. <https://doi.org/10.1037/a0015862>
- Page, S. E. (2007). *The difference: How the power of diversity creates better groups, firms, schools, and societies*. Princeton University Press.
- Pittman, K., & Irby, M. (2024). The XQ Competencies: A comparative review of leading student learning frameworks. XQ Institute. https://xqsuperschool.org/wp-content/uploads/2024/07/2024_XQ_Competencies_White_Paper.pdf
- Ratner, K., Porcelli, S. E., & Burrow, A. L. (2022). Purpose in life, urgency, and the propensity to engage in risky and self-destructive behaviors. *Motivation and Emotion*, 46(1), 59–73. <https://doi.org/10.1007/s11031-021-09915-0>
- Raufelder, D., & Kulakow, S. (2021). The role of the learning environment in adolescents' motivational development. *Motivation and Emotion*, 45(3), 299–311. <https://doi.org/10.1007/s11031-021-09879-1>
- Razzouk, R., & Shute, V. (2012). What is design thinking and why is it important? *Review of Educational Research*, 82(3), 330–348. <https://doi.org/10.3102/O034654312457429>
- Wang, M.-T., & Eccles, J. S. (2012). Social support matters: Longitudinal effects of social support on three dimensions of school engagement from middle to high school. *Child Development*, 83(3), 877–895. <http://www.jstor.org/stable/23255730>
- Willis, J. (2007). Review of research: Brain-based teaching strategies for improving students' memory, learning, and test-taking success. *Childhood Education*, 83(5), 310–315, doi.org/10.1080/00094056.2007.10522940
- Wirkala, C., & Kuhn, D. (2011). Problem-based learning in K–12 education: Is it effective and how does it achieve its effects? *American Educational Research Journal*, 48(5), 1157–1186. <http://www.jstor.org/stable/41306381>
- Yeager, D. S., & Dweck, C. S. (2012). Mindsets that promote resilience: When students believe that personal characteristics can be developed. *Educational Psychologist*, 47(4), 302–314. <https://doi.org/10.1080/00461520.2012.722805>
- Zeldin, S., Gauley, J. S., Barringer, A., & Chapa, B. (2018). How high schools become empowering communities: A mixed-method explanatory inquiry into youth-adult partnership and school engagement. *American Journal of Community Psychology*, 61(3–4), 358–371. <https://doi.org/10.1002/ajcp.12231>
- Zimmerman, B. (2000). Self-efficacy: An essential motive to learn. *Contemporary Educational Psychology*, 25(1), 82–91. <https://doi.org/10.1006/ceps.1999.1016>

Masters of All Fundamental Literacies

XQ Competencies —
Literature Review



Making Meaning

Introduction

In order to make meaning, students gather and synthesize information from multiple sources and disciplines, sorting and focusing on relevant content. They assess information generated in response to their own probing questions, and refine their questions accordingly. When they evaluate information, they understand the quality of their inputs drives the rigor of their interpretations.

Source 1: The Synthesizing Mind: Making Sense of the Deluge of Information

Citation: Gardner, H. (2007). The synthesizing mind: Making sense of the deluge of information. In M. Sánchez Sorondo, E. Malinvaud, & P. Léna (Eds.), *Globalization and education: Proceedings of the joint working group, The Pontifical Academy of Sciences, The Pontifical Academy of Social Sciences, 19 – 17 November 2005, Casino Pio IV* (pp. 3–18). De Gruyter.

Summary: Synthesis is the capacity to absorb information from various sources and domains, evaluate its importance and relevance, and organize it into a comprehensive and comprehensible picture of an event or situation. In today's information-saturated world, the ability to synthesize effectively is invaluable for navigating and making sense of diverse data and viewpoints. This article points out the difficulties in synthesis, such as biases, overwhelming data volumes, and the cognitive demand of integrating diverse information.

The author argues for the importance of explicitly teaching synthesis in educational settings, emphasizing project-based learning where students practice integrating diverse information sources. He suggests incorporating exercises that build synthesis skills, like summarizing complex texts or creating interdisciplinary projects that require integrating various types of knowledge.

The article highlights that synthesis is not only critical in academic contexts but is also essential in professional and everyday life, helping individuals make informed decisions and develop innovative solutions. As information continues to grow exponentially, the ability to synthesize will become even more critical, marking it as a key skill for future success in numerous fields.

Source 2: Disciplinary Literacies and Learning to Read for Understanding: A Conceptual Framework for Disciplinary Literacy

Citation: Goldman, S. R., Britt, M. A., Brown, W., Cribb, G., George, M., Greenleaf, C., Lee, C.D. & Project READI. (2016). Disciplinary literacies and learning to read for understanding: A conceptual framework for disciplinary literacy. *Educational Psychologist*, 51(2), 219–246.
<https://doi.org/10.1080/00461520.2016.1168741>

Summary: This study provides a comprehensive framework for enhancing literacy and learning in three key disciplines in middle and secondary school: literature, science, and history. Certain reading, reasoning, and argumentation skills are considered essential for engaging in knowledge generation across all disciplines, though the specifics of how these skills are applied can vary between disciplines.

For example, all disciplines require students to engage in close reading of texts, which includes understanding complex text structures, interpreting meaning, engaging in relevant inquiry (i.e., asking effective questions), and evaluating content critically. Close reading serves as a foundation for building knowledge in any disciplinary area. Another example is synthesizing information from multiple sources or texts, which is crucial in all disciplines to develop integrated understandings and to draw connections between concepts or findings.

The authors argue that these skills are particularly important in real-world learning situations. “In authentic learning situations, readers have specific inquiry tasks to accomplish and are often learning about complex situations and phenomena for which no single text provides a complete account. Competent readers must coordinate diverse—and sometimes contradictory—information and perspectives from multiple texts, accounting for authors’ intent, evaluating evidence presented in the text, and judging the relevance and usefulness of each text for the task at hand.”

Source 3: Teaching and Learning Argumentative Reading and Writing: A Review of Research

Citation: Newell, G.E., Beach, R., Smith, J., & VanDerHeide, J. (2011). Teaching and learning argumentative reading and writing: A review of research. *Reading Research Quarterly*, 46, 273–304. <https://doi.org/10.1598/RRQ.46.3.4>

Summary: Argumentative tasks enhance students' cognitive skills by requiring them to ask critical questions about texts, identify key arguments, analyze evidence, and synthesize information. These activities sharpen critical thinking and reasoning capabilities, important for academic success and effective communication.

In the process of forming arguments, students must ask questions about the information they read and write, which helps them to engage more deeply with the content and to construct more robust arguments. This aspect of inquiry is crucial for students to not only understand but also challenge and expand upon the ideas presented in academic texts or debates.

The ability to interpret information is also integral to understanding and constructing arguments. Students need to be able to understand and interpret the underlying argument, its claims, warrants, and evidence, in reading tasks. In writing, students must be able to organize their thoughts and articulate them clearly, using evidence effectively to support their claims.

Synthesis is also highlighted as a crucial cognitive skill in argumentation, as it involves drawing together diverse pieces of information, ideas, and evidence from multiple texts or discussions to construct well-supported arguments. This skill is crucial for developing strong, persuasive essays and presentations.

The article discusses the importance of creating supportive classroom environments that encourage interactive learning through debates and discussions. This approach helps students practice argumentation in a social context, improving their ability to articulate their thoughts and respond to others effectively. It stresses the need for educational practices that not only focus on the mechanics of argumentation but also on fostering a supportive and interactive learning environment.

Source 4: Teaching and Learning Information Synthesis: An Intervention and Rubric Based Assessment

Citation: Lundstrom, K., Diekema, A.R., Leary, H.M., Haderlie, S., & Holliday, W. (2015). Teaching and learning information synthesis: An intervention and rubric based assessment. *Communications in Information Literacy*, 9(1), 60–82. <https://doi.org/10.15760/comminfolit.2015.9.1.176>

Summary: This study explores effective strategies for teaching and evaluating information synthesis skills, testing an instructional method to teach and assess the skill of information synthesis in academic writing. Researchers created a lesson that broke the synthesis process into sequenced tasks involving group discussion and individual writing tasks. They developed a rubric to define student achievement within these tasks, and applied that rubric to evaluate the effectiveness of the instruction. The rubric assessed the level of synthesis in students' writings, focusing on how well they integrated information from multiple sources into coherent paragraphs. The intervention appeared to enhance students' ability to synthesize information, as evidenced by improved synthesis in their writing assignments post-lesson.

Despite the intervention, students generally showed low levels of synthesis, indicating the complexity of this skill and the need for further instructional support. The study found that rubrics can be a useful tool for assessing synthesis, identifying various levels of students' ability to integrate information effectively.

The findings suggest that teaching synthesis effectively requires more than a single lesson and should be integrated throughout the curriculum to better prepare students for complex writing tasks.

Source 5: To Build a Better Question

Citation: Kelley–Mudie, S., & Phillips, J. (2016). To build a better question. *Knowledge Quest*, 44(5), 14–19.

Summary: Good questions are the foundation of critical thinking. They initiate deeper inquiry and learning, contrasting with answers which are endpoints unless followed by further questions. Being “question-literate” means being able to construct and comprehend questions effectively.

Children naturally ask many questions, but this tends to decrease with age as formal education progresses. The authors advocate for reversing this trend by deliberately teaching questioning as a skill, both as an end in itself and as a means to deepen student engagement. Students need to understand the components of questions and how to assemble them to elicit the desired information. Like learning grammar for language, students must learn the “grammar” of questioning to become proficient.

The authors suggest practical strategies to foster questioning in the classroom, such as the “question formulation technique,” a structured approach where students generate, improve, and prioritize their questions, facilitating deeper engagement and understanding. By teaching students to ask better questions, educators equip them with a crucial tool for learning and adaptation in a complex world.

Conclusion

Being able to make meaning out of information is becoming increasingly important in our media-saturated world. Practicing information synthesis, effective questioning, and evaluating information for quality and clarity to inform one’s understanding are all important practices supporting students’ understanding of complex topics and events.

Persuasive Communication

Introduction

Persuasive communication is the strategic blend of logic, credibility, and emotion. More than choosing the right words, it involves connecting deeply with one's audience to turn information into influence and ideas into action. Mastering this competency means mastering the art of crafting messages that resonate and endure.

Source 1: Longitudinal Research on Perspective Taking in Adolescence: A Systematic Review

Citation: Hall, H.K., Millear, P.M.R., Summers, M.J., & Isbel, B. (2021). Longitudinal research on perspective taking in adolescence: A systematic review. *Adolescent Research Review*, 6, 125–150.
<https://doi.org/10.1007/s40894-021-00150-9>

Summary: In this review, perspective-taking, defined as the cognitive ability to understand others' viewpoints and emotions, is identified as a critical socio-cognitive skill during adolescence, a period marked by dynamic social and psychological growth.

Perspective-taking abilities increase gradually from ages 13 to 18, and can be developed in school settings. Adolescents with strong perspective-taking skills exhibit higher self-esteem, reduced depressive symptoms, and greater psychological well-being. Perspective-taking is positively associated with prosocial behaviors, such as altruism, collaboration, and moral reasoning, and negatively associated with antisocial behaviors such as aggression and prejudice.

High self-esteem, self-confidence, and a strong understanding of one's audience strengthen persuasive skills. Further, because perspective-taking skills are correlated with these traits, students benefit when they have opportunities to practice actively working to anticipate how audiences might receive their communications based on the way in which they communicate, and on their audience's specific backgrounds, experiences, and perspectives.

Source 2: Theoretical Perspectives on Interpersonal Adjustments in Language and Communication

Citation: Gasior, J. (2016). Theoretical perspectives on interpersonal adjustments in language and communication. In H. Giles (Ed.), *Communication accommodation theory: Negotiating personal relationships and social identities across contexts* (pp. 13–35). Cambridge University Press.
<https://doi.org/10.1017/CBO9781316226537.002>

Summary: This article synthesizes current research on communication adjustment across a range of disciplines. Although scholars in each of these fields are essentially observing the same behaviors and seeking to address many of the same basic questions about human interaction, they describe the phenomenon in different ways depending upon the academic discipline and theoretical frameworks within which they work. However, findings from across multiple research disciplines all support the importance of being able to adjust one's communication style.

Communication adjustment serves two functions: first, to establish common ground such that coherent interaction can take place, and second, to manage social distance between and among individuals in dyadic interaction and within groups. Communication adjustment may be either in response to others, or initiated by a speaker to achieve particular interactional or social ends. People have expectations about what constitutes "appropriate" adjustment in context. Many of these adjustment processes are unconscious and automatic, though they can also be consciously and deliberately practiced.

Source 3: Long-Term Persuasive Effects in Narrative Communication Research: A Meta-Analysis

Citation: Oschatz, C., & Marker, C. (2020). Long-term persuasive effects in narrative communication research: A meta-analysis. *Journal of Communication*, 70(4), 473–496.
<https://doi.org/10.1093/joc/jqaa017>

Summary: This meta-analysis aims to determine if the persuasive effects of narrative communication, known to be strong immediately after exposure, persist over time.

The analysis included 14 studies with immediate and delayed measurements, showing that narrative messages generally have a stronger persuasive impact than non-narrative messages both immediately and over time. Immediate effects were significant across beliefs, attitudes, and intentions, while delayed effects also included behaviors, suggesting that narratives can effectively influence long-term outcomes.

The greater immediate impact of narrative messages is partly attributed to "narrative involvement," which includes transportation into the story and identification with characters. This involvement reduces recipients' resistance (such as counterarguing) because they are engrossed in the narrative rather than the persuasive intent.

Contrary to what might be expected, the study found that the effects of narratives do not diminish faster than those of non-narrative messages. This suggests that narratives can have a sustainable impact, maintaining their persuasive power over time.

Source 4: The Neuroscience of Persuasion and Information Propagation: The Key Role of the Mentalizing System

Citation: Baek, E., Scholz, C., & Falk, E. B. (2020). [The neuroscience of persuasion and information propagation: The key role of the mentalizing system](#). In K. Floyd & R. Weber (Eds.), *Handbook of communication science and biology* (pp. 122–133). Routledge.

Summary: This review summarizes recent contributions from neuroscience on the neural correlates of information propagation, persuasion, and social influence. In particular, the authors consider *mentalizing*, the consideration of others' mental states, as a factor that has identified neural correlates and is recognized in earlier studies as supporting successful information dissemination.

The authors discuss the developmental trajectory of mentalizing, noting that while mentalizing ability initially develops in early childhood, the neural systems involved undergo significant structural and functional development in adolescence. Emerging research also explores how social context and social dynamics influence mentalizing. For example, adolescents with both greater mentalizing brain activity and less dense social networks (i.e., greater social isolation) are more susceptible to persuasion. The authors posit that the mechanisms of persuasion may not only involve engagement of the brain's mentalizing system but also implicate social networks and experiences.

Source 5: Effect of Scientific Argumentation on the Development of Critical Thinking

Citation: Giri, V., & Paily, M.U. (2020). Effect of scientific argumentation on the development of critical thinking. *Science & Education*, 29, 673–690. <https://doi.org/10.1007/s11191-020-00120-y>

Summary: This study examines the role of scientific argumentation in enhancing critical thinking skills within the context of secondary education in biology. Students who participated in a scientific argumentation-based approach showed significant improvement in their critical thinking abilities compared to those who experienced traditional teaching, indicating that the structured argumentation approach effectively develops higher-order thinking skills.

The process of scientific argumentation also inherently involves persuasive communication, where students learn to construct and present arguments that are persuasive in nature, supported by empirical evidence and logical reasoning. In the scientific argumentation process, students practice framing their arguments in ways that are coherent, logically structured, and persuasive. This involves making claims, supporting them with data, using warrants to connect the data with the claims, and considering counterarguments and rebuttals. This allows them to hone their ability to communicate persuasively in scientific contexts.

In addition to enhancing critical thinking skills, incorporating scientific argumentation into the curriculum can help foster a deeper understanding of scientific concepts, preparing students for more rigorous scientific inquiry and problem-solving in real-world contexts.

Conclusion

Persuasive communication skills are important, and are at their strongest when the communicator is able to understand the perspectives of their audience while knowing how to effectively leverage narratives and diverse forms of media. Such skills can be developed in high school classrooms, where practicing these skills can also contribute to deeper learning in domains such as science.

Critical Dialogue

Introduction

Learners question their own and others' reasoning with curiosity and kindness to explore different perspectives. They concisely explain their thought processes and respond to clarifying questions to deepen understanding. Using evidence and reasoning, learners design and justify claims, explaining how evidence supports causal links and refutes alternatives. Additionally, they provide constructive critique to enhance their own and others' understanding.

Source 1: Teaching and Learning Argumentative Reading and Writing: A Review of Research

Citation: Newell, G.E., Beach, R., Smith, J., & VanDerHeide, J. (2011). Teaching and learning argumentative reading and writing: A review of research. *Reading Research Quarterly*, 46, 273–304. <https://doi.org/10.1598/RRQ.46.3.4>

Summary: This article highlights how engaging students in argumentative reading and writing helps them develop empathy. By considering different viewpoints during discussions and debates, students enhance their ability to understand and share the feelings of others, fostering emotional growth.

Argumentative tasks enhance students' cognitive skills by requiring them to critically engage with texts, identify key arguments, analyze evidence, and synthesize information. These activities sharpen critical thinking and reasoning capabilities, important for academic success and effective communication.

The review emphasizes the importance of understanding the structural components of arguments—such as claims, evidence, and warrants—as essential for persuasive writing. It discusses how students learn to construct arguments that effectively persuade readers by logically and sequentially supporting their claims with evidence and clear reasoning. It highlights several pedagogical strategies aimed at improving students' persuasive skills. This includes teaching students how to recognize and use various argumentative structures/narratives.

The authors make a strong case for the pedagogical value of teaching argumentative reading and writing, highlighting how these skills contribute to students' emotional and cognitive development and stressing the need for educational practices that not only focus on the mechanics of argumentation but also on fostering a supportive and interactive learning environment.

Source 2: Socratic Pedagogy: The Importance of Argument

Citation: Nussbaum, M. (2017). Socratic pedagogy: The importance of argument. In *Not for profit: Why democracy needs the humanities* (pp. 47–78). Princeton University Press.
<https://doi.org/10.1515/9781400883509-007>

Summary: This chapter champions Socratic pedagogy, one of the original foundations of the notion of critical dialogue, as essential for civic education. Nussbaum asserts that critical questioning and intellectual independence are necessary for students to achieve high intellectual and civic outcomes, and that as such they are vital for a healthy democracy, where citizens must analyze and critique in order to self-govern.

Drawing on examples from Western and non-Western contexts, the article illustrates how Socratic principles have been integrated into educational practices across various cultures, highlighting their universal relevance and adaptability.

By promoting argumentative skills and critical thinking, Socratic pedagogy not only prepares students for democratic participation but also for personal growth and ethical reflection. It encourages students to examine their values and decisions, contributing to more thoughtful and engaged citizenship.

Source 3: School Mathematics Reform, Constructivism, and Literacy: A Case for Literacy Instruction in the Reform-Oriented Math Classroom

Citation: Draper, R.J. (2002). School mathematics reform, constructivism, and literacy: A case for literacy instruction in the reform-oriented math classroom. *Journal of Adolescent & Adult Literacy*, 45(6), 520–529.

Summary: Innovative and modern mathematics education aims to shift from traditional methods, which often emphasize rote memorization, to a more constructivist approach where knowledge is viewed as something students construct interactively rather than absorb passively. This approach suggests that learning is most effective when students are actively engaged in constructing their own knowledge through interaction with their environment and through solving problems that are meaningful to them.

This article argues that literacy skills—such as reading, writing, and reasoning—are integral to effective mathematics education, as these skills help students understand and communicate mathematical concepts more clearly. The article emphasizes the importance of fostering mathematical discourse in the classroom by encouraging students to articulate their mathematical thinking, engage in discussions, and debate mathematical ideas, which are essential aspects of critical dialogue.

Through constructivist teaching methods, teachers encourage students to question and explore mathematical concepts, which naturally leads to richer dialogues and debates about these topics. This process helps students construct deeper understanding and enhances their critical thinking skills. By integrating literacy into mathematics instruction, teachers can enhance students' understanding and engagement with mathematical concepts.

Source 4: [Peer Assessment](#)

Citation: Topping, K. J. (2009). Peer assessment. *Theory Into Practice*, 48(1), 20–27.

<https://doi.org/10.1080/00405840802577569>

Summary: Peer assessment involves students assessing the level, value, or quality of a product or performance of their peers. Implementation can vary widely in scale (i.e., individual or group assessments) and can cover a range of outputs such as writing, oral presentations, and portfolios.

The process of undertaking peer assessment supports the development of both cognitive and metacognitive skills among both assessors and the assessed. Topping emphasizes that peer assessment should involve not only providing feedback but also engaging in discussions about that feedback that require students to explain and justify their reasoning. This interaction requires students to consider their audience and refine their judgment using critical thinking skills.

Topping also highlights the importance of training students to provide feedback that is not only informative but also supportive and constructive. This includes teaching students how to phrase their feedback in a way that is helpful and encouraging, rather than critical or discouraging. The goal is to enhance learning by helping students identify strengths as well as areas for improvement without diminishing their motivation.

Peer assessment encourages active learning, increases engagement, and can lead to deeper understanding and retention of the material.

Source 5: [Establishing the Norms of Scientific Argumentation in Classrooms](#)

Citation: Driver, R., Newton, P., & Osborne, J. (2000). Establishing the norms of scientific argumentation in classrooms. *Science Education*, 84(3), 287–312.

[https://doi.org/10.1002/\(SICI\)1098-237X\(200005\)84:3<287::AID-SCE1>3.0.CO;2-A](https://doi.org/10.1002/(SICI)1098-237X(200005)84:3<287::AID-SCE1>3.0.CO;2-A)

Summary: This article posits that argument and argumentative practice are central activities in the scientific community, vital for the resolution of scientific controversies and the progression of scientific knowledge. It argues for the inclusion of argumentation in science education to reflect the true nature of scientific processes—contrary to the traditional view of science as merely the accumulation of facts, which fails to engage students in the interpretative and discursive practices that characterize real scientific work.

The authors argue that a number of important skills are practiced through argumentation, such as critical examination (in which students learn to critically examine scientific claims), dialogic argument (the practice of considering and reasoning through multiple viewpoints), and construction of knowledge (argumentation helps in the social construction of scientific knowledge, where students not only learn to formulate their arguments but also to evaluate and understand others' arguments, fostering a deeper understanding of science as a dynamically evolving field).

Conclusion

Critical dialogue not only helps students build deeper knowledge in traditional content areas such as math and science, but also supports the development of important cognitive skills such as critical thinking. Reasoning, explaining that reasoning, using evidence, and engaging in constructive debates and critiques about ideas all play a pivotal role in supporting students' growth in this area.

Computational Thinking

Introduction

Students think critically and solve problems creatively by understanding and leveraging computational tools. They express quantities in multiple ways, break down complex problems into simpler forms, analyze patterns for shortcuts and precision, and verify their results with different methods. As adaptable thinkers, students are capable of tackling diverse challenges, supporting their claims with clear reasoning, and communicating their strategies effectively.

Source 1: [Learning to Think Mathematically: Problem Solving, Metacognition, and Sense Making in Mathematics](#)

Citation: Schoenfeld, A. H. (2016). Learning to think mathematically: Problem solving, metacognition, and sense making in mathematics (Reprint). *Journal of Education*, 196(2), 1–38.
<https://doi.org/10.1177/002205741619600202>

Summary: In this seminal article, the author offers a comprehensive exploration of mathematical thinking, focusing on the processes that support effective problem solving and understanding in mathematics.

The article details various important components of mathematical problem solving, including expressing equivalencies (understanding and expressing mathematical relationships, which include equivalencies), using computational tools (applying formulas, functions, and algorithms flexibly and creatively), pattern recognition (the ability to analyze and generalize patterns in mathematical problem solving), and checking results (reflective thinking, a metacognitive strategy).

The article suggests that curricula should be designed to include opportunities for students to engage deeply with mathematical concepts through problem solving in real-world contexts, metacognitive reflection, and sense making. It calls for instructional approaches that prioritize understanding and critical thinking in mathematics over rote learning, encouraging educators to create learning environments that foster inquiry and reflection.

Source 2: [Computational Thinking](#)

Citation: Wing, J. (2006). Computational thinking. *Communications of the ACM*, 49(3), 33–35.
<https://doi.org/10.1145/1118178.1118215>

Summary: This article, which is largely credited with bringing the idea of computational thinking to the forefront of education discourse, emphasizes that computational thinking is an essential skill for everyone, not just computer scientists. It advocates adding computational thinking to the basic analytical abilities every child should develop, alongside reading, writing, and arithmetic. This skill builds on the power and limits of computing processes and is key to solving complex problems that would be unmanageable without the aid of computational techniques.

Computational thinking enhances problem-solving skills through various methods such as abstraction, automation, and algorithm design. It teaches students to approach problems like computer scientists, making them think at multiple levels of abstraction and consider both human and machine aspects of problem-solving. The article outlines several component skills of computational thinking, including problem decomposition, algorithm creation, evaluation of best solutions, and the use of heuristics. It explains that computational thinking involves understanding the capabilities and limitations of both humans and computers in solving problems.

The author advocates for the integration of computational thinking into all levels of education, suggesting that it should be taught from an early age, and that understanding computational thinking will enable individuals to tackle larger and more complex problems and design systems beyond current capabilities.

Source 3: [Defining Computational Thinking for Mathematics and Science Classrooms](#)

Citation: Weintrop, D., Beheshti, E., Horn, M., Orton, K., Jona, K., Trouille, L., & Wilensky, U. (2016). Defining computational thinking for mathematics and science classrooms. *Journal of Science Education and Technology*, 25(1), 127–147. <https://doi.org/10.1007/s10956-015-9581-5>

Summary: This article underlines the growing importance of computational thinking in the fields of science and mathematics, noting its inclusion in recent educational standards like the Next Generation Science Standards. The integration of computational thinking is seen as essential for preparing students for a technology-driven world, enhancing their problem-solving skills and readiness for future careers in STEM fields.

For high school students, developing computational thinking skills is vital as these skills are foundational for understanding and innovating within any scientific and mathematical context. Embedding computational thinking in science and mathematics education not only enhances domain-specific knowledge but also cultivates a deeper understanding of digital technologies and their applications.

The proposed taxonomy in the article breaks down computational thinking into specific component skills across four main categories: data practices, modeling and simulation practices, computational problem-solving practices, and systems thinking practices. These components are crucial for students to learn as they encapsulate the core aspects of computational thinking relevant to both scientific inquiry and problem-solving in mathematics.

Source 4: Exploring the Effects of Project-Based Learning in Secondary Mathematics Education

Citation: Holmes, V. L., & Hwang, Y. (2016). Exploring the effects of project-based learning in secondary mathematics education. *The Journal of Educational Research*, 109(5), 449–463. <https://doi.org/10.1080/00220671.2014.979911>

Summary: This study examines the effects of project-based learning (PBL) on secondary mathematics education. It finds that PBL significantly enhances academic skills, particularly in algebra and geometry, and fosters motivated learning, including cognitive, social, and motivational aspects. The article stresses the importance of PBL in making mathematics learning more relevant and engaging through real-world applications, which is vital for student success in a knowledge-based economy.

PBL is shown to be particularly effective during adolescence, a critical period for cognitive and social development. Adolescents benefit from PBL's emphasis on collaborative problem-solving and real-world relevance, which are key for deepening understanding and retaining knowledge. The findings of this study suggest that PBL helps bridge the gap between theoretical knowledge and practical application, fostering a deeper understanding of mathematical concepts during these formative years. Several other key skills are also enhanced by PBL, including critical thinking, problem-solving, and collaborative skills.

The article highlights the effectiveness of PBL in narrowing achievement gaps, particularly among at-risk and minority students, suggesting that PBL can serve as a powerful instructional approach to engage all students, regardless of their socio-economic status or ethnic background. The findings strongly support the inclusion of PBL in curricula to enhance engagement, motivation, and learning outcomes in mathematics, pointing out the need for educators to receive proper training to implement PBL effectively.

Source 5: The Role of Domain Knowledge in Creative Problem Solving

Citation: Mayer, R. E. (2016). The role of domain knowledge in creative problem solving. In J. C. Kaufman & J. Baer (Eds.), *Creativity and reason in cognitive development* (2nd ed., pp. 147–163). Cambridge University Press. <https://doi.org/10.1017/CBO9781139941969.008>

Summary: The author argues that successful problem-solving in mathematics requires extensive domain knowledge, categorizing this knowledge into five types: factual, conceptual, procedural, strategic, and metacognitive knowledge. The findings suggest a number of important component skills, including expressing equivalencies and using formulas/algorithms, restating problems, analyzing patterns, and checking results.

The article emphasizes that teaching should not only focus on procedural and factual knowledge, but should also cultivate a deeper understanding of conceptual, strategic, and metacognitive aspects to fully support all phases of problem-solving. This approach not only aids in solving routine problems but also enhances students' abilities to tackle non-routine, creative challenges in mathematics.

In conclusion, a well-rounded educational approach that integrates all five types of domain knowledge can significantly improve mathematical problem-solving skills.

Conclusion

Computational thinking is a fundamental literacy in the modern world. It is important for creative problem-solving in situations involving math and science and requires a range of component skills, such as using computational tools, expressing equivalencies, recognizing patterns, restating problems, and checking results. Project-based learning has been found to be an effective pedagogical practice for developing students' computational thinking.

Mathematical Modeling

Introduction

Students understand, develop, and refine models that represent real-world situations. They identify key quantities, define variables, and parse the relationships among them, assessing the models' accuracy and reliability. By explaining the assumptions behind models, justifying predictions, and communicating solutions effectively, students build a strong foundation for solving complex problems, explaining their solutions, and applying them in new contexts.

Source 1: [The Learning and Teaching of Mathematical Modelling](#)

Citation: Niss, M., & Blum, W. (2020). *The learning and teaching of mathematical modelling* (1st ed.). Routledge. <https://doi.org/10.4324/9781315189314>

Summary: Mathematical models are simplifications of reality that use mathematical language to describe, explain, predict, or control real phenomena. Niss and Blum emphasize the importance of modeling in mathematics education and its application across various learning levels, arguing that modeling serves not only as a tool for solving real-world problems, but also as a crucial element for developing deep mathematical understanding and competencies.

The authors describe the modeling process as a cycle that includes identifying a problem, translating it into mathematical terms, identifying and applying a correct mathematical model, and interpreting the results back in the context of the original problem. Key skills supporting competency in math modeling include problem structuring (defining and delineating the problem within a specific context, identifying the variables involved, and establishing relationships among these variables), making assumptions (simplifying real-world complexities into manageable mathematical forms, and validating these assumptions against the context of the problem), performing appropriate mathematical operations and techniques to derive a solution, and critically evaluating results.

Source 2: [A Systematic Literature Review of the Current Discussion on Mathematical Modelling Competencies: State-of-the-Art Developments in Conceptualizing, Measuring, and Fostering](#)

Citation: Cevikbas, M., Kaiser, G., & Schukajlow, S. (2022). A systematic literature review of the current discussion on mathematical modelling competencies: State-of-the-art developments in conceptualizing, measuring, and fostering. *Educational Studies in Mathematics*, 109, 205–236. <https://doi.org/10.1007/s10649-021-10104-6>

Summary: This review examines a wide range of literature on mathematical modeling competencies published in the past two decades, assessing developments in how these competencies are conceptualized, measured, and fostered. The review analyzed 75 peer-reviewed studies from prominent databases, focusing on the diversity of theoretical frameworks and the variety of research designs used in this field.

The review highlights a substantial richness in empirical approaches to fostering and measuring modeling competencies at various educational levels.

To foster mathematical modeling competencies, studies have found positive results by integrating mathematical modeling tasks directly into the curriculum, developing project-based learning environments where students engage in extended modeling tasks that mimic real-world scenarios, and providing teacher training in mathematical modeling.

Measuring the competencies most often takes place through task-based assessments (requiring the use of modeling skills to solve complex problems and often including open-ended questions to capture students' thought processes and strategies). Several studies use detailed rubrics or scoring guides to evaluate students' performance on modeling tasks. These rubrics often assess specific aspects of modeling, such as the ability to formulate problems, use appropriate mathematical methods, interpret results, and validate conclusions. The authors argue that there is a global consensus on the importance of mathematical modeling competencies in education, reflecting their critical role in solving real-world problems.

Source 3: [Modelling Competencies – Overview](#)

Citation: Greer, B., & Verschaffel, L. (2007). Modelling competencies – Overview. In: Blum, W., Galbraith, P.L., Henn, H.W., & Niss, M. (eds) *Modelling and Applications in Mathematics Education* (New ICMI Study Series, vol 10, pp. 219–224). Springer US. https://doi.org/10.1007/978-0-387-29822-1_22

Summary: This chapter serves as an introductory overview to the concept of modeling competencies within the context of mathematics education. The authors emphasize the importance of developing these competencies as essential elements of contemporary mathematical literacy, vital for both academic purposes and real-world problem solving. They categorize modeling activities into three distinct levels: implicit, explicit, and critical modeling.

In implicit modeling, students engage in modeling activities without explicit recognition that they are doing so. These activities are naturally integrated into various mathematical tasks, allowing students to apply their mathematical knowledge in contextual scenarios without a formal emphasis on the modeling process itself.

In explicit modeling, students engage in tasks where the modeling process is clearly identified and highlighted. Students are made aware of the steps they are taking within the modeling framework (including identifying the problem, formulating it mathematically by quantifying and assigning units to features, analyzing and solving the equations, and interpreting the results), helping them to understand the structure and purpose of their activities. This explicit recognition helps to solidify their understanding of how modeling fits into the broader mathematical and real-world contexts.

In critical modeling, students not only engage in modeling tasks but also critically evaluate the models themselves, the assumptions they are based on, the limitations they face, and the appropriateness of the models for various contexts. This level encourages a deeper reflection on the role of modeling in science, mathematics, and societal issues, promoting a critical understanding of the limitations and potentials of mathematical models.

Source 4: Mathematical Modeling of Tech-Related Real-World Problems for Secondary School-Level Mathematics

Citation: Kohen, Z., & Orenstein, D. (2021). Mathematical modeling of tech-related real-world problems for secondary school-level mathematics. *Educational Studies in Mathematics*, 107, 71-91. <https://doi.org/10.1007/s10649-020-10020-1>

Summary: This article argues for enhancing the relevance of mathematics education by connecting classroom learning to real-world applications in technology. It suggests that mathematical modeling serves as a bridge, making abstract mathematical concepts tangible and understandable while helping students understand the utility of mathematics in solving actual problems they might encounter in their daily lives and future careers, particularly in technology and engineering fields.

The study details a methodological framework for identifying, simplifying, and incorporating tech-related problems into the mathematics curriculum. This involves collaboration with professionals from the tech industry to ensure the problems are authentic yet accessible for secondary school students.

By solving real-world problems through mathematical modeling, students are likely to be more engaged and motivated. This approach also helps in developing higher-order thinking skills, such as problem-solving, critical analysis, and creative thinking.

Source 5: Beyond Motivation: Exploring Mathematical Modeling as a Context for Deepening Students' Understandings of Curricular Mathematics

Citation: Zbiek, R.M., & Conner, A. (2006.) Beyond motivation: Exploring mathematical modeling as a context for deepening students' understandings of curricular mathematics. *Educational Studies in Mathematics* 63, 89–112 . <https://doi.org/10.1007/s10649-005-9002-4>

Summary: This study demonstrates that mathematical modeling is an effective method for deepening students' comprehension of curricular mathematics. The focus is on examining how students apply mathematical modeling to solve real-world problems and how this process helps them understand and retain complex mathematical concepts.

Students who engage in mathematical modeling activities demonstrate a deeper understanding of mathematical concepts by applying them to complex, real-world scenarios. The activities prompt students to engage cognitively at a higher level, utilizing skills such as problem-solving, critical thinking, and application of knowledge. Mathematical modeling helps bridge the gap between abstract mathematical theories taught in the classroom and their practical applications, making the learning experience more relevant and impactful for students.

The findings support the integration of mathematical modeling into the standard curriculum to enhance the depth of students' mathematical understanding, not just to increase engagement or motivation. There is a call for teacher training programs to include strategies for implementing mathematical modeling effectively, ensuring that educators are prepared to guide students through complex modeling tasks.

The authors advocate for a broader perspective on the role of mathematical modeling in education, emphasizing its potential to deepen understanding and enhance practical application skills in students.

Conclusion

Mathematical modeling is a process that includes identifying a problem, translating it into mathematical terms, identifying and applying a correct mathematical model, and interpreting the results back in the context of the original problem. It is effective for developing students' understanding of the standard math curriculum, and has been found to be particularly effective when integrated directly into the curriculum and practiced in project-based learning environments focused on real-world scenarios.

Interpreting Data

Introduction

Students analyze, interpret, apply, and communicate data effectively while recognizing and addressing data limitations. To uncover patterns, make predictions, and support scientific claims, they leverage various data tools and techniques such as spreadsheets for data organization, statistical software for regression analysis, and coding languages for data modeling. Students also use data visualization tools to create visual representations that present complex information in clear, accurate, and compelling ways to inform designs, support decision-making, and communicate in the language of data.

Source 1: Future of Jobs Report 2025

Citation: World Economic Forum. (2022). Future of jobs report 2025. World Economic Forum. <https://www.weforum.org/reports/the-future-of-jobs-report-2025/>

Summary: The latest Future of Jobs Report from the World Economic Forum emphasizes the critical importance of data literacy and the ability to work effectively with data in the face of evolving job markets and technological advancements.

The report highlights how increased digital access and advancements in AI and big data are transforming industries. This shift necessitates a workforce that is proficient in data interpretation and analysis to leverage these technologies effectively.

The report predicts significant job growth in roles such as big data specialists and data analysts, indicating a strong demand for professionals who can interpret and manage data. This trend underscores the importance of data skills for career advancement and job security. The ability to work with data equips workers with the skills needed to adapt to rapid changes in the job market. Data literacy enhances an individual's ability to understand and respond to market trends, economic shifts, and technological transformations.

Organizations that foster a data-literate workforce can better strategize and maintain a competitive edge by harnessing analytical insights to forecast trends and optimize operations.

Overall, the Future of Jobs Report 2025 illustrates that the ability to interpret and utilize data is not only a valuable skill but is becoming essential across various industries and job roles.

Source 2: [Data Literacy: Real-World Learning Through Problem-Solving With Data Sets](#)

Citation: Erwin, R. W. (2015). Data literacy: Real-world learning through problem-solving with data sets. *American Secondary Education*, 43(2), 18–26. <http://www.jstor.org/stable/43694208>

Summary: This article stresses the importance of teaching students to interpret data through the lens of real-world applications. This approach helps students see the relevance of their learning and understand how data interpretation is crucial in various professional and daily life contexts.

The authors argue that data interpretation skills are often best developed through project-based learning, where students engage with real data sets to solve practical problems. This requires educators to provide adequate support and gradual introduction to complex data sets to build students' confidence and skills, especially when using complex real-world data sets. With proper support, this method encourages deep engagement with data and its implications, enhancing students' ability to draw meaningful conclusions from complex information. The article also highlights interdisciplinary approaches; by integrating data literacy across different subjects such as science, social studies, and mathematics, students can appreciate that data interpretation plays a role in various fields.

By learning to interpret data effectively, students enhance their decision-making skills, which are applicable in both academic and everyday contexts. The process of interpreting data also fosters critical thinking, as students must analyze information, discern patterns, and make inferences based on their analysis. A strong foundation in data interpretation prepares students for the demands of the modern, data-driven world, equipping them with essential skills for both higher education and future careers.

Source 3: [Getting Messy with Authentic Data: Exploring the Potential of Using Data from Scientific Research to Support Student Data Literacy](#)

Citation: Kjelvik, M. K., & Schultheis, E. H. (2019). Getting messy with authentic data: Exploring the potential of using data from scientific research to support student data literacy. *CBE—Life Sciences Education*, 18(2). <https://doi.org/10.1187/cbe.18-02-0023>

Summary: This article emphasizes the educational value of using authentic datasets, which are derived from actual scientific research and retain the complexity of real-world data. It discusses the importance of exposing students to data that may be “messy”—featuring variability, outliers, and missing values—to help them develop a realistic understanding of data work and improve their ability to evaluate data quality.

The integration of data science within educational settings involves using sophisticated tools for data analysis, including statistical software that can perform regression analysis and function fitting. The article suggests that effective data literacy involves not only understanding how to use these tools but also knowing when their application is appropriate in a given context.

The article points out that data visualization is a crucial aspect of data literacy. It helps students make sense of complex data sets and draw meaningful conclusions. Visualization tools and platforms can enable students to see patterns, trends, and outliers, making the data more accessible and interpretable.

The paper advocates for the integration of authentic data into the curriculum to make learning more relevant and to enhance students' ability to use data effectively in real-world contexts. Working with authentic data helps students develop a range of skills, from basic data handling and visualization to more advanced analytical skills using various data tools. This approach not only improves data literacy but also prepares students for data-driven environments in academic and professional settings.

Source 4: Data Seen Through Different Lenses

Citation: Konold, C., Higgins, T., Russell, S.J., & Khalil, K. (2015). Data seen through different lenses. *Educational Studies in Mathematics*, 88, 305–325. <https://doi.org/10.1007/s10649-013-9529-8>

Summary: This study discusses the progression of students' understanding of data from elementary through high school. High school students are more likely than younger students to exhibit the more sophisticated perspective of viewing data as an aggregate. This involves recognizing and interpreting broader patterns and trends within data sets, rather than focusing on individual data points or specific cases. Students' critical thinking and statistical reasoning skills are more developed in the high school years, allowing them to engage in more complex data interpretation tasks. However, they still benefit from structured guidance and teaching strategies that promote deeper understanding.

For high school students, the curriculum can be designed to challenge their thinking further and to solidify their understanding of data as an aggregate. This includes integrating more complex statistical analysis tasks that require synthesizing information from large data sets. The study encourages educators to focus lessons on recognizing and analyzing patterns and trends, which are crucial for preparing students for advanced mathematics and statistics courses in higher education.

By targeting high school students' ability to see data through the lens of aggregate analysis, educators can enhance their statistical reasoning skills and prepare them for more advanced studies and real-world data challenges.

Source 5: Studying the Opportunities Provided by an Applied High School Mathematics Course: Explorations in Data Science

Citation: Boaler, J., Conte, K., Cor, K., Dieckmann, J. A., LaMar, T., Ramirez, J., & Selbach-Allen, M. (2025). Studying the opportunities provided by an applied high school mathematics course: Explorations in data science. *Journal of Statistics and Data Science Education*, 33(1), 26–45. <https://doi.org/10.1080/26939169.2024.2333735>

Summary: Teaching data science in high school provides an avenue for students to engage with mathematical content relevant to real life, builds their math identity, and prepares students for college-level statistics. In this study, the authors examined outcomes for students in a high school data science course. Compared to students in traditional math pathways, students who take data science take more math courses overall during their high school careers, and express more positive attitudes toward a future in STEM. More data science students come from underrepresented demographic groups than students in traditional math courses.

Further, students in the data science course reported positive feelings about the course, felt the course was relevant, and appreciated the opportunity to apply math content to real-world problems in creative ways. They also outperformed peers who took the traditional algebra sequence on an assessment of content covered in both courses.

Data science courses can enhance students' experience of math coursework as well as improve their mastery of math content. The authors argue for continued policy support for data science pathways in addition to the traditional math scope and sequence. Their findings suggest that data science pathways are more effective and more deeply engaging for students, especially those who are historically underrepresented in STEM.

Conclusion

The full complement of data interpretation skills includes important elements that are currently missing from conventional math coursework. High school students are at a developmental stage where they are able to engage in more complex data interpretation tasks. Studies highlight that supporting students to work with real-world datasets helps them learn to use data tools, examine data quality, and create meaningful data visualizations. Compared with traditional math coursework, students find real-world data projects more engaging, which in turn supports students' development of math identity and their affinity for a future in STEM majors and careers.

Scientific Investigation

Introduction

Students master the scientific method, from identifying key variables and testing hypotheses to collecting and analyzing high-quality data. They evaluate relationships between variables, distinguishing correlation from causation, and use this knowledge to model complex systems. With these skills, students design rigorous investigations that yield reliable evidence for informed predictions and decision-making.

Source 1: Intrinsic Motivation, Self-Efficacy, and Interest in Science/ Fostering Students' Identification With Mathematics and Science

Citation: Glynn, S. M., Bryan, R. R., Brickman, P., & Armstrong, N. (2015). Intrinsic motivation, self-efficacy, and interest in science. In K. A. Renninger, M. Nieswandt, & S. Hidi (Eds.), *Interest in mathematics and science learning* (pp. 189–202). American Educational Research Association. <http://www.jstor.org/stable/j.ctt1s474j0.17>

Jones, B. D., Ruff, C., & Osborne, J. W. (2015). Fostering students' identification with mathematics and science. In K. A. Renninger, M. Nieswandt, & S. Hidi (Eds.), *Interest in mathematics and science learning* (pp. 331–352). American Educational Research Association. <http://www.jstor.org/stable/j.ctt1s474j0.25>

Summary: These book chapters emphasize the role of intrinsic motivation (motivation for learning science driven by personal curiosity, without reliance on external rewards), self-efficacy (students' belief in their capability to learn and understand science effectively), and interest (both a component of intrinsic motivation and a distinct, persistent trait that directs attention to scientific phenomena and activities) in fostering scientific literacy among high school and college students. The authors argue that these elements are crucial not only for aspiring scientists but for all students to enhance their ability to engage with scientific concepts effectively.

Findings indicate that intrinsic motivation and self-efficacy are mutually reinforcing and strongly associated with students' science achievement. Intrinsic motivation is also supported by fostering students' incorporation of mathematics and science into their sense of identity (e.g., "I am a person who cares about and/or enjoys science"), which is critical for nurturing a sustained interest and commitment to these fields.

Improving science and math outcomes for all learners requires the creation of educational environments that promote personal engagement, personal relevance, and personal identification with mathematical and scientific concepts. This involves integrating real-world applications and problems that students find meaningful and engaging, and includes the use of projects, collaborative work, and opportunities for students to explore their own scientific questions.

Source 2: Effects of Inquiry-Based Approaches on Students' Higher-Order Thinking Skills in Science: A Meta-Analysis

Citation: Antonio, R.P. & Prudente M.S. (2024). Effects of inquiry-based approaches on students' higher-order thinking skills in science: A meta-analysis. *International Journal of Education in Mathematics, Science, and Technology*, 12(1), 251–281. <https://doi.org/10.46328/ijemst.3216>

Summary: This meta-analysis finds that inquiry-based science learning significantly enhances students' higher-order thinking skills. The inquiry process typically involves several key phases: posing questions, conducting investigations, interpreting data, and drawing conclusions regarding causality. Each phase is designed to engage students actively and promote a deeper understanding of scientific concepts. Students are encouraged to explore their own questions and hypotheses, conduct experiments, and use scientific tools and techniques.

This hands-on approach helps students make connections between theory and practice while developing skills such as critical thinking (by evaluating information critically, assessing the validity of their findings, and considering alternative explanations), problem solving (by challenging students to use scientific knowledge to solve complex problems), collaborative skills (through inquiry-based activities that involve students collaborating to design experiments, collect data, and share findings), and data interpretation (interpreting the data gathered during experiments, which helps students learn to apply statistical tools, recognize patterns, and make data-driven decisions).

Inquiry-based science not only develops specific cognitive skills but also increases students' engagement and interest in science. It makes learning more relevant and enjoyable by linking educational content with real-world applications. The article underscores that inquiry-based science is not just about conducting experiments but involves a comprehensive approach to learning that fosters a wide range of cognitive, social, and professional skills.

Source 3: Using Cooperative Learning to Enhance Students' Learning and Engagement During Inquiry-Based Science

Citation: Gillies, Robyn M. (2023). Using cooperative learning to enhance students' learning and engagement during inquiry-based science. *Education Sciences*, 13(12), 1242. <https://doi.org/10.3390/educsci13121242>

Summary: This review explores the effectiveness of cooperative learning strategies in enhancing student engagement and learning outcomes within inquiry-based science education, ultimately emphasizing the importance of active student participation in learning through cooperative group activities that involve inquiry-based science tasks.

Inquiry-based learning is an approach that prompts students to ask questions, explore solutions, and engage deeply with scientific content, thereby enhancing their understanding of scientific concepts. Cooperative learning benefits from carefully structured group work featuring clear roles and responsibilities, shared goals, and mutual support among group members.

The review highlights the critical role of teachers in facilitating cooperative learning by providing the necessary scaffolding, guiding the inquiry process, and fostering an environment conducive to group discussions and collaborative problem-solving.

Students who engage in cooperative learning within an inquiry-based framework tend to develop a better understanding of scientific processes and concepts. Such settings also improve critical thinking, problem-solving, and communication skills, crucial for scientific inquiry and understanding.

The use of cooperative learning in inquiry-based science has been linked to higher academic achievement, greater engagement, and more positive attitudes towards science among students. The article concludes with recommendations for more widespread adoption of cooperative learning in science education, emphasizing its potential to transform science teaching and learning by making it more engaging and effective.

Source 4: Alignment of Hands-On STEM Engagement Activities with Positive STEM Dispositions in Secondary School Students

Citation: Christensen, R., Knezek, G., & Tyler-Wood, T. (2015). Alignment of hands-on STEM engagement activities with positive STEM dispositions in secondary school students. *Journal of Science Education and Technology*, 24(6), 898–909. <http://www.jstor.org/stable/43867753>

Summary: This study investigates a variety of hands-on STEM activities, ranging from laboratory experiments in science classes to engineering design projects and mathematical modeling exercises. The study further assesses how different STEM activities influence students' dispositions toward STEM fields, including their interest, motivation, and potential career aspirations in science, technology, engineering, and mathematics. The findings suggest that hands-on, interactive activities significantly increase student engagement and interest in STEM. Students who participate in these activities demonstrate a deeper understanding of the content and a stronger connection to STEM careers. There is a notable alignment between the type of engagement activities and the development of positive STEM dispositions. Activities that involve real-world applications, problem-solving, and collaboration are particularly effective.

The impact of these activities varies by discipline within STEM. For instance, hands-on science experiments tend to increase interest and understanding in scientific principles, while technology and engineering projects boost creativity and practical skills. The study emphasizes the importance of developing both hard skills, like technical proficiency and data literacy, and soft skills, such as teamwork and problem-solving, through these activities.

The study concludes that engaging students in hands-on STEM activities not only enhances their immediate educational experience but also prepares them for future educational and career paths in STEM fields. Educators are encouraged to design and implement more interactive and interdisciplinary STEM activities that cater to diverse student interests and learning styles.

Source 5: Effect of Scientific Argumentation on the Development of Critical Thinking

Citation: Giri, V., & Paily, M.U. (2020). Effect of scientific argumentation on the development of critical thinking. *Science & Education*, 29, 673–690. <https://doi.org/10.1007/s11191-020-00120-y>

Summary: This study examines the role of scientific argumentation in enhancing critical thinking skills within the context of secondary education in biology. Students who participated in a scientific argumentation-based approach showed significant improvement in their critical thinking abilities compared to those who experienced traditional teaching, indicating that the structured argumentation approach effectively develops higher-order thinking skills.

The study underscores scientific argumentation not just as a learning activity but as a vital pedagogical approach that helps students understand the nature of science as a discipline that evolves through evidence-based discussions and debates. The authors advocate for a shift in science teaching methodologies from traditional rote learning to approaches that involve students in the active construction and evaluation of arguments based on empirical evidence.

Incorporating scientific argumentation into the curriculum can help foster a deeper understanding of scientific concepts and enhance critical thinking skills, preparing students for more rigorous scientific inquiry and problem-solving in real-world contexts.

Conclusion

Science and scientific thinking will always be important. Improving science and math outcomes requires the creation of educational environments that promote personal engagement, personal relevance, and personal identification with mathematical and scientific concepts for all learners. Cooperative learning environments and hands-on activities have been found effective for supporting personal identification with science.

Holders of Foundational Knowledge

XQ Competencies —
Literature Review



Governments

Introduction

Students explore why governments came to exist, how they function, how individual rights are defined, and how policies shape societies. They understand the historic, philosophical, and legal precedents of different government institutions, analyze diverse perspectives, evaluate policy impacts, and advocate for equitable change. These skills prepare students to navigate and influence the complexities of governance, becoming informed citizens and effective leaders in their personal and professional lives.

Source 1: Civic Purpose: An Integrated Construct for Understanding Civic Development in Adolescents

Citation: Malin, H., Ballard, P. J., & Damon, W. (2015). Civic purpose: An integrated construct for understanding civic development in adolescence. *Human Development*, 58(2), 103–130.
<https://www.jstor.org/stable/26765100>

Summary: This study examines the development of “civic purpose” in adolescence and its role in reducing antisocial behavior through emotional and cognitive growth. It highlights that early indicators of purpose often manifest when young people experience empathy in response to the suffering or negative experiences of others, and then take action to alleviate that suffering. This empathetic response not only motivates moral actions but also fosters a sense of civic responsibility and purpose.

Cognitive processes also play a crucial role in the development of effective civic purpose. As adolescents' cognitive abilities mature, they become better equipped to understand moral ambiguity and make ethical decisions, which further contributes to their ability to understand governments, public policies, and perspectives on the rights and responsibilities of individuals.

The integration of emotional empathy and cognitive development can lead to the formulation of a civic purpose, which is defined as a sustained intention to contribute to the world beyond oneself through civic or political action. This purpose is supported by engagement in civic activities, future-oriented civic intentions, and motivation to contribute to the greater good. The development of civic purpose, underpinned by emotional and cognitive growth, contributes to a reduction in antisocial behavior among adolescents.

By engaging in civic activities and harboring a purpose that extends beyond personal gain, adolescents develop a stronger sense of social responsibility and moral identity, which are inversely related to antisocial behaviors.

Source 2: Teaching for Citizenship in 12th Grade Government Classes

Citation: Pace, J. L., & Parker, W. C. (2008). Teaching for citizenship in 12th grade government classes. In J. L. Pace & J. S. Bixby (Eds.), *Educating democratic citizens in troubled times: Qualitative studies of current efforts* (pp. 25–57). State University of New York Press.
<https://doi.org/10.2307/jj.18254005.5>

Summary: This chapter examines the role of high school government classes in fostering political engagement among students. It explores various instructional approaches and their impact on student engagement and understanding of democratic principles.

The chapter emphasizes the importance of teaching the fundamental principles and values of democracy, including the rule of law, the rights and responsibilities of citizens, and the structure and function of government. The authors suggest that government classes should include content that enables students to critically analyze different political systems and their impact on society, and how different political systems around the world address similar issues differently.

The authors conclude by stressing the importance of reevaluating and enhancing the way government classes are taught. They advocate for educational reforms that prioritize civic engagement and critical thinking over rote memorization.

Source 3: The Political Classroom: Evidence and Ethics in Democratic Education

Citation: Hess, D.E., & McAvoy, P. (2014). *The political classroom: Evidence and ethics in democratic education* (1st ed.). Routledge. <https://doi.org/10.4324/9781315738871>

Summary: This book, which is based on a study of 1,001 students and 35 teachers from 21 high schools in Illinois, Indiana, and Wisconsin, makes the case that democratic education is both deeply important for the long-term well-being of democracy and a crucial focus for schools.

Political polarization complicates the teaching of controversial issues, and the authors highlight how societal divides—especially in terms of political ideology—can influence classroom dynamics. The chapter argues that teachers face significant challenges but can play a crucial role in mediating these contextual pressures by fostering a classroom environment conducive to open and respectful political discussion and diversity of thought.

Well-implemented discussions enhance students' interest in politics, improve their understanding of democratic processes, and increase their ability to engage in civil discourse.

The authors emphasize the importance of preparing students not just academically but also as informed and engaged citizens. They argue for the importance of aligning classroom practices with democratic aims to foster students' ability to participate thoughtfully in democratic processes.

Source 4: Preparing the Next Generation for Electoral Engagement: Social Studies and the School Context

Citation: Callahan, R., Schiller, K., & Muller, C. (2010). Preparing the next generation for electoral engagement: Social studies and the school context. *American Journal of Education*, 116(4), 525–556. <https://doi.org/10.1086/653627>

Summary: This study examines the interaction between social studies learning in secondary school and civic participation (voter registration and voting behavior) in late adolescence. The authors argue that social studies programs aim to impart basic civic knowledge, covering fundamental aspects of governance, democracy, and civic responsibilities. This foundational knowledge is crucial for students to understand their roles within a democratic society. But social studies education can also seek to prepare students for active participation in their communities and the broader political landscape. The study investigated the connections among enrollment in social studies courses, connectedness to school, connectedness of peers to school, and civic participation.

The study found that students with higher social studies GPAs are more likely to register to vote and actually vote than those with lower GPAs. The study's findings also indicate that eventual civic participation is influenced by students' sense of connectedness to school, and even more so by peers' sense of connectedness to school. That is, high schools with highly connected students in general produce more civically engaged graduates. Further, students in schools with higher average social studies GPAs are more likely to vote even when their individual sense of connectedness is low. That is, the school context and culture have great influence on students' later participation in civic activities, even when individual factors might suggest lower participation for any particular student.

The authors argue that these findings illuminate two critical paths for educators. First, educators and policymakers should channel resources into social studies programming, and support students academically to succeed in that programming. Second, educators should develop programming of any type that enhances youth connectedness to school. A sense of connectedness among students within a high school has a strong positive impact on later civic behavior.

Source 5: Liberty: Thriving and Civic Engagement Among America's Youth

Citation: Lerner, R. (2004). *Liberty: Thriving and civic engagement among America's youth*. SAGE Publications.

Summary: Building upon evidence from research on biopsychosocial development and evolutionary biology, this book contends that thriving youth, civil society, and vibrant democracy are fundamentally interconnected, which suggests that positive youth development can contribute to addressing global challenges.

The mutual dependency between individuals and their social contexts ideally leads to the flourishing of both the individuals and their social environments when conditions are right. Nurturing and supportive environments are crucial for healthy individual development. These environments also contribute to maintaining and enhancing social institutions that are just, equitable, and democratic. Individuals who develop in such supportive environments exhibit positive characteristics such as commitment to contributing to social justice, propelled by a societal commitment to ensure justice and opportunity for all.

The author argues that in a system characterized by liberty, it is the moral duty of its members to sustain and enhance this system. Possession of such a “noble purpose” can be seen in the alignment between a person’s actions and the broader goals of fostering positive development and supporting a thriving civil society.

Conclusion

The continued participation in civic processes of each new generation is crucial for the maintenance of democracy. Adolescence is a key period in which individuals can begin to develop civic identities, and a good way to support this development is through classroom discussions on topics such as the fundamental principles and values of democracy, the rights and responsibilities of citizens, and the structure and function of government.

COMPETENCY

Politics

Introduction

Students learn to understand and navigate governmental structures, social structures, and economic systems. They analyze power dynamics, differences and similarities among groups, and strategies that drive meaningful change. Learning from social movements of the past, students develop the knowledge, judgment, and leadership skills required to shape the future of their communities and societies.

Source 1: [Civic Engagement and the Transition to Adulthood](#)

Citation: Flanagan, C., & Levine, P. (2010). Civic engagement and the transition to adulthood. *The Future of Children*, 20(1), 159–179. <https://doi.org/10.1353/foc.0.0043>

Summary: This review provides a comprehensive analysis of the civic engagement patterns of young adults, and explores how these patterns have evolved and how they differ across various demographics.

The authors find that civic engagement is critical for democratic societies and for the personal development of young adults. Young adults today are less likely than previous generations to participate in traditional forms of civic engagement, and opportunities for such engagement vary across the economic spectrum and by ethnic group.

Colleges serve as crucial sites for civic integration of young adults, but there are no equivalent institutions for those who do not attend college. Given that the transition to adulthood is marked by significant changes in civic engagement, influenced by broader social, economic, and personal factors, effective civic education during these years is pivotal for fostering engaged and informed citizens. The authors suggest that teachers should focus on creating and enhancing opportunities for civic engagement, particularly in structured settings such as Americorps.

Source 2: [Different Pedagogy, Different Politics: High School Learning Opportunities and Youth Political Engagement](#)

Citation: Kahne, J., Crow, D., & Lee, N.-J. (2013). Different pedagogy, different politics: High school learning opportunities and youth political engagement. *Political Psychology*, 34(3), 419–441. <https://doi.org/10.1111/j.1467-9221.2012.00936.x>

Summary: This study assesses how different pedagogical approaches affect students' engagement with civic and political issues. The authors analyzed two main practices: open discussions of societal issues and service learning. Each has distinct impacts on different types of civic and political engagement.

Open discussion of societal issues tends to promote engagement with traditional political processes (termed "big P" politics), such as voting and political discussions. Service learning is more effective in enhancing community-based and expressive actions (referred to as "little p" politics), like volunteer work and community service.

In light of the variety of learning opportunities that support the development of civic and political engagement among students, the authors encourage teachers to leverage these strategies to promote students' engagement in both "big P" and "little p" politics. This diversity in pedagogical approaches holds significant promise for cultivating a politically and civically active populace.

Source 3: Beyond Idealized Citizenship Education: Embodied Cognition, Metaphors, and Democracy

Citation: Fischman, G. E., & Haas, E. (2012). Beyond idealized citizenship education: Embodied cognition, metaphors, and democracy. *Review of Research in Education*, 36(1), 169–196. <https://doi.org/10.3102/O091732X11420927>

Summary: This article proposes a conceptual framework for education about government and politics that accounts for the cognitive aspects of citizenship. By bringing new research from the cognitive sciences to bear on how we think about civics education, the authors argue for a civic and social studies pedagogy that more closely resembles the way young people experience citizenship, i.e., through a sense of identity, shared goals, and shared experiences with others in their nation and community.

In particular, the authors note that the foundations of civics education have historically referenced a Cartesian, rational actor-type understanding of how to behave as a citizen (defined as a member of society, not as a legal status classification). Civic education of this historic bent "assumes that after a guided pedagogical process, a new identity will emerge" in which students understand, embrace, and will behave in accordance with the civic identity bestowed upon them by this pedagogical process. But, the authors note, identity as a psychological construct is neither developed through direct instruction nor ever truly finished in a developmental sense.

As such, the authors argue, schools should provide information about how American democracy works and what responsibilities students will hold as civic actors in American society in ways that intersect with and leverage the developmental science around students' burgeoning identities. The authors suggest that education in civics, politics, and government must draw linkages between the current state of the global and national community, students' own lives—including in-school and out-of-school experiences—and opportunities for "active participation in authentic democratic activities." This, in turn, can provide a more impactful education in politics and civics than direct instruction alone.

Source 4: Education For Democracy by Young People in Community-Based Organizations

Citation: Checkoway, B. (2013). Education for democracy by young people in community-based organizations. *Youth & Society*, 45(3), 389–403. <https://doi.org/10.1177/0044118X11419535>

Summary: This article explores how community-based organizations are implementing educational programs to foster civic competencies among young people. Amidst declining civic engagement and the diminishing civic mission of public schools, such programs in various regions (South Bronx, Mississippi Delta, Albuquerque, East Oakland) are stepping up to fill the gap by educating youth in democracy.

The author discusses the broader spectrum of civic engagement beyond just voting, emphasizing active participation in community issues, and describes "education for democracy" or "new civics" as crucial for fostering civic competencies among youth, which includes understanding how to navigate and influence political and social systems effectively.

The article references experiential learning and active participation as key pedagogical practices, in contrast to the often passive learning environments found in traditional history education. By participating in these programs, youth develop crucial civic competencies that empower them to effectively engage with their communities.

Source 5: [Teaching Politics in the U.S. History Classroom](#)

Citation: Journell, W. (2014). Teaching politics in the U.S. history classroom. *The History Teacher*, 48(1), 55–69. <http://www.jstor.org/stable/43264378>

Summary: The author advocates for an interdisciplinary approach to teaching U.S. history using a curriculum that not only covers key historical events but also explores their political implications and the policymaking processes involved. This helps students understand historical events within the broader socio-political frameworks of their times, helping them make connections to contemporary issues, and deepening their understanding of electoral processes, civil rights, and how governmental decisions impact society.

Teachers can use simulations and role-play exercises to mimic governmental and political processes. This hands-on approach allows students to experience the complexities of political decision-making and legislative processes. Debates and discussions on historical and current political issues can foster critical thinking and articulate reasoning among students.

Integrating politics into history teaching can enhance students' critical thinking and analytical skills by providing an additional complex domain in which to practice evaluating sources, understanding biases, and developing reasoned arguments.

Conclusion

In addition to knowledge of government, students need an awareness of the nature of politics, and adolescence is a prime phase in which to develop this awareness. To help students more deeply understand how policies are developed and implemented, how policies impact individuals and communities, and how they can be influenced or changed, teachers can give students the chance to engage with and work to resolve community issues or participate in simulations of governmental and political processes.

Economics

Introduction

Students gain a deep understanding of how economic forces, resource access, and competition shape the world. They explore how individual choices, institutions, and policies impact individual behaviors that, in turn, influence markets, societies, and the environment. Learners examine complex economic concepts such as globalization, resource competition, inflation, employment, and economic inequality. As students analyze these issues, they understand the nature of global economic systems and develop solutions to real-world problems.

Source 1: Economic Understanding in US High School Courses

Citation: Walstad, W. B. (2013). Economic understanding in US high school courses. *American Economic Review*, 103(3), 659–663. <https://doi.org/10.1257/aer.103.3.659>

Summary: This study examines how well high school courses prepare students to understand the principles of economics. It assesses the impact of various courses, such as Advanced Placement Economics, honors economics, general economics, and combined government and economics courses.

The results indicate that economic understanding varies considerably across different student demographics and course types. Students enrolled in AP Economics generally showed the highest levels of understanding, followed by those in honors and general economics courses. AP Economics often utilizes a variety of teaching methods, including case studies, real-world application projects, and data analysis activities, which help students understand complex concepts more effectively. The assessment methods in AP Economics are rigorous and designed to test a wide range of skills, from knowledge recall to application and analysis, encouraging students to engage deeply with the material.

The article emphasizes the need for improved economic education in high schools and suggests that a robust economic curriculum is essential for preparing students to understand and navigate the increasingly complex economic issues they will face as adults.

Source 2: Neuroeconomics

Citation: Glimcher, P. W., Camerer, C. F., Fehr, E., & Poldrack, R. A. (2009). *Neuroeconomics*. Academic Press.

Summary: In this authoritative text on the emerging field of neuroeconomics, the authors lay out the history of how the sciences of psychology, economics, and neurology have converged to understand the neural correlates of economic decision-making. Glimcher et al. also examine the current state of the field, providing new findings about how economic decision making happens at the neurological level.

Authored chapters in this book examine the neurobiology and neurochemistry of economic decision making, including issues of risk and reward, assigning value, collaboration and competition, and the impact of context on decision making. Because the field is evolving quickly, the authors note that identifying implications for policy and pedagogy is challenging. Some chapters deal with social cognition and the neurology of emotional and cognitive phenomena such as empathy and bias, which have potential implications for how classroom teachers and students understand and learn about economic theory and practice.

Source 3: Perspectives on Economics in the School Curriculum: Coursework, Content, and Research

Citation: Walstad, W. B., & Watts, M. (2015). Perspectives on economics in the school curriculum: Coursework, content, and research. *The Journal of Economic Education*, 46(3), 324–339. <https://doi.org/10.1080/00220485.2015.1040185>

Summary: The number of high school students taking economics is growing, largely because some states now require the subject. However, the teaching of economics and the content covered vary considerably from state to state and classroom to classroom. Research cited in this review indicates that high-quality economics instruction in high school positively impacts college performance, knowledge about economics in adulthood, and use of financial services such as bank accounts. However, too few students receive high-quality instruction and master content in economics, as indicated by national test scores such as NAEP.

To remedy this, the authors advocate for the integration of foundational economic concepts—such as supply and demand, market equilibrium, and the role of prices—into subjects such as language arts, mathematics, science, history, civics, and business. This interdisciplinary approach ensures that students can see the relevance of economics in various contexts and learn to think critically and ethically about economic issues. Students should also learn to apply economic reasoning to real-world situations and to understand how individuals, businesses, and governments make choices based on resource scarcity and opportunity costs.

The authors conclude with reflections on the necessity of maintaining strong economic education in schools to prepare students for real-world economic challenges and decisions, advocating for sustained efforts to improve the quality and reach of economic education across all grades.

Source 4: Problem-Based Learning and High School Macroeconomics: A Comparative Study of Instructional Methods

Citation: Maxwell, N. L., Mergendoller, J. R., & Bellisimo, Y. (2005). Problem-based learning and high school macroeconomics: A comparative study of instructional methods. *The Journal of Economic Education*, 36(4), 315–329. <https://doi.org/10.3200/JECE.36.4.315-331>

Summary: This study examines the effectiveness of problem-based learning (PBL) compared with traditional lecture-discussion methods, aiming to assess whether PBL enhances high school students' understanding of macroeconomics more effectively than traditional teaching methods.

The results indicate that PBL can improve students' learning of macroeconomics, but that this improvement is likely to vary significantly based on the teacher's effectiveness and familiarity with both PBL and economic concepts.

Source 5: [Recent Research on the Teaching and Learning of Pre-Collegiate Economics](#)

Citation: Miller, S., & VanFossen, P. (2008). Recent research on the teaching and learning of pre-collegiate economics. In: Levstik, L.S., & Tyson, C.A. (Eds.), *The handbook of research in social studies education*, (pp. 284–306). Routledge..

Summary: This chapter highlights the importance of economic literacy, defining it as a critical skill that enables students to understand and participate effectively in the economy. It emphasizes that economic literacy is essential for informed citizenship and personal financial management.

A significant challenge identified is the preparation of teachers. The research indicates that teachers' understanding of economics profoundly impacts students' learning outcomes. Better-prepared teachers are associated with better student performance in economics. Further, economics curricula often lack a consistent standard across different schools and states, making it difficult to implement effective teaching strategies universally. The authors also highlight the need for improved assessment tools that accurately reflect students' understanding of economics.

Conclusion

Economics has long been recognized as an important area of knowledge for a thriving and civically engaged society. Key concepts for high schoolers include microeconomic behaviors and decision making, macroeconomic principles related to market behaviors, the roles and impacts of governmental economic policies, and the impact of resource availability and renewability. Pedagogies that utilize problem-based learning and integrate economics content into other content areas are particularly effective for supporting the development of economic knowledge.

Cultures

Introduction

Students explore the world's cultures, learning about social identities and their historical and global connections. They apply formal methods of inquiry to analyze both the historical origins and present-day function of cultures. Students evaluate critical modern challenges from different viewpoints and in a global context. Ultimately, students make informed, culturally aware decisions and learn to advocate for their community and others.

Source 1: [An Attainable Global Perspective](#)

Citation: Hanvey, R.G. (1982). An attainable global perspective. *Theory Into Practice*, 21, 162–167.
<https://doi.org/10.1080/00405848209543001>

Summary: In this frequently cited article, Hanvey argues that a global perspective is not something one simply has or doesn't have, but rather a blend of various elements and capacities that can be developed over time by focusing on a number of dimensions related to cultural understanding.

For example, recognizing that one's view of the world is shaped by specific cultural and informational influences and that others have different, equally valid perspectives can help individuals accept and appreciate the way cultures themselves are developed and sustained. Understanding global conditions and trends helps individuals understand the interconnected dynamics that define cultures in the globalized world. Acknowledging the range of possibilities available to individuals and societies in responding to global challenges helps build awareness of the impacts of human choices.

The article presents a compelling case for the deliberate cultivation of a global perspective as a necessary component of education in an increasingly interconnected world.

Source 2: [Historical Reasoning: Towards a Framework for Analyzing Students' Reasoning About the Past](#)

Citation: van Drie, J., & van Boxtel, C. (2008). Historical reasoning: Towards a framework for analyzing students' reasoning about the past. *Educational Psychology Review*, 20(2), 87–110.
<https://doi.org/10.1007/s10648-007-9056-1>

Summary: The authors of this review propose a framework to analyze historical reasoning in secondary education, defining it as the ability to organize information about the past to describe, explain, and compare historical phenomena. When learning history, students need to do more than acquire content knowledge about the past; they also need to learn skills such as asking historical questions (encouraging critical engagement with historical material, promoting deeper analysis and understanding), contextualizing (placing events within a broader historical framework to understand their significance and impact), and utilizing substantive and meta-concepts of history (applying specific historical terms and concepts like causation and change to analyze and discuss historical events).

The authors advocate for incorporating these skills into history education to enhance students' understanding and reasoning skills.

Source 3: Critical Consciousness: A Critique and Critical Analysis of the Literature

Citation: Jemal, A. (2017). Critical consciousness: A critique and critical analysis of the literature. *The Urban Review*, 49, 602–626.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC5892452/pdf/nihms893449.pdf>

Summary: Critiquing Freire's seminal work on critical consciousness, Jemal calls for a new socio-cognitive construct of *transformative potential*. Transformative potential combines metacognitive awareness of one's potential to have an impact on the world with the ability and motivation to take actions that create that impact.

This reformulation of students' situational awareness and agency provides educators a clearer theoretical framework for cultivating students' knowledge of historical and current cultures, students' ability to identify complex similarities and differences across cultures, and students' translation of knowledge to action. Pedagogically, the transformative potential framework requires educators to create cycles of reflection and action in order for students to develop an understanding of the origins and histories of world cultures, as well as how they themselves can change or be changed by those cultures. Tools such as critical dialogue, reflective questions, and real-world projects support the development of transformative potential.

Source 4: Agency, Choice and Historical Action: How History Teaching Can Help Students Think About Democratic Decision Making

Citation: Barton, K. C. (2012). Agency, choice and historical action: How history teaching can help students think about democratic decision making. *Citizenship Teaching & Learning*, 7(2), 131–142.

https://www.las2.se/wp-content/uploads/2019/10/Barton_2012.pdf

Summary: This commentary argues that students need deeply refined methods of historical inquiry in order to interpret the past and inform choices in the present. Noting that no culture or nation is truly monolithic, Barton argues that social studies teaching must expose students to historic debates and differences of opinion in order to connect the agency of historic actors to choices students face in their lives today.

Barton suggests that social studies pedagogy should support students in developing not just historical knowledge about culture and society but also a metacognitive awareness that they are agentic individuals within culture and society. This metacognitive awareness helps students contribute to social change and continuity, anticipate the consequences of their actions, and enhance their participation in society.

Conclusion

The study of different societies and cultures enhances students' abilities in critical inquiry, historical reasoning, and reflective action. By examining how perspectives are shaped and cultures are interconnected, students learn to recognize and evaluate different worldviews. Education that emphasizes metacognition, critical dialogue, and the analysis of historical and contemporary challenges equips students with the ability to think critically, make reasoned decisions, and take meaningful action informed by current and historical experience. This approach fosters a sense of agency and responsibility, empowering students to contribute to their communities and navigate an increasingly complex and interconnected world.

COMPETENCY

Artistic Expression

Introduction

Students explore, create, and appreciate art as a powerful tool for personal and community expression and connection. They develop their own artistic abilities, analyze how art reflects lived experiences, and collaborate to support and produce works of art. This skill set fosters creativity, empathy, and cultural awareness.

Source 1: Why We Need Arts Education

Citation: Roege, G. B., & Kim, K. H. (2013). Why we need arts education. *Empirical Studies of the Arts*, 31(2), 121–130. <https://doi.org/10.2190/EM.31.2.EOV.1>

Summary: Arts education correlates with higher academic performance, as measured by indicators such as SAT scores, and has been shown to be linked to creativity, entrepreneurial skills, self-confidence, better mental health, and other life skills.

The article highlights the role of the arts in promoting emotional well-being and social skills, offering students methods for expression and conflict resolution, and discusses the therapeutic uses of art in improving psychological well-being and recovery in clinical settings, emphasizing non-verbal communication and self-expression.

Overall, the article advocates for a balanced educational approach that includes arts education, suggesting that it equips students with critical emotional and cognitive skills necessary for personal and professional success.

Source 2: What is Community-Based Art Education?

Citation: Ulbricht, J. (2005). What is community-based art education? *Art Education*, 58(2), 6–12. <https://doi.org/10.1080/00043125.2005.11651529>

Summary: Community-based arts education encompasses a wide array of programs and initiatives outside traditional K-12 curriculum that aim to enhance art skills, empower students, and integrate learning about local art and culture. This form of education extends beyond the classroom to include informal and community settings, engaging learners in real-world contexts.

This article outlines various objectives of community-based art education, including improving art skills, educating the public about art, and addressing social issues through artistic efforts. Methods range from traditional teaching in community centers to innovative projects that involve public art installations and community engagement. Several examples illustrate the range of community-based art education, from mural projects in urban neighborhoods to ethnographic art studies that involve students in researching and creating art that reflects their community's culture and social issues.

Community-based arts education expands the focus of traditional arts education to include not only the acquisition of art skills but also engagement with the community.

Source 3: Philosophical and Practical Foundations of Artistic Inquiry: Creating Paradigms, Methods, and Presentations Based in Art / Aesthetic-Based Research as Pedagogy: The Interplay of Knowing and Unknowing Toward Expanded Seeing

Citation: McNiff, S. (2018). Philosophical and practical foundations of artistic inquiry: Creating paradigms, methods, and presentations based in art. In: P. Leavy (Ed.), *Handbook of arts-based research* (pp. 22–37).. The Guilford Press.

Bresler, L. (2018). Aesthetic-based research as pedagogy: The interplay of knowing and unknowing toward expanded seeing. In: P. Leavy (Ed.), *Handbook of arts-based research*, (pp. 649–673). The Guilford Press.

Summary: McNiff argues that artistic inquiry should be recognized as a legitimate and vital way of constructing knowledge. He advocates for the unique insights that artistic practices can bring to researchers, emphasizing that these insights are often inaccessible through traditional scientific methods, and suggests that art provides a dynamic and integrative way to engage with the world, allowing for a deeper exploration of subjects that may be complex or abstract. McNiff encourages the use of creative practices within learning processes, arguing that these practices are not just supplementary; they are essential for deepening inquiry and engaging with subject matter in transformative ways.

These methodological innovations highlight the unique contributions of arts-based research, offering fresh ways to approach and understand complex content. Bresler then discusses how aesthetic experiences can significantly enrich educational processes. The chapter emphasizes the role of arts-based learning in enhancing perceptual understanding and expanding the cognitive and emotional engagement of students.

Bresler identifies a number of mechanisms making this pedagogical practice powerful, including that it encourages “multisensory learning” (which can help to cement knowledge more firmly in students’ memory) and emotional and cultural connections to what is being learned, while requiring students to practice applying creative thinking in their problem-solving efforts. Common applications include projects that integrate visual arts, music, and drama into core subject teachings. For example, a history lesson might include creating period-specific art or dramatizing important historical events, helping students to better understand and retain the information by engaging multiple senses.

Source 4: STEAM by Another Name: Transdisciplinary Practice in Art and Design Education

Citation: Costantino, T. (2017). STEAM by another name: Transdisciplinary practice in art and design education. *Arts Education Policy Review*, 119(2), 100–106.
<https://doi.org/10.1080/10632913.2017.1292973>

Summary: The author argues that the integration of arts into STEM (science, technology, engineering, and math) learning enhances students' ability to think creatively and apply knowledge in innovative ways, preparing them for complex problem-solving in real-world contexts. This article underscores that a STEAM approach represents a return to traditional modes of knowledge production, in which arts and sciences are intertwined.

Pedagogies in art and design—such as critical making and object-based learning—are highlighted as effective for STEAM education. Critical making refers to a hands-on approach where the process of creating something helps in understanding complex concepts. This method fosters deep engagement with material and conceptual elements of both art and science. These pedagogies are particularly useful in a STEAM context because they emphasize active, experiential learning, which is crucial for understanding and innovation across disciplines. They help students grasp abstract concepts in tangible ways and encourage an explorative, inquiry-based approach to learning.

Source 5: [Educating for Cultural Citizenship: Reframing the Goals of Arts Education](#)

Citation: Kuttner, P. J. (2015). Educating for cultural citizenship: Reframing the goals of arts education. *Curriculum Inquiry*, 45(1), 69–92. <https://doi.org/10.1080/03626784.2014.980940>

Summary: The author advocates for the importance of cultural citizenship, which involves recognizing and nurturing the cultural capacities and rights of individuals within a community through participation, self-expression, and the development of cultural identity. The article argues that the traditional goals of arts education should be reframed to focus more on developing students' abilities to engage with and contribute to cultural life in meaningful ways. This shift involves moving beyond purely aesthetic appreciation or technical skill acquisition.

Discussing how the arts can encourage social engagement, the author suggests specific educational strategies, including integrating community-based learning experiences, promoting interdisciplinary projects, and encouraging critical reflection on cultural issues through artistic expression.

Conclusion

Artistic expression supports higher academic achievement and psychological and emotional well-being. It is a recognized practice for strengthening individuals' construction of knowledge, to the point that it has been widely integrated into STEM education. It holds further promise for encouraging students' community engagement through participation in local art programs.

COMPETENCY

Art Analysis

Introduction

Students recognize the deep connections among art, history, and society. By analyzing artistic movements, techniques, and trends through multiple lenses—cultural, political, mathematical, scientific, and social—students uncover how art shapes and reflects communities. Learners apply this understanding to create, curate, and evaluate art in ways that inspire their communities.

Source 1: [The Shaping and Reshaping the Aesthetic Brain: Emerging Perspectives on the Neurobiology of Embodied Aesthetics](#)

Citation: Kirsch, L. P., Urgesi, C., & Cross, E. S. (2016). The shaping and reshaping the aesthetic brain: Emerging perspectives on the neurobiology of embodied aesthetics. *Neuroscience and Biobehavioral Reviews*, 62, 56–68. <https://doi.org/10.1016/j.neubiorev.2015.12.005>

Summary: Neuroscientific studies have identified neurobiological underpinnings to aesthetic experiences that are responsible for how sensory information is processed by the brain to create what is perceived as beautiful or pleasing in art and nature. Such studies have found that aesthetic experiences can shape and reshape the brain, indicating that repeated engagement with aesthetically pleasing objects or environments can lead to long-term changes in the neural networks associated with sensory, cognitive, and emotional processing.

For example, regular exposure to art can influence the limbic system, which is involved in emotional regulation. This can enhance an individual's resilience against stress and improve overall emotional well-being. Engaging with art can also stimulate neural networks associated with complex cognitive functions like creativity, problem-solving, and critical thinking. This stimulation can lead to improved cognitive flexibility and enhanced capacity for creative thinking. Art engagement can increase the connectivity between different brain regions, particularly those involved in sensory, cognitive, and emotional processing. Increased connectivity often correlates with higher levels of creativity and better problem-solving abilities. Artistic activities also promote neuroplasticity—the brain's ability to reorganize itself by forming new neural connections throughout life. This capability is crucial for adaptation and learning, allowing the brain to compensate for injury and adjust to new situations or changes in the environment.

Source 2: [Studio Thinking 3: The Real Benefits of Visual Arts Education](#)

Citation: Sheridan, K.M., Veenema, S., & Winner, E., Hetland, L. (2022). *Studio thinking 3: The real benefits of visual arts education*. Teachers College Press.

Summary: Chapters eight and nine of this book emphasize the importance of teaching students to observe and reflect on art more deeply.

Observing details that are not immediately apparent to the untrained eye is crucial in art, where noticing subtle differences in color, form, light, and texture can significantly impact the quality of interpretation. The habit of observation is linked to both scientific and artistic practices. It involves a disciplined way of looking that artists share with scientists, such as observing phenomena or details critically and with curiosity. Observational skills developed in art classes are applicable in various contexts outside the arts, such as in science (observing natural phenomena), humanities (noticing cultural details), and everyday life (enhanced awareness of surroundings).

Reflecting on art requires students to learn to question and explain their own artistic decisions and evaluate their successes and areas for improvement. This reflective practice is crucial for artistic growth. Reflecting also involves developing the ability to participate in critiques of their own and others' works. Through reflection, students develop an awareness of their learning processes, understanding how they create and how they can continue to develop their skills and artistic voice.

These habits help students not only in their artistic endeavors but also enrich their personal and academic growth by fostering skills that are applicable across many areas of life and learning, such as critical thinking and self-awareness.

Source 3: The Sense of Art History in Art Education

Citation: McKeon, P. (2002). The sense of art history in art education. *Journal of Aesthetic Education*, 36(2), 98. <https://doi.org/10.2307/3333760>

Summary: This article examines the integration of art history into art education, advocating for an approach that transcends traditional methods by engaging students as active participants in arts and arts-adjacent fields of inquiry and creation. It introduces the concept of the "artworld," which includes artists, audiences, and artworks, as a holistic framework for understanding art history. This approach is aimed at making art history accessible and relevant to all students, not just those pursuing academic or artistic careers.

The author suggests that art history education should foster the ability to participate actively in the artworld by developing students' skills in historical interpretation, critical analysis, and theoretical application.

The author also advocates for the inclusion of art history and theory in art education. She argues that such an approach enhances the potential for a comprehensive understanding of art and avoids limiting the educational experience to technical skill development.

The ultimate goal, as presented in the article, is to prepare students to be well-rounded individuals who can appreciate, interpret, and engage with art in a meaningful way. This involves a shift from viewing art education as merely a skill-based endeavor to recognizing it as a critical component of general education.

Source 4: Active Learning in Art History: A Review of Formal Literature

Citation: Gasper-Hulvat, M. (2017). Active learning in art history: A review of formal literature. *Art History Pedagogy & Practice* 2(1). <https://academicworks.cuny.edu/ahpp/vol2/iss1/2>

Summary: This review defines active learning as learning that involves student engagement through activities such as discussions, problem-solving, and group projects that go beyond passive listening. A wide range of studies have demonstrated the impact of active learning techniques in enhancing student understanding and retention of art history content. Active learning in art history also supports the development of critical thinking, collaborative skills, and a greater appreciation of art.

The review details specific active learning methods used in art history courses, such as visual analysis exercises, collaborative research projects, and the use of digital tools to create interactive learning experiences. While this review is focused on higher education settings, it is well understood that active learning is invaluable for adolescents, suggesting that its findings are transferable to high school settings.

Source 5: Artifact-Oriented Learning: A Theoretical Review of the Impact of the Arts on Learning

Citation: Peppler, K., Davis-Soylu, H. J., & Dahn, M. (2021). Artifact-oriented learning: A theoretical review of the impact of the arts on learning. *Arts Education Policy Review*, 124(1), 61–77. <https://doi.org/10.1080/10632913.2021.1925609>

Summary: This review introduces the concept of artifact-oriented learning, which highlights the importance of creating and interacting with physical and conceptual artifacts such as artworks in learning.

The authors argue that learning is not an isolated cognitive activity but a socially and culturally embedded process. Because artifacts carry cultural significance, they can be leveraged to contribute to learners' social and intellectual development and to encourage student engagement.

The authors conclude that artifact-oriented learning can provide a rich framework for understanding and enhancing learning through the arts. They emphasize the model's potential to foster a deeper integration of arts into education, advocating for a holistic view of arts education within the learning sciences.

Conclusion

Art has always inspired people, but it has only been recently that scientists have been able to show that this aspect of human nature can go so far as to productively shape and reshape specific parts of the human brain. By actively examining artworks and artistic trends, and taking part in the impacts of art on cultures and communities, students develop observational and analytical skills that support their development while building skills in areas such as science and social studies.

Original Thinkers for an Uncertain World

XQ Competencies —
Literature Review



Creative Process

Introduction

An effective creative process begins with seeking answers and identifying complex new questions. It requires identifying, considering, and generating new ideas or combining existing ones to create something novel. Creativity can also take the form of finding new ways and new places to apply existing ideas. Staying open-minded, iterating, and innovating throughout the process leads to original thoughts, artistic output, and effective solutions.

Source 1: The Development and Enhancement of Adolescent Creativity

Citation: Hoffmann, J. D., & Hills, E. (2021). The development and enhancement of adolescent creativity. In S. W. Russ, J. D. Hoffmann, & J. C. Kaufman (Eds.), *The Cambridge Handbook of Lifespan Development of Creativity* (pp. 139–158). Cambridge University Press.

<https://doi.org/10.1017/9781108755726.010>

Summary: Research suggests that creativity will be a high-demand skill in the coming decades, and also one of the least likely skills to be automated. At an individual level, creativity has been likened to a prerequisite for survival, prosperity, and problem solving as an adult. On a societal level, creativity and innovation are central to the health of the global economy and societal improvement.

Past research has shown that creativity in adults is not simply an extension of childhood creativity, and that the presence of creativity during childhood does not guarantee creativity as an adult. Rather, adolescence is a time when individuals who demonstrated creativity as children may lose their passion for creative pursuits.

The authors conclude that adolescence is a crucial window in the development of creativity, and that school is a critical context for supporting adolescent development of creativity. Educators can nurture adolescent creativity by positing positive beliefs about the nature of creativity (e.g., that creativity is possible and a valuable skill for all people), focusing on the nature of the creative process rather than the outcome (e.g., asking questions, practicing innovative thinking or trying new ideas, and creating new things that build off other work or contexts), and cultivating a belief that peoples' creative skills can develop with practice and effort.

Source 2: Revisiting the Relationship Among Schooling, Learning, and Creativity

Citation: Beghetto, R. A., & Plucker, J. A. (2016). Revisiting the relationship among schooling, learning, and creativity. *Creativity and Reason in Cognitive Development*, 72–91.

<https://doi.org/10.1017/cbo9781139941969.005>

Summary: The authors review research demonstrating that creativity serves as a mechanism for both developing and demonstrating one's understanding. Unless a student can do both, it is impossible to differentiate between memorization and meaningful knowing. Further, they refine the working definition of "creativity" for educators, noting that research definitions of creativity include elements of both novelty and utility, while colloquial definitions often de-emphasize utility. But it is this element of utility that renders creativity so important a skill for students today.

To address this challenge, the authors recommend creating learning environments that de-emphasize rote memorization and time constraints, which can deter teachers from fostering opportunities for creativity. Educators and schools should provide opportunities for student agency and work to ensure that students develop a personally meaningful understanding of the problem, the solution, and the strategy used to arrive at the solution.

Learning and creativity complement each other, and creativity undergirds other critical competencies necessary for success in college, work, and life.

Source 3: Domain-Generality Versus Domain-Specificity of Creativity

Citation: Sternberg, R. (2009). Domain-general versus domain-specificity of creativity. In P. Meusburger, J. Funke, & E. Wunder (Eds.), *Milieus of creativity: An interdisciplinary approach to spatiality of creativity* (pp. 25–38). Springer. https://doi.org/10.1007/978-1-4020-9877-2_3

Summary: There is a distinction between creativity that is domain-general (the broad ability to think creatively across a variety of contexts and types of problems) and creativity that is domain-specific (specific to particular areas or tasks, such as academic or interpersonal problem-solving). Although there is ongoing debate about whether creativity is predominantly domain-general or domain-specific, it is most important to recognize that creativity as a cognitive phenomenon most frequently contains elements of both.

Unpacking this idea, the author discusses studies showing that environments play a significant role in fostering or suppressing creativity, that broad knowledge across disciplines can enhance creativity by introducing diverse perspectives, and that emphasis should be placed not only on acquiring knowledge but also on learning how to think critically and creatively about it.

Source 4: Predictors of Change in Creative Thinking Abilities in Young People: A Longitudinal Study / Predictors of Creativity in Young People: Using Frequentist and Bayesian Approaches in Estimating the Importance of Individual and Contextual Factors

Citation: Asquith, S. A., Wang, X., Quintana, D. S., & Abraham, A. (2024). Predictors of change in creative thinking abilities in young people: A longitudinal study. *Journal of Creative Behavior*, 58(2), 262–278. <https://doi.org/10.1002/jocb.647>

Asquith, S., Wang, X., Quintana, D., & Abraham, A. (2022). Predictors of creativity in young people: Using frequentist and Bayesian approaches in estimating the importance of individual and contextual factors. *Psychology of Aesthetics, Creativity, and the Arts*, 16(2), 209–220.
<https://doi.org/10.1037/aca0000322>

Summary: These two related studies explore how personal traits and activities influence creative potential in young people ages 14–20. The researchers found that openness—curiosity and willingness to try new things—is not just related to creativity, but actually predicts future creative development. They also found that participating in creative hobbies supports the ongoing development of creative thinking. In fact, creative fluency—the ability to generate ideas—improved over two years when individuals participated regularly in creative activities that they found personally interesting.

These studies affirm that both individual traits and contextual factors impact the development of creativity. Creativity can be cultivated by creating conditions that provide young people with regular opportunities to try new things (e.g., through asking questions and/or making new connections) and practice creative activities (e.g., through regularly creating original works and/or developing new ideas).

Source 5: [Rethinking Creativity: Inside-the-Box Thinking as the Basis for Innovation](#) (Chapter 1)

Citation: Weisberg, R. W. (2020). *Rethinking creativity: Inside-the-box thinking as the basis for innovation*. Cambridge University Press. <https://doi.org/10.1017/9781108785259>

Summary: The author argues that, contrary to the "genius view" of creativity—which involves sudden leaps of inspiration—creativity is actually often incremental, leveraging "in-the-box thinking" that involves logical steps and continuous refinement. To put the argument simply, even the most novel creations build on ideas and work that already exist.

The creative process relies in large part on analogical thinking, a cognitive phenomenon in which a creator transfers knowledge across domains based on shared features, contexts, and/or structures and applies that knowledge to new domains in order to generate wholly novel solutions. Educators can foster creativity by giving students opportunities to practice connecting relevant ideas in addition to supporting them in the type of *sui generis* ideation we more typically think of as "creativity."

The research suggests that developing creativity in school contexts is likely to be most effective when integrated across curriculum areas, and when supported by a school culture prioritizing curiosity and asking questions, creative thinking across disciplines, and openness to trying innovative new ideas.

Conclusion

Creativity is an important skill for both students' future success and their learning in school. Creativity can be nurtured in school settings by de-emphasizing rote memorization and encouraging the creative application of knowledge in real-world contexts, especially in ways that prompt students to draw connections between ideas, ask questions, and share original works and ideas.

Sharing Ideas

Introduction

Intentionally seeking and embracing feedback with curiosity and sensitivity is vital to effective idea sharing. Actively engaging diverse groups, processing their input with an open mind, and thoughtfully considering the broader implications of ideas while crafting solutions ensures maximum impact. True innovation lies in considering ideas that challenge norms, strategically seeking feedback to anticipate potential impact, and explaining how these ideas benefit the field or society.

Source 1: Creativity, Critical Thinking, Communication, and Collaboration: Assessment, Certification, and Promotion of 21st Century Skills for the Future of Work and Education

Citation: Thornhill–Miller, B., Camarda, A., Mercier, M., Burkhardt, J. –M., Morisseau, T., Bourgeois–Bougrine, S., Vinchon, F., El Hayek, S., Augereau–Landais, M., Mourey, F., Feybesse, C., Sundquist, D., & Lubart, T. (2023). Creativity, critical thinking, communication, and collaboration: Assessment, certification, and promotion of 21st century skills for the future of work and education. *Journal of Intelligence*, 11(3), 54. <https://doi.org/10.3390/jintelligence1030054>

Summary: Communication is ranked as one of the most critical skills for the future workforce, with a direct impact on employability and personal success. Further, communication is a widely agreed-upon essential competency for success in education and civic participation.

Communication includes linguistic or verbal skills, appropriate use of language, and social skills such as empathy and understanding others' perspectives. Effective communication is interlinked with—that is, supportive of and supported by—other essential skills such as collaboration, critical thinking, and the ability to generate and share creative ideas.

While this interrelatedness among communication and other essential skills can enhance teaching and learning, it can also pose a challenge when assessing these skills. Strategies such as small group discussions, role-playing, project-based learning, and simulations can improve students' communication competency and support the development of their cognitive potential.

Source 2: We Feel, Therefore We Learn: The Relevance of Affective and Social Neuroscience to Education

Citation: Immordino–Yang, M. H., & Damasio, A. (2007). We feel, therefore we learn: The relevance of affective and social neuroscience to education. *Mind, Brain, and Education*, 1(1), 3–10. <https://doi.org/10.1111/j.1751-228X.2007.00004.x>

Summary: This article explores the connection between emotion, social functioning, and cognitive activities such as learning, paying attention, and decision-making. The authors demonstrate that cognitive functions are deeply integrated with emotional processes, suggesting that emotional states are not just influential but foundational to cognitive operations in educational settings. Further, meta-cognition and emotional awareness are both cornerstones of effective self-management, which in turn supports effective learning and decision-making. Self-awareness of both one's cognitive and one's emotional states is essential for students to effectively navigate and utilize their emotional and cognitive resources in academic and social environments.

The authors discuss the role of emotion regulation in cognitive phenomena such as decision-making and creativity. They emphasize that emotions are fundamentally intertwined with cognition, affecting learning, attention, memory, decision-making, and social functioning. Emotions, particularly social emotions like compassion, embarrassment, and guilt, all play a critical role in enabling or hindering both rational and creative thinking.

Source 3: Longitudinal Research on Perspective Taking in Adolescence: A Systematic Review

Citation: Hall, H.K., Millear, P.M.R., Summers, M.J. & Isbel, B. (2021). Longitudinal research on perspective taking in adolescence: A systematic review. *Adolescent Research Review*, 6, 125–150. <https://doi.org/10.1007/s40894-021-00150-9>

Summary: Perspective-taking, defined as the cognitive ability to understand others' viewpoints and emotions, is a critical socio-cognitive skill during adolescence, a period marked by dynamic social and psychological growth.

Perspective-taking abilities increase gradually from ages 13 to 18 and can be developed in school settings. Adolescents with strong perspective-taking skills exhibit higher self-esteem, reduced depressive symptoms, and greater psychological well-being. Perspective-taking is positively associated with prosocial behaviors, such as altruism, cooperation, and moral reasoning, and negatively associated with antisocial behaviors like aggression and prejudice.

Source 4: The Impact of Peer Assessment on Academic Performance: A Meta-Analysis of Control Group Studies / The Power of Feedback

Citation: Double, K.S., McGrane, J.A., & Hopfenbeck, T.N. (2020). The impact of peer assessment on academic performance: A meta-analysis of control group studies. *Educational Psychology Review*, 32, 481–509. <https://doi.org/10.1007/s10648-019-09510-3>

Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research*, 77(1), 81–112. <https://doi.org/10.3102/003465430298487>.

Summary: These two studies explore the meaning and role of feedback in high school learning. In “The Impact of Peer Assessment,” the authors provide a meta-analysis examining 54 experimental and quasi-experimental studies that evaluate the effect of peer assessment on academic performance in primary, secondary, and tertiary students across subjects and domains. Peer assessment—seeking feedback from fellow classmates, whose perspectives may differ from those of a teacher or one another—has the potential to offer student insights into the learning process, which in turn can enhance later performance.

For high school students, peer assessment is more effective for encouraging learning than receiving no assessment or teacher-driven assessment (although peer assessment does not have a significantly different impact than self-assessment). Its effectiveness remains stable regardless of the mode of delivery (online or offline), anonymity, or whether the feedback was scaffolded or freeform. Repeated peer assessment yielded better outcomes compared to a single instance.

The authors of “The Power of Feedback” argue further that effective feedback can motivate students, clarify their understanding, and improve learning outcomes by addressing specific aspects of their performance. When feedback is effectively targeted, it can increase students’ effort, attention, and engagement, helping them refine their approach to tasks and improve their performance.

To create a productive culture of feedback, teachers should establish classroom climates that encourage peer and self-assessment and foster open discussions about learning goals. The paper identifies different levels at which feedback can be directed—task-level, process-level, self-regulation, and personal feedback—and highlights that feedback is most beneficial when it guides students in developing strategies for self-regulation and self-assessment, helping them become more independent and reflective learners.

Source 5: Projection and Anticipation: The Forward-Looking Nature of Embodied Communication

Citation: Streeck, J., & Jordan, J. S. (2009). Projection and anticipation: The forward-looking nature of embodied communication. *Discourse Processes*, 46(2–3), 93–102.
<https://doi.org/10.1080/01638530902728777>

Summary: This article looks at how both individual factors, such as body movements and the way we talk, and contextual factors, such as where we hold a conversation, influence what will happen next in a conversation. This helps people interact smoothly and understand each other better. For example, we often use gestures or change our tone to give clues about what we’re going to say next, helping the other person get ready for their turn to speak. Similarly, people often copy each other’s body language without thinking, which helps them get along and communicate better. Hand movements and other gestures are not just for emphasis—they actually help make the conversation clearer and guide the flow of dialogue.

This subconscious interaction of the individual, their conversation partner, and their context enhances the effectiveness of communication. Students need to understand how to move fluidly from one context to another and from interlocutor to interlocutor, which requires educational environments that mimic complex, real-world contexts and obligate students to practice communicating with different partners.

Conclusion

The competency of sharing ideas, which includes seeking feedback and considering others' perspectives, plays a critical role in enhancing learning. Studies consistently highlight the value of seeking diverse perspectives, refining ideas through feedback, and strategically considering the impact of one's contributions. This process not only supports individual learning but also drives innovation and societal benefit.

Interpreting Information

Introduction

Today, access to information is almost unlimited. But that information only becomes useful when it is analyzed, evaluated, and related to what one already knows. Effective information interpretation reveals what's true and relevant, helps connect the dots to see the bigger picture, and leads to smarter decisions and problem-solving.

Source 1: Strategies for Teaching Students to Think Critically: A Meta-Analysis

Citation: Abrami, P. C., Bernard, R. M., Borokhovski, E., Waddington, D. I., Wade, C. A., & Persson, T. (2015). Strategies for teaching students to think critically: A meta-analysis. *Review of Educational Research*, 85(2), 275–314. <https://doi.org/10.3102/0034654314551063>

Summary: This meta-analysis looks at strategies for teaching critical thinking skills, reviewing empirical evidence to assess instructional approaches and their effectiveness. The authors define critical thinking as purposeful, self-regulatory judgment encompassing skills such as analysis, evaluation, and inference. They find that instructional interventions can help students develop critical thinking skills across educational levels and disciplines, that effective critical thinking instruction often integrates subject matter content with explicit training, and that leveraging real-world scenarios to contextualize learning enhances critical thinking skills.

The authors include skills such as interpretation (i.e., categorization, decoding significance, clarifying meaning) and analysis (examining ideas and identifying arguments) as core components of critical thinking.

Source 2: Teaching and Learning Argumentative Reading and Writing: A Review of Research

Citation: Newell, G.E., Beach, R., Smith, J., & VanDerHeide, J. (2011). Teaching and learning argumentative reading and writing: A review of research. *Reading Research Quarterly*, 46, 273–304. <https://doi.org/10.1598/RRQ.46.3.4>

Summary: The authors underscore that the ability to identify main claims and supporting arguments is crucial for engaging with complex texts that require analysis and synthesis. They also argue that students need to be able to evaluate the relevance and strength of the evidence presented in support of arguments, which includes assessing whether the evidence logically supports the claims and considering counterarguments, both of which are essential for deep comprehension and critical engagement with texts.

Students benefit from opportunities to learn strategies to manage information, such as using outlines or templates. These aids can help students organize their thoughts and evidence coherently, which is vital for effective argumentation and critical reading.

Educational practices that foster both cognitive understanding of argument structures and social interaction, where students learn to use these skills in collaborative and real-world contexts, can help improve students' ability to critically engage with texts and articulate well-reasoned arguments, enhancing their overall academic literacy.

Source 3: Using Messy, Authentic Data to Promote Data Literacy & Reveal the Nature of Science

Citation: Schultheis, E. H., & Kjellvik, M. K. (2020). Using messy, authentic data to promote data literacy & reveal the nature of science. *The American Biology Teacher*, 82(7), 439–446. <https://doi.org/10.1525/abt.2020.82.7.439>

Summary: The authors argue that students should be given the chance to work with “messy data”—defined as data that include natural variations, unexpected outliers, and incomplete information, reflecting the quality one typically finds when examining real-life data—to enhance the relevance and impact of their learning experience. Working with authentic data also helps demystify the scientific process, showing that science is not a linear path but rather a complex and iterative process of discovery.

The article discusses strategies for introducing students to messy data gradually, using scaffolding techniques to build their skills and confidence in data handling. Specific examples and activity sequences are provided to help educators integrate messy data into their curricula, starting with simple tasks and progressing to more complex data analysis.

The complexity and unpredictability of messy data can be daunting for both students and educators. The article suggests methods to mitigate these challenges through structured learning experiences and supportive teaching practices. The article concludes by stressing the necessity of data literacy in modern science education, advocating for a systemic inclusion of authentic data experiences across all levels of science education to prepare students for data-driven futures.

Source 4: Data Literacy: Real-World Learning Through Problem-Solving With Data Sets

Citation: Erwin, R. W. (2015). Data literacy: Real-world learning through problem-solving with data sets. *American Secondary Education*, 43(2), 18–26. <http://www.jstor.org/stable/43694208>

Summary: The author argues that data literacy is crucial for preparing students for modern, data-driven environments, both in higher education and the workforce, and advocates for incorporating data literacy into secondary education through project-based learning (PBL).

The article suggests that engaging students with authentic data sets enhances their ability to apply mathematical and scientific concepts in real-world contexts. Using PBL, teachers can engage students in using data to address complex, real-world problems, which fosters deeper learning and engagement.

Handling data sets allows for an interdisciplinary learning experience, connecting mathematics, science, and technology with global issues and social studies. The article discusses the need for projects to be developmentally appropriate, ensuring that the complexity of data and analysis tools aligns with students' educational levels.

Source 5: Recognizing and Addressing the Barriers to Adolescents' "Reading Like Historians"

Citation: Nokes, J. D. (2011). Recognizing and addressing the barriers to adolescents' "reading like historians." *The History Teacher*, 44(3), 379–404. <http://www.jstor.org/stable/41303991>

Summary: Reading like a historian involves not just understanding textual information, but also analyzing, interpreting, and synthesizing multiple sources to build a cohesive historical understanding.

Students are often challenged by key elements of reading like a historian. For example, they may struggle to identify arguments and themes in historical texts, especially primary sources, when the language and contextual references are unfamiliar. As a result, they frequently default to trusting textbooks over primary sources. Students also face significant challenges in managing and synthesizing information from various texts, which is compounded by a lack of historical background knowledge.

The article discusses several educational strategies to improve students' ability to read like historians, such as direct instruction on historians' heuristics, scaffolding their learning experiences, and encouraging the use of multiple sources to enhance depth of understanding and critical thinking.

Conclusion

Extracting main ideas, analyzing them for connection to a central question or thesis, and organizing them effectively for future use and/or targeted communication are key skills for both high school students and adults. Research paints interpreting information as a cornerstone of critical thinking in content areas such as social studies and science, where competing claims often vie for recognition and acceptance. Further, this skill supports information literacy, which is widely identified as particularly important in today's landscape of social media.

Logical Thinking

Introduction

Logical thinkers critically analyze assumptions, recognize patterns, and use reason to make effective decisions. They identify and evaluate what beliefs or biases may be informing an idea and support their reasoning with evidence. Logical thinking leads to well-informed choices, helps manage bias, and builds confidence to meet and overcome challenges at school, the workplace, or in one's personal life.

Source 1: Learning How to Manage Bias: A Case Study of Youth Participatory Action Research

Citation: Kirshner, B., Pozzoboni, K., & Jones, H. (2011). Learning how to manage bias: A case study of youth participatory action research. *Applied Developmental Science*, 15(3), 140–155.
<https://doi.org/10.1080/10888691.2011.587720>

Summary: This case study focuses on the cognitive challenge involved in balancing emotional investment and the need for objectivity when handling data that could confirm or disprove one's initial assumptions.

The study takes place in a project where youth study problems that directly impact them, blending activism with inquiry. The process involved requiring participants to use personal experience for insight while remaining open to data that might challenge their views, in an effort to help them learn to distinguish between personal beliefs and external evidence.

The authors highlight that participants were successful in learning to navigate their personal biases, including in cases where their findings contradicted their expectations or desired outcomes. The paper suggests that learning to manage biases in a research setting can have broader implications for educational practices and developing critical thinking skills among youth.

Source 2: Creativity and Reason: Friends or Foes?

Citation: Guignard, J.-H., & Lubart, T. (2016). Creativity and reason: Friends or foes? In J. C. Kaufman & J. Baer (Eds.), *Creativity and reason in cognitive development* (2nd ed., pp. 282–301). Cambridge University Press. <https://doi.org/10.1017/CBO9781139941969.015>

Summary: This article examines the intricate relationship between logical thinking and creativity, particularly focusing on how these cognitive processes influence each other throughout development.

Logical thinking requires a learner to develop and support assertions based on a set of arguments, providing the basis for action or decisions. The development of logical thinking in adolescence, including formal processes and strategies that involve hypothetical and abstract thinking, is crucial for both complex problem-solving and creativity. Mature reasoning involves the integration of subjective and objective thinking, where logical reasoning processes help form a deeper understanding and can generate novel insights. Logical thinking involves, among other cognitive skills, the ability to analyze multiple possibilities. This capability allows individuals to explore various solutions and engage in divergent thinking, a key component of creativity.

Emphasizing the development of reasoning as an essential educational goal, the article suggests that fostering both creative and logical abilities can lead to more robust cognitive development. It advocates for educational approaches that do not merely focus on rote learning but encourage an integration of logical reasoning with creative thinking.

Source 3: [Don't They Teach That in High School? Examining the High School to College Information Literacy Gap](#)

Citation: Saunders, L., Severyn, J., & Caron, J. (2017). Don't they teach that in high school? Examining the high school to college information literacy gap. *Library & Information Science Research*, 39, 276–283. <https://doi.org/10.1016/j.lisr.2017.11.006>

Summary: College librarians and faculty frequently express concerns about incoming students' information literacy skills, which include the ability to locate, evaluate, and use information effectively. This study aimed to determine if information literacy skills are taught in high schools, retained by students, and aligned with college expectations. The researchers conducted parallel surveys among high school and college librarians across the United States to gather insights on the teaching and perception of information literacy skills.

While information literacy is part of high school curricula, the study finds a gap in the retention and application of these skills when students transition to college. Both high school and college librarians agree on the importance of certain information literacy skills, but they observe that students often fail to transfer these skills to college-level work, and that there is a notable disconnect between the skills taught in high school and those applied or expected in college settings. High school and college librarians have different priorities and emphases in teaching information literacy, which may contribute to the perceived gap.

The study underscores the critical role of information literacy in academic success and calls for strategic interventions to align high school teaching with college expectations.

Source 4: [The Second Decade: What Develops \(and How\)](#)

Citation: Kuhn, D., & Franklin, S. (2006). The second decade: What develops (and how). In D. Kuhn, R. S. Siegler, W. Damon, & R. M. Lerner (Eds.), *Handbook of child psychology: Cognition, perception, and language* (6th ed., pp. 953–993). John Wiley & Sons, Inc..

Summary: Adolescence is a stage during which individuals develop and refine both deductive and inductive reasoning. Adolescents improve in their ability to use deductive reasoning, a core component of logical thinking, as they move from concrete operations to more abstract formal operations. This transition involves not only greater abstraction but also less egocentrism in thought. Adolescents also develop inductive reasoning skills, learning to draw conclusions based on evidence and to integrate theory with evidence. Inductive reasoning involves interpreting data to form logical conclusions about causality and other relationships.

There is significant individual variability in the development of logical thinking during adolescence. Some adolescents may develop these skills earlier or more completely than others, influenced by factors such as personal interests and educational settings/opportunities.

The authors emphasize the need to support students in developing logical thinking through teaching strategies that encourage active engagement with logical problems and promote the integration of abstract reasoning skills.

Source 5: Cognitive and Affective Theory of Mind in Adolescence: Developmental Aspects and Associated Neuropsychological Variables

Citation: Gabriel, E. T., Oberger, R., Schmoeger, M., Deckert, M., Vockh, S., Auff, E., & Willinger, U. (2021). Cognitive and affective Theory of Mind in adolescence: developmental aspects and associated neuropsychological variables. *Psychological Research*, 85(2), 533–553.
<https://doi.org/10.1007/s00426-019-01263-6>

Summary: Theory of Mind (ToM), the ability to understand that others have unique thoughts and perspectives that are different from one's own, is an important foundation for many cognitive skills. A fundamental component of cognitive ToM is the ability to recognize that others can hold false beliefs that differ from reality. Another component is perspective taking, where adolescents learn to distinguish between their own views and the views of others, understanding that different perspectives can coexist.

While the emergence of ToM in early childhood is well understood, this study explores the development of ToM during adolescence to examine how these capabilities are linked to various neuropsychological functions. The authors find connections among memory, attention, language comprehension, verbal fluency and flexibility, affective intelligence—all neuropsychological phenomena known to increase in sophistication throughout adolescence—and ToM. These findings suggest that social cognition in adolescents, and in particular ToM, develops via the neuropsychological integration and maturation of cognitive and emotional processes.

The substantial refinement of ToM during adolescence indicates ways to support adolescents in both academic work and in navigating social contexts. The ability to take other perspectives, and to understand that perspectives may stem from false beliefs, is a necessary precursor for skills ranging from productive collaboration to historical analysis. The maturation of ToM also supports young people in developing thinking and interpersonal skills that allow them not just to understand that others have different thoughts and perspectives, but to examine and analyze these other perspectives in order to better understand their own perspectives and biases.

Conclusion

Adolescence is a key period for the development of logical thinking, including the ability to inductively and deductively reason about thoughts and ideas, consider the merits and weaknesses of arguments and evidence, and consider one's own and others' potential biases. Pedagogical practices that support the development of logical thinking include exercises that integrate abstract reasoning, analysis of information, and divergent perspectives on issues that are relevant to students' lives.

Synthesis

Introduction

Think of synthesis as the art of weaving together strands of information to form a rich tapestry of understanding. It's not just about gathering facts from here and there; it's about using the varied information threads to create a vibrant, more colorful expression. The act of synthesis transcends subjects, as learners actively question, "What do I know? How do I know it? How are these ideas connected?" It elevates thinking to new heights, enabling new levels of innovation and creation.

Source 1: The Synthesizing Mind: Making Sense of the Deluge of Information

Citation: Gardner, H. (2007). The synthesizing mind: Making sense of the deluge of information. In M. Sánchez Sorondo, E. Malinvaud, & P. Léna (Eds.), *Globalization and Education: Proceedings of the Joint Working Group, The Pontifical Academy of Sciences, The Pontifical Academy of Social Sciences, 19 – 17 November 2005, Casino Pio IV* (pp. 3–18). De Gruyter.
<https://doi.org/10.1515/9783110207019-004>

Summary: The author defines synthesis as the capacity to absorb information from various sources, judge its importance and relevance, and integrate it into a comprehensive and comprehensible structure. Synthesizing involves critical thinking, selecting relevant data, and creatively combining insights to form a new, holistic understanding. The difficulties of synthesis include biases, overwhelming data volumes, and the cognitive demand of integrating diverse information.

It is important, the author argues, to teach synthesis explicitly in educational settings, particularly through project-based learning where students practice integrating diverse information sources. He suggests incorporating exercises that build synthesis skills, like summarizing complex texts or engaging in interdisciplinary projects that require integrating various types of knowledge.

The article highlights that synthesis is essential not only in academic contexts but also in professional and everyday life, helping individuals make informed decisions and develop innovative solutions. Gardner concludes that as information continues to grow exponentially, the ability to synthesize will become even more critical, marking it as a key skill for future success in numerous fields.

Source 2: Factors Affecting Students' Information Literacy as They Transition from High School to College

Citation: Varlejs, J., Stec, E., & Kwon, H.E. (2014). Factors affecting students' information literacy as they transition from high school to college. *School Library Research*, 17.

Summary: This study investigates why students entering college often lack adequate information literacy skills, despite efforts to teach these skills in high school. College educators have noted poor information literacy among incoming students, suggesting a gap between high school training and college expectations.

The study explores this gap by examining the instructional environments in high schools and their impact on students' preparedness for college-level research. In particular, the integration of librarians into high school curriculum and skill-building lags behind that of post-secondary librarianship, creating a barrier for high schoolers in accessing a key resource for information literacy development. Many librarians report insufficient collaboration with teachers, which is critical for effective information literacy instruction. Challenges such as budget limitations and staffing issues were common, affecting the quality of library services. There is also significant variation in how information literacy is taught across different schools, with no consistent standard.

The study highlights the need for better cooperation between high school librarians and teachers to improve students' information literacy and prepare them for the demands of college research.

Source 3: Teaching and Learning Information Synthesis: An Intervention and Rubric Based Assessment

Citation: Lundstrom, K., Diekema, A.R., Leary, H.M., Haderlie, S., & Holliday, W. (2015). Teaching and learning information synthesis: An intervention and rubric based assessment. *Communications in Information Literacy*, 9, 60–82.

Summary: The ability to synthesize information from multiple sources is essential for academic writing. To explore effective strategies for teaching information synthesis skills, the authors developed a lesson that broke the synthesis process into sequenced tasks involving group discussion and individual writing tasks and applied a rubric to evaluate the effectiveness of the instruction.

The intervention appeared to enhance students' ability to synthesize information, as evidenced by improved synthesis in their writing assignments post-lesson. Yet students generally showed low levels of synthesis, indicating the complexity of this skill and the need for further instructional support.

The findings suggest that teaching synthesis effectively requires more than a single lesson and should be integrated throughout the curriculum to better prepare students for complex writing tasks.

Source 4: Relations Between Adolescents' Text Processing and Reasoning

Citation: Wolfe, M. B. W., & Goldman, S. R. (2005). Relations between adolescents' text processing and reasoning. *Cognition and Instruction*, 23(4), 467–502. https://doi.org/10.1207/s1532690xc2304_2

Summary: This study investigates how adolescents process and reason with text, particularly when they encounter multiple sources with conflicting accounts of an event. The complexity of their reasoning was associated with their ability to integrate information across texts and relate it to prior knowledge. The ability to make coherent connections between different sources of information predicted deeper historical understanding and reasoning. The study suggests that students who are better able to integrate conflicting information from multiple texts into a coherent understanding demonstrate more sophisticated historical reasoning.

The authors advocate for educational practices that encourage students to actively integrate and synthesize information from multiple sources, thereby improving their critical thinking and reasoning skills in learning history. This research underscores the importance of teaching strategies that foster an integrated approach to reading and reasoning, particularly in handling complex, real-world issues presented through diverse texts.

Source 5: [Educational Neuroscience for Reading Researchers](#)

Citation: Hruby, G. G., & Goswami, U. (2018). Educational neuroscience for reading researchers. In *Theoretical models and processes of literacy*, 252–277. Routledge.

Summary: While the neuroscience of reading is still in its infancy as an area of study, researchers are making strides in identifying neural correlates of the cognitive processes that constitute reading, such as determining genre, identifying idiomatic writing and metaphor, and understanding written text in relation to pre-existing knowledge.

The authors summarize some of the latest findings regarding reading comprehension skills, such as fMRI work that distinguishes the neural correlates of sentence- versus passage-level comprehension. They also note overlap in areas of the brain that are activated when synthesizing paragraphs into a holistic narrative structure and those that support self-awareness and self-monitoring. This finding in particular may have future implications for classroom teachers who seek strategies to support students simultaneously in reading comprehension, task persistence, and physioemotional regulation.

Conclusion

Synthesis is the capacity to absorb information from various sources, judge its importance and relevance, and integrate it into a comprehensive and comprehensible structure. It is important both for academic contexts as well as professional and everyday life, helping individuals make informed decisions and develop innovative solutions to challenges and problems. Adolescence is a key period in which individuals become able to make more complex connections between ideas, suggesting that high school is a key setting for the development of synthesis skills.

Problem Seeking

Introduction

Effective problem seeking begins with the ability to diagnose a problem and to differentiate between symptoms and root causes. The opinions and perspectives of anyone impacted by the problem are a primary consideration, and the problem seeker should consider the ethical and practical implications of any solution before taking action.

Source 1: Finding the Problem: How Students Approach Problem Identification

Citation: Rubenstein, L. D., Callan, G. L., Speirs Neumeister, K., & Ridgley, L. M. (2020). Finding the problem: How students approach problem identification. *Thinking Skills and Creativity*, 35, 100635. <https://doi.org/10.1016/j.tsc.2020.100635>

Summary: This study examines how students identify and articulate problems when tasked with open-ended questions in a classroom setting. The authors found that students employ a range of strategies to identify problems, which often depend on their academic background and personal experiences. Some common strategies include breaking down the problem into smaller components, seeking patterns or anomalies, and using brainstorming techniques. Many students struggle with defining the scope of the problem. They often either oversimplify complex problems or get overwhelmed by the complexity, struggling to find a starting point.

The authors suggest that explicitly teaching problem-identification skills can significantly improve students' ability to engage with complex problems effectively. Incorporating structured problem-solving exercises into the curriculum can help students practice and refine their problem-identification skills. The authors also found that the clarity and structure of the instructions provided significantly influenced students' ability to effectively identify problems. Clear, concise, and well-structured prompts led to better problem identification.

Source 2: The Role of Empathy in Problem Construction and Creative Problem Solving

Citation: Kripal, S.J., & Reiter-Palmon, R. (2024). The role of empathy in problem construction and creative problem solving. *Learning and Individual Differences*, 114, 102501. <https://doi.org/10.1016/j.lindif.2024.102501>

Summary: This study examines the relationship between empathy, defining problems, and solving problems. In the study, participants were given a real-world problem to solve using an online platform. They were asked to reframe the problem in different ways and then suggest one solution.

The findings showed a strong link between empathy and the quality and originality of how participants defined the problem. The authors suggest that empathy can improve problem solving by helping people think about others' views when figuring out unclear problems.

Source 3: [The KEEN Framework](#): A Guide for Entrepreneurial Mindset

Citation: *The KEEN framework*. (n.d.). Engineering Unleashed. Retrieved June 3, 2025, from <https://engineeringunleashed.com/card/3362>

Summary: KEEN, a respected voice in the engineering education community, has built an innovative framework for engineering mindsets that places problem seeking and understanding on a level of importance equal to that of actual problem solving and design. Placing such emphasis on this phase of the engineering process, the organization has found, yields entrepreneurial mindsets for engineering students that help them ensure that their solutions are well-aligned to the problems they are trying to solve.

KEEN's framework includes components such as identifying opportunities, investigating markets, testing concepts with potential customers, and evaluating societal benefits.

Source 4: [The Role of Problem Construction in Creative Production](#)

Citation: Reiter-Palmon, R. (2017). The Role of problem construction in creative production. *The Journal of Creative Behavior*, 51(4), 323–326. <https://doi.org/10.1002/jocb.202>

Summary: Problem construction is a crucial, initial step in the creative process, one that involves recognizing a gap or problem in a given scenario and creatively determining how to address it. The elements of problem construction include understanding the problem's context, identifying core drivers of the problem, defining its boundaries, and establishing the criteria that potential solutions should meet.

The article explains that the empirical research on problem construction demonstrates clear connections to creativity. Firstly, when individuals actively engage in problem construction, they increase their creativity. Similarly, individuals identified as creative thinkers are more likely to engage in problem construction than non-creatives.

Additionally, problem construction often requires taking multiple perspectives into account to comprehensively define the problem. The article also highlights how diverse team members contribute different viewpoints, leading to a richer understanding of the problem and more innovative solutions.

Conclusion

Effective problem solving depends on ensuring that the problem itself is thoroughly understood. More and more, fields such as engineering, where professionals are often required to design solutions for people they may have never met, are placing a high value on problem diagnosis and understanding. Problem-identification skills are often best taught through structured problem-solving exercises that require students to understand the perspectives of others.

Problem Solving

Introduction

Problem solving is the ability to identify, analyze, and overcome obstacles. Critical thinking, creativity, and knowing how to put a solution into practice are all key aspects of problem solving. Successful leaders use problem solving to enhance productivity, teamwork, and project outcomes.

Source 1: Education for the Age of AI

Citation: Fadel, C., Black, A., Taylor, R., Slesinski, J., Dunn, K., Heinonen, O.-P., & Fullan, M. (2024). *Education for the age of AI*. The Center for Curriculum Redesign.

Summary: AI's ability to process and generate extensive amounts of information should lead to a reevaluation of the knowledge that is taught in schools. As AI handles more routine tasks, the ability to solve complex, unstructured problems that require human ingenuity becomes more valuable.

Curriculum should be designed to challenge students with real-world problems that encourage them to apply interdisciplinary knowledge and critical thinking skills. This approach helps students develop the capacity to navigate the complex problems they will face in their personal and professional lives. Collaborative problem-solving skills are of particular importance, as future challenges will often require collective efforts and the ability to work effectively in teams, sometimes in partnership with AI systems.

The process of teaching knowledge will be most effective when crossing traditional disciplinary boundaries, as this approach can foster greater transferability, creativity, and innovation among students. Further, it will become increasingly important for schools to develop not just intellectual competencies but also character qualities such as empathy and resilience. These aspects are increasingly important as AI shifts the landscape of human jobs toward those requiring social interaction and reskilling.

Source 2: The Neuroscience of Adolescent Decision-Making

Citation: Hartley, C. A., & Somerville, L. H. (2015). The neuroscience of adolescent decision-making. *Current Opinion in Behavioral Sciences*, 5, 108–115. <https://doi.org/10.1016/J.COBEHA.2015.09.004>

Summary: Adolescence is marked by significant neurodevelopmental changes that influence decision-making abilities, including the maturation of the prefrontal cortex, which is crucial for impulse control, working memory, and complex reasoning. These cognitive capacities are important foundations that support problem solving, as they facilitate effectively assessing options and planning actions, engaging in strategic planning, and observing the impacts of solutions and changes.

The authors of this review discuss the value of experience-based learning in adolescence, where taking actions and observing their outcomes can help students develop effective decision-making skills. This kind of learning, facilitated by real-world problem-solving, enables adolescents to make beneficial choices by seeing and understanding the consequences of their actions.

Source 3: Strategies for Teaching Students to Think Critically: A Meta-Analysis

Citation: Abrami, P. C., Bernard, R. M., Borokhovski, E., Waddington, D. I., Wade, C. A., & Persson, T. (2015). Strategies for teaching students to think critically: A meta-analysis. *Review of Educational Research*, 85(2), 275–314. <https://doi.org/10.3102/0034654314551063>

Summary: The development of the prefrontal cortex, which is crucial for higher cognitive functions like planning and solving complex problems, continues into adolescence. The authors of this study find that specific instructional strategies that enhance critical thinking skills include dialogue and discussion, exposure to real-world problems, and mentoring and guidance that support students in navigating complex problem-solving processes such as managing constraints and strategic planning.

The authors conclude that high schools should intentionally integrate learning experiences that foster critical thinking skills. Integrating problem-solving tasks into the curriculum can help meet the developmental needs of adolescents, who are still honing their cognitive abilities for complex thought processes.

Source 4: Adolescents' Development of Skills for Agency in Youth Programs: Learning to Think Strategically

Citation: Larson, R., & Angus, R. (2011). Adolescents' development of skills for agency in youth programs: Learning to think strategically. *Child Development*, 82(1), 277–294. <https://doi.org/10.1111/j.1467-8624.2010.01555.x>

Summary: Participation in arts and leadership-focused learning experiences can help adolescents develop strategic thinking skills, which are crucial for problem solving.

The authors of this study demonstrate that learning experiences that span several weeks or months and immerse students in real-world contexts require adolescents to engage in planning, monitoring progress, adjusting plans, and evaluating outcomes—activities that inherently boost logical reasoning and strategic thinking. This form of thinking integrates higher-order logical reasoning with practical, real-world applications, enhancing adolescents' ability to solve complex problems effectively.

Source 5: What Is Design Thinking and Why Is It Important?

Citation: Razzouk, R., & Shute, V. (2012). What is design thinking and why is it important? *Review of Educational Research*, 82(3), 330–348. <https://doi.org/10.3102/0034654312457429>

Summary: The authors describe design thinking as an iterative process that teams use to understand users, challenge assumptions, redefine problems, and create innovative solutions to prototype and test. It involves five phases—empathize, define, ideate, prototype, and test.

The approach has been found in numerous instances and settings to support innovation, particularly in technology and business, as it fosters creativity and leads to the development of breakthrough solutions that deeply resonate with users. Design thinking is also seen as a powerful tool for addressing complex problems that are ill-defined or unknown by reframing these problems in human-centric ways.

The article discusses the application of design thinking in educational settings, highlighting its potential to enhance learning by engaging students in real-world problem-solving. Problem solving fosters a range of other skills, including critical thinking, collaboration, and creativity, which are essential for success in the modern economy.

Conclusion

New technologies and a plethora of other factors are changing the skills required for personal and career success at an ever-increasing rate. Such changes demand problem-solving skills that can be applied creatively in different circumstances. Educators can help students build problem-solving skills by offering learning experiences that involve real-world learning and opportunities to solve problems related to their interests and communities.

Generous Collaborators for Tough Problems

XQ Competencies —
Literature Review



Diverse Perspectives

Introduction

Collaborating with individuals from different backgrounds enriches the learning experience, strengthens students' ability to tackle complex challenges, and leads to better outcomes. By seeking out people with varied viewpoints and critically identifying, assessing, and challenging their own biases, students foster self-awareness and become more inclusive thinkers and communicators, better prepared to navigate and contribute to an increasingly diverse and interconnected world.

Source 1: Longitudinal Research on Perspective Taking in Adolescence: A Systematic Review

Citation: Hall, H.K., Millea, P.M.R., Summers, M.J., & Isbel, B. (2021). Longitudinal research on perspective taking in adolescence: A systematic review. *Adolescent Research Review*, 6, 125–150. <https://doi.org/10.1007/s40894-021-00150-9>

Summary: Perspective-taking is a critical socio-cognitive skill during adolescence, a period marked by dynamic social and psychological growth. Perspective-taking abilities increase gradually from ages 13 to 18, and the development of these abilities is significantly supported by parents and educators who model effective perspective-taking themselves.

Strong perspective-taking skills serve as a protective factor, meaning that adolescents with strong perspective-taking skills have higher self-esteem, reduced depressive symptoms, and greater psychological well-being. Perspective-taking is also positively associated with prosocial behaviors, such as altruism, cooperation, and moral reasoning, and negatively associated with antisocial behaviors like aggression and prejudice. Longitudinal studies reveal that increased perspective-taking correlates with better conflict resolution and greater use of problem-solving strategies. Adolescents with advanced perspective-taking skills experience improved relationships with parents, peers, and teachers.

Source 2: Creativity, Critical Thinking, Communication, and Collaboration: Assessment, Certification, and Promotion of 21st Century Skills for the Future of Work and Education

Citation: Thornhill-Miller, B., Camarda, A., Mercier, M., Burkhardt, J. -M., Morisseau, T., Bourgeois-Bougrine, S., Vinchon, F., El Hayek, S., Augereau-Landais, M., Mourey, F., Feybesse, C., Sundquist, D., & Lubart, T. (2023). Creativity, critical thinking, communication, and collaboration: Assessment, certification, and promotion of 21st century skills for the future of work and education. *Journal of Intelligence*, 11(3), 54. <https://doi.org/10.3390/jintelligence11030054>

Summary: This article explores the critical role of the "4Cs"—creativity, critical thinking, communication, and collaboration—in preparing individuals for the demands of the 21st century. Collaboration is defined as mutual engagement in coordinated efforts to achieve shared goals. It requires knowledge sharing, task coordination, communication, and fostering trust among group members. The authors argue that diverse perspectives enhance collaboration by introducing varied ways of understanding problems, framing solutions, and generating innovative ideas. Inclusion of diverse perspectives ultimately results in deeper engagement and more innovative problem-solving, although it also requires managing differences effectively to avoid conflicts and misunderstandings.

Schools can support the development of collaboration skills by incorporating team-based projects into curricula. Exposure to varied viewpoints in structured settings helps students produce stronger work in real time, and fosters respect for others, adaptability, and the social regulation skills necessary for teamwork in any setting.

Source 3: [Social Cognitive Development During Adolescence](#)

Citation: Choudhury, S., Blakemore, S-J., Charman, T. (2006). Social cognitive development during adolescence. *Social Cognitive and Affective Neuroscience*, 1(3), 165–174, <https://doi.org/10.1093/scan/nsl024>

Summary: Adolescence is a pivotal time for the development of neurocognitive functions, including plasticity, executive functioning, and emotional processing. Together, these functions support the complex skills of recognizing, understanding, and reconciling perspectives that differ from each other, and that differ from one's own perspective.

In this study, the authors gave individuals of different ages a perspective-taking task and noted the relative ease (as determined by reaction time) with which children, adolescents, and adults could complete the task. As predicted, perspective-taking improves throughout adolescence and into adulthood, in alignment with changes in the neurological structures implicated in social cognition.

These results suggest that the efficiency and strategy of perspective-taking improves during the adolescent years, as young people transition from egocentric (self-focused) to allocentric (other-focused) viewpoints. Further, the authors note other literature that suggests that adolescents who excel in perspective-taking are better equipped for leadership, conflict resolution, and other social roles.

Source 4: [The Difference: How the Power of Diversity Creates Better Groups, Firms, Schools, and Societies](#)

Citation: Page, S. E. (2007). *The difference: How the power of diversity creates better groups, firms, schools, and societies*. Princeton University Press.

Summary: This seminal book explores the transformative power of diversity in groups, arguing that diverse perspectives, skills, and experiences often lead to better problem-solving, decision-making, and innovation, particularly in complex and uncertain environments. The author emphasizes that the modern economy is increasingly complex and interconnected, requiring skills like adaptability, problem-solving, and cross-cultural communication. This is the world our high schoolers will be competing in.

In both problem-solving and prediction tasks, diverse teams consistently outperform homogenous teams, provided the tasks are complex and require collaborative contributions. In educational contexts, diversity can enhance learning by introducing students to varied perspectives and fostering critical thinking. A diverse classroom helps students develop empathy, adaptability, and collaborative skills, all of which are crucial for thriving in an interconnected world.

Source 5: Does Diversity Matter? Measuring the Impact of High School Diversity on Freshman GPA

Citation: Tam, M.Y.S., & Bassett, G.W., Jr. (2004). Does diversity matter? Measuring the impact of high school diversity on freshman GPA. *Policy Studies Journal*, 32, 129–143.
<https://doi.org/10.1111/j.0190-292X.2004.00056.x>

Summary: Research suggests that students learn more and think in deeper, more complex ways when they are part of a diverse educational environment. This cognitive engagement is thought to be stimulated by exposure to varied perspectives and ideas, which challenges students to think critically and adaptively about the perspectives of others. By interacting with peers from different backgrounds, students gain skills in communication, empathy, and cultural understanding, which are valuable in the workforce and broader social contexts.

Students from more diverse high schools tend to have higher GPAs during their first semester in undergraduate studies. Specifically, students from highly diverse high schools have an expected GPA that is one-fourth to one-half point higher than students from less diverse schools. The impact of diversity is more pronounced for students from minority groups and at the lower end of the GPA distribution.

The study concludes that high school diversity plays a significant role in enhancing academic performance at the university level, particularly for students from minority or less represented groups and for female students.

Conclusion

Teams and groups that incorporate individuals with different perspectives often yield better problem-solving strategies and decision-making, as diverse perspectives enhance collaboration by introducing varied ways of understanding problems, framing solutions, and generating innovative ideas. Adolescence is a key developmental stage in which to develop the ability to understand others' perspectives and engage in collaboration with those whose views differ from one's own.

COMPETENCY

Navigating Power

Introduction

Learners are able to identify those who hold power in social, political, cultural, and economic contexts, and can explain how that power impacts individual experiences and institutional relationships. They can think critically about current power structures and whether and how those might be changed.

Source 1: Civic Education, Civic Engagement, and Youth Civic Development

Citation: Dudley, R. L., & Gitelson, A. R. (2003). Civic education, Civic engagement, and youth civic development. *Political Science and Politics*, 36(2), 263–267.
<https://doi.org/10.1017/s1049096503002191>

Summary: This article discusses the critical role of political knowledge in supporting democratic citizenship and civic engagement, asserting that a well-informed citizenry is crucial for the health of democracy.

Tracing the origins of civic education back to ancient political theory, the article highlights the enduring importance of intentionally nurturing civic responsibility and engagement. The authors suggest that civic education must encompass both the mechanics of government and political processes as well as an understanding of the power structures and dynamics within society. This understanding is crucial for fostering effective civic engagement and empowering young people to actively participate in their communities and political systems.

This study, and others the authors cite, show that political knowledge not only supports civic engagement but also fosters a broader understanding and appreciation of democratic principles.

Source 2: Unraveling the Complexity of Critical Consciousness, Political Efficacy, and Political Action Among Marginalized Adolescents

Citation: Diemer, M. A., & Rapa, L. J. (2016). Unraveling the complexity of critical consciousness, political efficacy, and political action among marginalized adolescents. *Child Development*, 87(1), 221–238. <https://doi.org/10.1111/cdev.12446>

Summary: This study examines how critical consciousness affects political engagement among marginalized adolescents. The authors break down critical consciousness into three related sub-phenomena—critical reflection, political efficacy, and critical action—and then investigate the ways critical reflection and political efficacy impact participants’ critical action, i.e. civic and political behavior.

The findings suggest that perceptions of inequality and beliefs about societal equality influence various forms of political participation, such as voting, conventional political actions, and social activism. By integrating discussions about structural inequalities and power dynamics into the curriculum, educators can help students develop a deeper understanding of their social environments and empower them to participate actively in civic and political life.

Source 3: Civic Engagement and the Transition to Adulthood

Citation: Flanagan, C., & Levine, P. (2010). Civic engagement and the transition to adulthood. *The Future of Children*, 20(1), 159–179. <https://doi.org/10.1353/foc.0.0043>

Summary: Civic engagement plays an important role in sustaining democracy and supporting the personal growth and maturation of young adults. Civic engagement activities provide young people with opportunities to learn about and critically analyze social, political, and economic power dynamics. Civic activities also give young people a sense of belonging and purpose beyond personal gain, and allow young adults to contribute to their communities, thereby gaining a stake in society's future.

Opportunities for civic engagement are unevenly distributed among young people. Young adults from lower socioeconomic backgrounds, particularly those who do not attend college, often have fewer opportunities for civic involvement compared to their peers in higher education. The authors advocate for programs and policies that allow all youth to participate in civic activities.

Source 4: Building Power, Learning Democracy: Youth Organizing as a Site of Civic Development

Citation: Rogers, J., Mediratta, K., & Shah, S. (2012). Building power, learning democracy: Youth organizing as a site of civic development. *Review of Research in Education*, 36(1), 43–66. <https://doi.org/10.3102/0091732X11422328>

Summary: This chapter focuses on the role of youth organizing in civic development, particularly as it provides a platform for young people, including those from low-income communities, to engage in meaningful civic actions and develop a deeper understanding of democracy. Youth organizing is often centered around community-based organizations that focus on addressing local issues. These organizations provide a supportive environment where young people can learn about and practice democracy. Youth organizing often involves direct engagement with specific policy issues that affect young people's lives and communities.

Through engagement in such learning experiences, participants gain firsthand experience in understanding how policies are formed, implemented, and can be influenced or changed. Pedagogical strategies include educational workshops, practical experience in campaigns and advocacy efforts, mentorship and guidance, and collaboration and networking.

Youth organizing can bring the voices of young people into public debates and decision-making processes. Through organizing, youth have successfully influenced policies that directly affect their lives and communities, showcasing the potential of organized civic engagement to bring about substantive change.

Conclusion

Power structures are inherent to governance. Being able to identify the holders of power in a situation supports political efficacy among youth, which in turn encourages them to take political action such as voting and pursuing change while building their critical thinking and collaboration skills.

Healthy Relationships

Introduction

Students balance their own needs and boundaries with those of others, making thoughtful choices to nurture and sustain relationships. By continually refining their communication and social skills, students build strong, positive connections with peers, teachers, and mentors. These skills not only support their personal growth but also lead to meaningful, lasting relationships that enhance their future opportunities.

Source 1: Peer Relationships in Adolescence

Citation: Brown, B. B., & Larson, J. (2009). Peer relationships in adolescence. In R. M. Lerner & L. Steinberg (Eds.), *Handbook of adolescent psychology: Contextual influences on adolescent development* (3rd ed., pp. 74–103). John Wiley & Sons, Inc..
<https://doi.org/10.1002/9780470479193.adlpsy002004>

Summary: One of the predominant concerns among those who care for adolescents is the impact of “bad influences.” This review seeks to reframe that idea, arguing that peer relationships are crucial during adolescence, specifically because they are essential to various aspects of personal development during this phase.

In adolescence, peer relationships become more complex. New types of relationships, like romantic relationships, emerge, and the peer system becomes layered, involving different levels like crowds and cliques. The influence of peers also becomes more pronounced, competing with adults in shaping adolescent attitudes, behaviors, and emotional well-being.

Ultimately, these findings underscore the importance of supporting adolescents in developing healthy and productive peer relationships.

Source 2: Achievement Motivation in Adolescents: The Role of Peer Climate and Best Friends

Citation: Nelson, R. M., & DeBacker, T. K. (2008). Achievement motivation in adolescents: The role of peer climate and best friends. *Journal of Experimental Education*, 76(2), 170–189.
<https://doi.org/10.3200/JEXE.76.2.170-190>

Summary: This study finds that a positive peer climate, where adolescents feel valued and respected by their classmates, leads to higher motivation to achieve. Adolescents who perceive positive support and engagement from their peers are more likely to exhibit adaptive motivation strategies.

The quality of peer relationships significantly influences achievement motivation. Friendships where young people exhibit mutual respect and support encourage positive academic motivations, whereas negative peer interactions can reduce motivation.

Educators and schools should consider the significant role that peer relationships and friendships play in motivating students. Creating a classroom environment that promotes positive interactions and recognizes the importance of social relationships can enhance students' academic engagement and motivation.

Source 3: The Potential of Schools to Facilitate and Constrain Peer Relationships

Citation: Juvonen, J. (2018). The potential of schools to facilitate and constrain peer relationships. In W. M. Bukowski, B. Laursen, & K. H. Rubin (Eds.), *Handbook of peer interactions, relationships, and groups* (2nd ed., pp. 491–509). The Guilford Press.

Summary: Students' peer groups—and the ways students interact with those peers—are in part determined by who attends their school and how their school approaches its students. For example, students attending schools with more heterogeneity show less exclusionary and biased behaviors. Further, contextual factors such as school composition, policies, and pedagogy influence individual students (especially those whose individual characteristics present a poor fit for their school environment) in ways that may be attributed to the student themselves, rather than to the context within which the student is operating. A school's strategies for classroom management, inclusion of students with learning differences, bullying interventions, and general approach to social emotional learning can all support or detract from students' experiences with peers, and therefore require intentionality on the part of a school's adults.

Schools and teachers can pay particular attention to dynamics between individual students and their peers, in both social and academic spaces, in order to foster productive, reciprocal relationships among peers. The author notes that school leaders and teachers can manage demographic, intellectual, and behavioral diversity in their schools and classrooms in ways that “unite rather than separate” students, so that all students have the experience of healthful peer relationships.

Source 4: The Teenage Brain: Sensitivity to Social Evaluation

Citation: Somerville, L. H. (2013). The teenage brain: Sensitivity to social evaluation. *Current Directions in Psychological Science*, 22(2), 121–127. <https://doi.org/10.1177/0963721413476512>

Summary: Ongoing brain development during adolescence, especially in socioaffective circuitry involving areas like the amygdala, striatum, and medial prefrontal cortex, contributes to teenagers' acute sensitivity to social cues. These brain changes, coupled with hormonal shifts during puberty, may enhance adolescents' emotional responses to social interactions.

Research using tasks that simulate social evaluations (e.g., receiving feedback on social likability) shows that adolescents display strong emotional reactions to both positive and negative feedback compared to other age groups. These reactions are linked to specific patterns of brain activity that suggest adolescents experience these social evaluations more intensely.

The intense social sensitivity experienced during adolescence can have long-lasting effects, influencing social behaviors and potentially contributing to the risk of mood and anxiety disorders. Social experiences during these years are crucial and can impact an individual's psychological development well into adulthood.

Source 5: The Brain Basis for Integrated Social, Emotional, and Academic Development: How Emotions and Social Relationships Drive Learning

Citation: Immordino-Yang, M.H., Darling-Hammond, L., & Krone, C. (2018). *The brain basis for integrated social, emotional and academic development: How emotions and social relationships drive learning*. The Aspen Institute National Commission on Social, Emotional and Academic Development.

Summary: Adolescents' brain development is significantly influenced by their social experiences. Engaging actively and safely with enriching social relationships is crucial for optimal brain development during these years. These relationships provide emotional support and cognitive opportunities that are vital for adolescents' learning and overall development.

Healthy relationships during adolescence are characterized by emotional safety, mutual respect, and cognitive stimulation. These relationships are supportive and nurturing, allowing adolescents to explore their identities and express themselves without fear of judgment or rejection, offering constructive feedback, and encouraging problem-solving and critical thinking, which are important for academic and personal growth. Additionally, the emotional connection in these relationships helps in developing empathy and understanding towards others, which are crucial social skills during adolescence.

The authors explain that brain development and learning are interdependent processes that are deeply influenced by the quality of social relationships. These relationships help in shaping the neural circuits that underlie key cognitive functions such as attention, memory, and problem-solving. Positive social interactions and relationships stimulate the brain's neural pathways, enhancing the brain's plasticity and its ability to learn new information and skills.

Conclusion

In adolescence, the brain develops in ways that cause individuals to more strongly define themselves based on their relationships with their peers and communities. Positive social influences and experiences during adolescence can have significant positive effects on motivation and academic achievement, while negative social influences and experiences can lead to long-term issues in psychological health and well-being. Healthy relationships during adolescence are not just beneficial, but essential, for enabling adolescents to thrive emotionally, socially, and academically.

Negotiating Conflict

Introduction

Transforming a disagreement into an opportunity for growth hinges on effective conflict negotiation. Start by identifying the core issue and actively listening to all perspectives. Share your own viewpoint respectfully, and work towards collaboration and compromise. Even when tensions escalate, maintaining calm and seeking common ground is crucial. By mastering – or simply improving – your conflict negotiation skills, you can significantly enhance relationships and resolve issues constructively, whether in the workplace or your personal life.

Source 1: Collaboration, Intragroup Conflict, and Social Skills in Project-Based Learning

Citation: Lee, D., Huh, Y., & Reigeluth, C. M. (2015). Collaboration, intragroup conflict, and social skills in project-based learning. *Instructional Science*, 43(5), 561–590.

<http://www.jstor.org/stable/43575307>

Summary: This case study explores how social skills and intragroup conflicts impact collaborative project-based learning (PBL) in high school settings.

The authors emphasize that collaborative skills are crucial in modern educational settings, particularly when using PBL strategies. Effective collaboration skills and the ability to resolve conflicts are essential for successful teamwork and project completion. The authors categorize intragroup conflicts into three types: task-related, process-related, and relationship-related conflicts. Task-related conflicts can encourage critical thinking and problem-solving among students. Process-related conflicts arise from disagreements over group processes and responsibilities, often leading to delays and dissatisfaction. Relationship-related conflicts stem from personal disagreements; these conflicts can severely hinder group cohesion and collaboration.

The study finds that strong group-level social skills are crucial in minimizing process and relationship-related conflicts, which are detrimental to collaboration. Social skills at the group level are more influential than one-on-one social skills in promoting effective collaboration and reducing conflicts. Groups with high collective social skills experience less conflict and more productive collaboration.

Source 2: The Impact of Enhancing Students' Social and Emotional Learning: A Meta-Analysis of School-Based Universal Interventions

Citation: Durlak, J. A., Weissberg, R. P., Dymnicki, A. B., Taylor, R. D., & Schellinger, K. B. (2011). The impact of enhancing students' social and emotional learning: A meta-analysis of school-based universal interventions. *Child Development*, 82(1), 405–432.

<https://doi.org/10.1111/j.1467-8624.2010.01564.x>

Summary: This meta-analysis looks at the results of studies undertaken with more than 200,000 K-12 students across more than 200 schools. Students participating in social-emotional learning (SEL) programs showed improved academic performance, social behaviors, and emotional well-being. Specifically, the meta-analysis found an average 11-percentile-point gain in academic achievement among students involved in SEL programs, as well as a significant reduction in behavioral issues and emotional distress. The authors highlight the successful implementation of SEL programs by school teaching staff, indicating that such programs can be effectively conducted within the school environment by existing personnel.

The authors highlight that conflict resolution is an essential social competency that can be effectively taught as part of broader SEL programs. Conflict resolution can be integrated into the curriculum through explicit instruction, role-playing, and other interactive activities that encourage students to practice these skills. Teaching conflict resolution is associated with improved social behaviors and reduced conduct problems among students. It also supports a more positive social environment within schools, which contributes to better academic outcomes.

The findings support the incorporation of evidence-based SEL programs into standard educational practice. The positive outcomes associated with SEL programs suggest that they are not only beneficial for individual student development but also contribute to a healthier school environment and society.

Source 3: [Now More Than Ever: Why Youth Organizing Work Belongs in Schools and What that Means in Theory and Practice](#)

Citation: Arthurs, S. G. (2018). Now more than ever: Why youth organizing work belongs in schools and what that means in theory and practice. *The High School Journal*, 102(1), 46–71.
<https://www.jstor.org/stable/26757714>

Summary: Adolescence is a critical period for identity development, self-efficacy, and civic awareness. Programs that integrate youth organizing activities during this stage capitalize on students' growing capacity for critical thinking and their desire to influence the world around them. This study demonstrated that such activities also raise natural and authentic opportunities for students to develop and hone conflict resolution skills.

Youth organizing encourages students to engage in critical thinking and open dialogue about community issues, which are crucial skills for conflict resolution. By understanding and discussing various viewpoints and the societal factors that contribute to community tensions, students can work towards resolving these conflicts constructively.

Educators should integrate experiential learning opportunities that promote conflict resolution and organizing skills within the curriculum. Students can gain these skills by identifying community issues, analyzing systemic problems, and proposing actionable solutions. Schools can foster a culture of collaboration and civic engagement by supporting student-led initiatives and creating safe spaces for discussions about power and justice. Partnerships with local organizations can also enhance the effectiveness of these programs.

Source 4: The Prosocial Classroom: Teacher Social and Emotional Competence in Relation to Student and Classroom Outcomes

Citation: Jennings, P. A., & Greenberg, M. T. (2009). The prosocial classroom: Teacher social and emotional competence in relation to student and classroom outcomes. *Review of Educational Research*, 79(1), 491–525. <https://doi.org/10.3102/0034654308325693>

Summary: This article highlights the critical role of teachers' social and emotional competence (SEC) in fostering positive classroom environments and effective classroom management. Teachers with high SEC build stronger relationships with students, which improves classroom management, allows for productive SEC modeling, and enhances the overall learning climate. Positive classroom climates fostered by emotionally competent teachers are associated with better student academic performance, engagement, and social-emotional development.

The authors note that socially and emotionally competent teachers are better equipped to navigate and resolve conflicts in the classroom by employing skills such as emotional regulation, perspective-taking, and effective communication. Teachers with high SEC can manage conflict situations in ways that promote cooperation and intrinsic motivation among students, fostering a more supportive and less disruptive classroom environment.

Teachers with higher SEC are more likely to use proactive strategies to address classroom conflicts, such as understanding the dynamics of peer interactions and addressing underlying issues rather than resorting to punitive measures. Conflict resolution skills contribute to creating a positive classroom climate, which in turn supports both social-emotional and academic outcomes for students.

Source 5: Trajectories of Antisocial Behavior and Psychosocial Maturity From Adolescence to Young Adulthood

Citation: Monahan, K. C., Steinberg, L., Cauffman, E., & Mulvey, E. P. (2009). Trajectories of antisocial behavior and psychosocial maturity from adolescence to young adulthood. *Developmental Psychology*, 45(6), 1654–1668. <https://doi.org/10.1037/a0015862>

Summary: Antisocial behavior (like breaking rules or acting aggressively) changes as young people grow from teenagers into young adults. As cognitive abilities mature, adolescents gain a better capacity to think about the long-term consequences of their actions, which plays a significant role in curbing impulsivity and reconsidering the potential negative impacts of their behaviors on themselves and others.

With growth in emotional and cognitive skills, such as empathy and conflict management, there is a natural decline in behaviors that harm or create disagreement with others. This change is part of the broader developmental trajectory from adolescence into adulthood, where individuals become more socially responsible and less inclined toward disruptive or harmful behaviors.

The authors suggest that adolescence is a key stage in which to provide interventions that can help teenagers grow in positive ways that will help them navigate and reduce conflict in their lives. Schools and programs that help teenagers improve their emotional and cognitive skills can reduce antisocial behavior. Teaching teens to control their impulses and consider the consequences of their actions can be particularly effective.

Conclusion

Conflicts happen. When collaborating with others, differences of opinion or perspective often lead to tension, which can build into outright conflict that has the potential to harm students' progress toward their goals or their relationships with peers. Being able to communicate one's perspective respectfully, and consider the perspectives of others, provides a foundation for adolescents to reflect, rethink, and ultimately adjust or bolster their own points of view. These skills are critical to negotiating conflict, which in turn means that they are crucial to related interpersonal skills such as empathy-building, effective communication, collaboration, and contextualizing one's own perspective.

Building Empathy

Introduction

Students actively consider and question their own assumptions, showing care for others' emotions and perspectives and building meaningful connections across different identities. By reflecting on how their words and actions impact others, students create more inclusive, respectful, and supportive environments.

Source 1: Empathy and Positive Aspects of Adolescent Peer Relationships: A Scoping Review

Citation: Portt, E., Person, S., Person, B., Rawana, E., & Brownlee, K. (2020). Empathy and positive aspects of adolescent peer relationships: A scoping review. *Journal of Child and Family Studies*, 29(9), 2416–2433. <https://doi.org/10.1007/s10826-020-01753-x>

Summary: This review explores the relationship between empathy and positive aspects of adolescent interpersonal functioning, noting its relevance to improving prosocial behavior and overall well-being among adolescents. Empathy, especially as adolescents interact with peers, can enhance understanding and support within friendships and broader peer interactions, potentially leading to more positive social environments.

During adolescence, peer relationships and friendships become more critical, characterized by greater intimacy and reliance on emotional support from peers rather than parents. This period is significant for the development of empathy, which can profoundly affect these relationships by facilitating deeper understanding and connection. The authors highlight two separate types of empathy: cognitive empathy, which involves understanding another person's perspective or mental state, and affective empathy, which relates to sharing or understanding the emotions of others, sometimes expressed through empathic concern or emotional responsiveness.

The findings suggest that fostering empathy in educational settings could significantly enhance the quality of peer relationships among adolescents. Educational strategies that promote empathy could include curriculum elements that encourage students to engage in perspective-taking and empathetic interactions.

Source 2: The Neural Correlates of Empathy: Experience, Automaticity, and Prosocial Behavior

Citation: Rameson, L. T., Morelli, S. A., & Lieberman, M. D. (2012). The neural correlates of empathy: Experience, automaticity, and prosocial behavior. *Journal of Cognitive Neuroscience*, 24(1), 235–245. doi.org/10.1162/jocn_a_00130

Summary: This study underscores the fundamental importance of empathy, a critical aspect of human emotion which influences individual behavior and societal functioning. The authors focus on understanding the neural bases of empathy and its relationship with real-world prosocial behavior, addressing gaps in the current knowledge about the automaticity of empathic responses and the connection between neural activity and daily helping behaviors. The research provides a neuroscientific basis for enhancing empathy through targeted educational and social interventions.

When educators recognize that empathy can be influenced by cognitive load and is not entirely automatic, they can create opportunities for young people to strengthen their empathic skills, particularly in environments where cognitive demands are high. The study supports integrating activities that promote empathy into educational curricula to foster prosocial behavior and improve interpersonal interactions among students.

Source 3: We Feel, Therefore We Learn: The Relevance of Affective and Social Neuroscience to Education

Citation: Immordino-Yang, M. H., & Damasio, A. (2007). We feel, therefore we learn: The relevance of affective and social neuroscience to education. *Mind, Brain, and Education*, 1(1), 3–10. <https://doi.org/10.1111/j.1751-228X.2007.00004.x>

Summary: This review explores the critical interplay between emotions, social functioning, and educational processes. Recent neuroscience research shows a profound integration of emotional and cognitive processes: emotions are not just additions to the learning process, but are integral, influencing attention, memory, decision-making, and social functioning. Emotional processes are crucial for transferring knowledge from structured educational settings to real-world decision-making, acting as an "emotional rudder" that guides judgment and action.

The article explains that brain regions involved in emotional processing are also crucial for cognitive functions, offering a neurobiological explanation for the integration of emotion and cognition.

Educational practices that embrace the role of emotions can improve learning outcomes, and also ensure that students can apply their knowledge and skills in practical, real-world contexts. The authors call for a holistic approach to education that recognizes the emotional dimensions of learning.

Source 4: The Role of Empathy in Problem Construction and Creative Problem Solving

Citation: Kripal, S.J., & Reiter-Palmon, R. (2024). The role of empathy in problem construction and creative problem solving. *Learning and Individual Differences*, 114, 102501. <https://doi.org/10.1016/j.lindif.2024.102501>

Summary: This study explores the relationship between empathy, defining problems, and solving problems creatively. In the study, participants were given a real-world problem to solve using a digital platform. They were asked to reframe the problem in different ways and then suggest one solution.

The findings showed a strong link between empathy and the quality and originality of how participants defined the problem. The authors suggest that empathy can improve problem solving by helping people think about others' views when figuring out unclear problems.

Source 5: Trajectories of Antisocial Behavior and Psychosocial Maturity From Adolescence to Young Adulthood

Citation: Monahan, K. C., Steinberg, L., Cauffman, E., & Mulvey, E. P. (2009). Trajectories of antisocial behavior and psychosocial maturity from adolescence to young adulthood. *Developmental Psychology*, 45(6), 1654–1668. <https://doi.org/10.1037/a0015862>

Summary: This study looks at how antisocial behavior, such as breaking rules or acting aggressively, changes as young people grow up from teenagers into young adults. Empathy is an important component of psychosocial maturity, and adolescence is a key stage in which to provide interventions that can help teenagers develop in ways that support growth in psychosocial maturity.

The article suggests that as cognitive abilities mature, adolescents gain a better capacity to think about the long-term consequences of their actions, which plays a significant role in curbing impulsivity and reconsidering the potential negative impacts of their behaviors on themselves and others.

With growth in emotional and cognitive skills such as empathy and foresight, there is a natural decline in behaviors that harm or create disagreement with others. This change is seen as part of the broader developmental trajectory from adolescence into adulthood, where individuals become more socially responsible and less inclined toward disruptive or harmful behaviors that can negatively impact productive collaboration. On the other hand, individuals who persist in antisocial behavior into adulthood tend to exhibit decreased empathy, a deficit that can contribute to continued antisocial behaviors.

The study suggests that schools and programs that help teenagers improve their emotional and cognitive skills can reduce antisocial behavior and help students build empathy.

Conclusion

Empathy is fundamental to individuals' ability to collaborate and solve problems. The ability to empathize develops substantially over the course of adolescence, and school settings are a prime opportunity to support that development. The development of empathy in adolescence supports healthy pro-social habits in the long-term, and supports individuals in more effectively applying their knowledge to real-world settings.

Productive Collaboration

Introduction

Productive collaborators set shared goals, communicate well, and ensure everyone knows their responsibilities. Trust is key. Team members should respect others' viewpoints and feel comfortable sharing ideas. An effective team checks in regularly to assess how well they're working together, and makes changes if necessary to stay focused and on track.

Source 1: Teaching Adolescents To Become Learners: The Role of Noncognitive Factors in Shaping School Performance: A Critical Literature Review

Citation: Farrington, C.A., Roderick, M., Allensworth, E., Nagaoka, J., Keyes, T.S., Johnson, D.W., & Beechum, N.O. (2013). *Teaching adolescents to become learners. The role of noncognitive factors in shaping school performance: A critical literature review*. The University of Chicago Consortium on Chicago School Research.

Summary: Noncognitive factors such as collaboration play an important role in students' academic performance. Collaboration skills include interpersonal qualities such as cooperation, assertion, responsibility, and empathy. These skills are essential for both academic and future career success. For adolescents, these skills are particularly vital as they correlate with career readiness.

Students' ability to engage effectively in group work and class discussions can support their academic performance, although these skills may not be directly measured by grades, especially in environments where traditional lecture-based teaching predominates.

The authors advocate for educational strategies that promote social skills development as part of a broader approach to enhancing student success. These skills can contribute to a positive school environment and better learning outcomes, particularly as students transition into roles that require more collaboration and teamwork.

Source 2: Collaboration, Intragroup Conflict, and Social Skills in Project-Based Learning

Citation: Lee, D., Huh, Y., & Reigeluth, C. M. (2015). Collaboration, intragroup conflict, and social skills in project-based learning. *Instructional Science*, 43(5), 561–590.
<http://www.jstor.org/stable/43575307>

Summary: This case study explores how social skills and intragroup conflicts impact collaborative project-based learning (PBL) in high school settings. Collaborative skills and the ability to resolve conflicts are deemed essential for successful teamwork and project completion during PBL activities.

The authors categorize intragroup conflicts into three types: task-related, process-related, and relationship-related conflicts. Task-related conflicts encourage critical thinking and problem solving among students. Process-related conflicts arise from disagreements over group processes and responsibilities, often leading to delays and dissatisfaction. Relationship-related conflicts stem from personal disagreements; these conflicts can severely hinder group cohesion and collaboration.

The study finds that strong group-level social skills are crucial in minimizing process and relationship-related conflicts, which are detrimental to collaboration. Social skills at the group level are more influential than one-on-one social skills in promoting effective collaboration and reducing conflicts. Groups with high collective social skills experience less conflict and more productive collaboration.

Source 3: [Impactful Learning Environments: A Humanistic Approach to Fostering Adolescents' Post Industrial Social Skills.](#)

Citation: Leach, N. (2018). Impactful learning environments: A humanistic approach to fostering adolescents' postindustrial social skills. *Journal of Humanistic Psychology*, (62)3
doi:10.1177/0022167818779948

Summary: This longitudinal ethnographic case study examined the development of "post industrial social skills," i.e., the social emotional skills commonly identified as necessary for success in a 21st century economy, such as communication, collaboration, and interpersonal skills.

Over three years observing and interviewing students, teachers, and parents at a mastery-based high school, the author found that the elements of humanistic learning environments, such as a strong sense of community and a culture of respect, resulted in strong collaboration skills among students. In particular, a commitment to collaboration as a core operating principle at the school was articulated for students starting at orientation and reinforced throughout their high school years in advisory and during collaborative learning experiences. This created an environment within which students reported feeling comfortable in collaborative settings because they had developed bonds with fellow students through their shared learning.

Similarly, a culture of respect helped students "reject judgmental attitudes that hindered collaboration" and created a baseline expectation that diversity of experience and opinion would and should be part of a collaborative learning environment. Students felt respected and valued, which in turn increased their comfort with leading projects and informed their own understanding of collaboration skills as an asset.

Source 4: [Using Problem-Centered Learning for Teaching Collaboration in a General Methods Course](#)

Citation: Memory, D. M., Yoder, C. Y., & Williams, R. O. (2003). Using problem-centered learning for teaching collaboration in a general methods course. *The Clearing House: A Journal of Educational Strategies, Issues and Ideas*, 77(2), 67–72. <https://doi.org/10.1080/00098650309601231>

Summary: This study examines how collaboration skills can be effectively taught in educational settings with the support of problem-centered professional learning for teachers.

The course described in the article integrates problem-centered learning to foster collaboration among student teachers. This approach uses real-world scenarios that require student teachers to work in teams, solving problems that educators commonly face. Student teachers are placed in different teams, where each team member takes turns leading a problem-solving activity. These activities are designed to simulate real teaching and collaboration challenges. This setup encourages student teachers to engage deeply with the content, promotes leadership skills, and helps them understand the dynamics of working in diverse teams.

By participating in problem-centered learning experiences that require teamwork and collaboration, teachers themselves gain a deeper understanding of these processes. This experience is not only beneficial for their professional growth but also crucial for their role in modeling these skills to students.

By engaging in these activities, teachers develop a more profound understanding of the dynamics of collaboration, including communication, conflict resolution, and joint decision-making. These insights allow them to teach these skills more effectively to their students, fostering a classroom environment that values and promotes collaborative learning.

Source 5: Using Cooperative Learning to Enhance Students' Learning and Engagement During Inquiry-Based Science

Citation: Gillies, R. M. (2023). Using cooperative learning to enhance students' learning and engagement during inquiry-based science. *Education Sciences*, 13(12), 1242.
<https://doi.org/10.3390/educsci13121242>

Summary: This review examines the effectiveness of cooperative learning strategies in enhancing the outcomes of inquiry-based science education. Inquiry-based learning, which encourages students to actively investigate and engage with academic concepts, is highlighted for its potential to increase student curiosity, participation, and understanding. Cooperative learning involves students working in groups to explore and solve problems collectively, which can enhance understanding and retention of scientific concepts. This approach not only aids in learning complex scientific ideas but also fosters important social skills and group dynamics.

Various studies and meta-analyses have shown that cooperative inquiry-based learning has a positive impact on student achievements, surpassing traditional methods. The success of these programs often hinges on the level of active guidance provided by teachers, with structured inquiry yielding better results than open inquiry scenarios. Successfully implementing inquiry-based learning requires significant changes in classroom management and teaching strategies, with teachers needing to adapt to more facilitative roles.

Effective inquiry-based learning requires a dialogic approach where teachers and students engage in thoughtful discussions that challenge existing ideas and encourage deep thinking. For inquiry-based methods to be effective, teachers must create an environment that encourages scientific discourse and critical examination of ideas. Ongoing training and support for teachers are critical for effectively mastering the pedagogical skills needed for the effective implementation of inquiry-based and cooperative learning strategies.

Conclusion

Collaboration is a crucial life skill, and it can also be a strong mechanism for enhancing student learning. It is a key skill for enabling effective project-based learning, and contributes to a positive school environment and better learning outcomes. By practicing effective collaboration with their peers, teachers can gain experience that helps them facilitate collaboration for students. Teachers in subject areas such as science can utilize collaboration-focused curriculum designs to support student learning.

COMPETENCY

Community Advocacy

Introduction

Students understand power dynamics and choose effective ways to advocate for their communities. They reflect on how they can best get involved to represent and serve the interests of their communities, aiming to make a bigger impact. Students also learn to communicate their ideas clearly across mediums, refining their message to better support their beliefs.

Source 1: [The Psychology of Citizenship and Civic Engagement](https://doi.org/10.1093/acprof:oso/9780199752126.001.0001).

Citation: Pancer, S. M. (2015). *The psychology of citizenship and civic engagement*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199752126.001.0001>

Summary: This book discusses a number of factors influencing civic engagement, including the role of schools and neighborhoods in fostering civic engagement among individuals. It emphasizes that these environments often offer the first real community experiences that influence lifelong civic participation.

Schools can initiate and sustain civic participation through programs such as service-learning and community service, where students take part in activities they may continue into adulthood. Neighborhood organizations also play a crucial role by providing a sense of community and opportunities for engagement.

Young people who are actively involved in civic activities are less likely to engage in problem behaviors such as alcohol and drug use, delinquency, and school dropout. Civic engagement provides them with a sense of purpose and belonging, which can deter negative behaviors and promote positive outcomes.

Participation in civic activities often correlates with better academic performance and enhanced social skills. Engaged youth tend to perform better in school and have higher educational aspirations, likely due to the skills and networks they develop through their civic involvement.

The book underscores the importance of early civic engagement as a foundational element for healthy adolescent development and a successful transition into adult roles within society. Structured programs and supportive environments in schools and communities can foster significant benefits, influencing youths' attitudes towards civic responsibility and their future participation in societal activities.

Source 2: Citizenship, Curriculum, and Critical Thinking Beyond the Four Walls of the Classroom: Linking the Academic Content with Service-Learning

Citation: Ponder, J., Veldt, M. V., & Lewis-Ferrell, G. (2011). Citizenship, curriculum, and critical thinking beyond the four walls of the classroom: Linking the academic content with service-learning. *Teacher Education Quarterly*, 38(4), 45–68. <http://www.jstor.org/stable/23479630>

Summary: The authors of this study discuss the importance of service-learning in educational settings, emphasizing that such programs not only enhance academic achievement but also promote civic responsibility and community engagement among students.

As part of the study, teachers were tasked with creating service-learning projects that align with social justice themes, allowing for the practical application of civic knowledge and the development of critical thinking skills. The students showed higher levels of engagement, enhanced understanding of social issues, and the ability to apply academic content in real-world contexts.

By participating in service-learning projects, students develop a robust civic identity and are encouraged to participate actively in their communities. The process involves students in meaningful problem solving that connects academic learning to real-world issues.

The authors advocate for educators to allow students to take leadership roles in service-learning projects, suggesting that student-directed activities enhance engagement and learning outcomes. The authors recommend building partnerships with local organizations and community members to extend the impact and relevance of service-learning projects.

Source 3: Different Pedagogy, Different Politics: High School Learning Opportunities and Youth Political Engagement

Citation: Kahne, J., Crow, D., & Lee, N.-J. (2013). Different pedagogy, different politics: High school learning opportunities and youth political engagement. *Political Psychology*, 34(3), 419–441. <https://doi.org/10.1111/j.1467-9221.2012.00936>

Summary: This study explores how different pedagogical approaches in high schools affect students' civic and political engagement. The study finds that open classroom discussions of societal issues tend to promote engagement with traditional political activities (referred to as "big P" politics), such as voting and following political issues. In contrast, service learning opportunities are more effective at encouraging community-based and expressive actions (referred to as "little p" politics), such as volunteerism and community service.

Both types of educational opportunities increase students' commitment to participatory citizenship, highlighting the importance of including diverse civic learning opportunities within educational settings to foster a broad range of civic competencies. The research revisits the concept of political socialization, emphasizing the significance of formal civic education in fostering democratic participation among youth.

The authors argue that the distinction between "big P" and "little p" politics is crucial in understanding how specific educational strategies align with broader societal trends in political engagement, particularly among youth. The findings suggest that schools should diversify their pedagogical approaches to enhance both political and civic outcomes effectively. This diversification should ideally reflect an integration of both discussion-based and experiential learning opportunities.

Source 4: Engaging Youth in Communities: A Framework for Promoting Adolescent and Community Health

Citation: Ballard, P. J., & Syme, S. L. (2016). Engaging youth in communities: A framework for promoting adolescent and community health. *Journal of Epidemiology and Community Health*, 70(2), 202–206. <https://doi.org/10.1136/jech-2015-206110>

Summary: This article focuses on the significant impact of community engagement on adolescent development and community health. The article starts by underscoring that community factors play a crucial role in shaping individual health outcomes, recognizing communities not just as settings but as active determinants of health status. Adolescence is highlighted as a pivotal period in which civic engagement can greatly influence both personal and social development. Engaging in community activities provides adolescents meaningful opportunities to contribute, promoting their social skills and sense of efficacy.

The authors discuss the movement within health research that emphasizes the social causes of health, suggesting that community-based methods are essential for understanding and addressing complex health issues and that effective health interventions must consider the community context. From a developmental psychology perspective, the authors argue that understanding the bidirectional interactions between adolescents and their communities can significantly enhance health promotion strategies.

The article concludes with a call for a more integrated approach that considers the reciprocal relationship between youth and their communities, suggesting that such engagement not only supports adolescent development but also strengthens community health.

Source 5: Entering Politics: General Self-Efficacy and Voting Behavior Among Young People

Citation: Condon, M. and Holleque, M. (2013). Entering politics: General self-efficacy and voting behavior among young people. *Political Psychology*, 34, (167–181). <https://doi.org/10.1111/pops.12019>

Summary: This study explores how general self-efficacy influences the voting behaviors of young first-time voters, particularly highlighting the impact among youth from low socioeconomic backgrounds. Self-efficacy is defined as an individual's belief in their capability to execute behaviors necessary to produce specific performance attainments across various domains, with "political efficacy" focused on beliefs related to political agency, such as the ability to influence political processes.

The study finds that general self-efficacy positively influences the likelihood of voting among young adults, particularly those from low socioeconomic-status families. Young people from disadvantaged backgrounds, who often have fewer political role models and less exposure to political engagement, rely more heavily on their general self-efficacy when deciding to vote.

The study illustrates how socioeconomic status affects the development of efficacy beliefs and subsequent political engagement, and suggests that increasing general self-efficacy among youth, particularly those from lower socioeconomic backgrounds, can bolster political participation.

Conclusion

Communities suffer when their voices go unheard, which can lead youth from such communities to lack political self-efficacy, which in turn often leads to civic disengagement. Adolescence is an important window in which individuals can develop and practice advocating for themselves and others in the face of neglect or injustice, and such activities lead to healthier communities, stronger communication skills, and a higher rate of political participation.

Community Mobilization

Introduction

Students can create meaningful improvements in their communities, pursuing efforts that are both effective and inclusive. They explore the skills and mindsets necessary to develop, evaluate, and refine their advocacy efforts. Building trusting relationships, collaborating across diverse groups, and using data-driven strategies to remove systemic barriers, students effectively drive action.

Source 1: Civic Purpose: An Integrated Construct for Understanding Civic Development in Adolescents

Citation: Malin, H., Ballard, P. J., & Damon, W. (2015). Civic purpose: An integrated construct for understanding civic development in adolescence. *Human Development*, 58(2), 103–130.
<https://www.jstor.org/stable/26765100>

Summary: This article examines the development of civic purpose in adolescence and its role in reducing antisocial behavior through emotional and cognitive growth.

Early indicators of purpose often manifest when young people experience empathy in response to the suffering or negative experiences of others and then take action to alleviate that suffering. This empathetic response not only motivates moral actions but also fosters a sense of civic responsibility and purpose.

Cognitive processes also play a significant role in the development of civic purpose. Cognitive skills such as perspective-taking, moral reasoning, and judgment facilitate moral development throughout adolescence and influence the trajectory of purpose development. As adolescents' cognitive abilities mature, they become better equipped to understand moral ambiguity and make ethical decisions, which further contribute to their civic engagement and reduce tendencies towards antisocial behavior.

The integration of emotional empathy and cognitive development leads to the formulation of civic purpose, which is defined as a sustained intention to contribute to the world beyond oneself through civic or political action. This purpose is supported by a combination of current engagement in civic activities, future-oriented civic intentions, and motivation to contribute to the greater good.

The development of civic purpose, underpinned by emotional and cognitive growth, contributes to a reduction in antisocial behavior among adolescents. By engaging in civic activities and harboring a purpose that extends beyond personal gain, adolescents develop a stronger sense of social responsibility and moral identity, which are inversely related to antisocial behaviors.

Summary 2: The Psychology of Citizenship and Civic Engagement

Citation: Pancer, S. M. (2015). *The psychology of citizenship and civic engagement*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199752126.001.0001>

Summary: This book discusses a number of factors influencing civic engagement, including the role of schools and neighborhoods in fostering civic engagement among individuals. It emphasizes that these environments often offer the first real community experiences that influence lifelong civic participation.

Schools can initiate and sustain civic participation through programs such as service-learning and community service, where students take part in activities they may continue into adulthood. Neighborhoods also play a crucial role by providing a sense of community and opportunities for engagement.

Young people who are actively involved in civic activities are less likely to engage in problem behaviors such as alcohol and drug use, delinquency, and school dropout. Civic engagement provides them with a sense of purpose and belonging, which can deter negative behaviors and promote positive outcomes.

Participation in civic activities often correlates with better academic performance and enhanced social skills. Engaged youth tend to perform better in school and have higher educational aspirations, likely due to the skills and networks they develop through their civic involvement.

The book underscores the importance of early civic engagement as a foundational element for healthy adolescent development and a successful transition into adult roles within society. Structured programs and supportive environments in schools and communities can foster significant benefits, influencing youths' attitudes towards civic responsibility and their future participation in societal activities.

Source 3: Civic Engagement and the Transition to Adulthood

Citation: Flanagan, C., & Levine, P. (2010). Civic engagement and the transition to adulthood. *The Future of Children*, 20(1), 159–179. <https://doi.org/10.1353/foc.0.0043>

Summary: Civic engagement helps maintain a functioning democracy and supports the personal growth and maturation of young adults. Engaging in civic activities meets human needs for belonging and purpose beyond personal gain, and it allows young adults to contribute to their communities, thereby gaining a stake in society's future.

Opportunities for civic engagement are unevenly distributed across social classes and racial and ethnic groups. Young adults from lower socioeconomic backgrounds, particularly those who do not attend college, often have fewer opportunities for civic involvement compared to their peers in higher education.

Organizations like City Year offer alternative opportunities for civic engagement to young adults from disadvantaged backgrounds. The article emphasizes the need for institutions and policies to better incorporate young adults into civic life.

Source 4: Developing Social Entrepreneurs and Social Innovators: A Social Identity and Self-Efficacy Approach

Citation: Smith, I. H., & Woodworth, W. P. (2012). Developing social entrepreneurs and social innovators: A social identity and self-efficacy approach. *Academy of Management Learning & Education*, 11(3), 390–407. <https://doi.org/10.5465/amle.2011.0016>

Summary: The authors of this article emphasize the role of social identity and self-efficacy in cultivating “social entrepreneurs.” Social identity theory suggests that people define themselves by the groups to which they belong, and self-efficacy involves belief in one’s capability to achieve a goal. When taught effectively, social entrepreneurship education can convey knowledge and also foster personal identification with social entrepreneurship.

While this study was focused on higher education settings, its findings are relevant for high school learning. Group projects that address real-world problems—whether in the high school or the college classroom— encourage active learning and the application of theoretical concepts to practical situations, enhancing the students’ sense of self-efficacy. Exposing students to case studies and role models in social entrepreneurship helps them understand the attributes and behaviors of successful social entrepreneurs, fostering a deeper connection to the field.

The authors argue for a structured approach in social entrepreneurship education, emphasizing the importance of developing a strong social identity and self-efficacy. This approach not only prepares students to launch successful ventures but also embeds a lifelong commitment to social innovation.

Source 5: Adolescent Civic and Political Engagement: Associations Between Domain-Specific Judgments and Behavior

Citation: Metzger, A., & Smetana, J. G. (2009). Adolescent civic and political engagement: Associations between domain-specific judgments and behavior. *Child Development*, 80(2), 433–441. <http://www.jstor.org/stable/29738625>

Summary: This study investigated how adolescents differentiate various forms of civic involvement—such as community service, political activities, and social movements—based on moral, conventional, and personal judgments.

Utilizing social domain theory, the study found, for example, that adolescents often viewed community service activities as moral obligations, emphasizing the importance of helping others, while they viewed standard political involvement like voting as more obligatory based on conventional reasons tied to societal expectations. The findings suggest that educational programs that promote a nuanced understanding of civic duties could foster more informed and committed civic participation among youth.

The study concludes that adolescents’ conceptualizations of civic involvement are influenced by their social and educational experiences and that these perceptions shape their engagement in civic activities. The results underscore the importance of educating young people about the various dimensions of civic participation to encourage more active and informed involvement in civic life.

Conclusion

Strong communities are an important cornerstone of both democracy and civic engagement. People who feel a sense of self-efficacy and purpose are most likely to engage with and attempt to strengthen their communities. Schools can initiate and sustain this type of civic participation, helping students learn the value of community involvement and increasing the likelihood that they continue these activities into adulthood.

Learners for Life

XQ Competencies —
Literature Review



Wellness

Introduction

Students can understand and manage their emotions, thoughts, and physical reactions, helping them navigate challenges and grow from new experiences. They recognize their personal needs, identify patterns in how different situations affect them, and develop strategies to meet those needs effectively. By understanding how their social identities shape who they are, students build confidence and resilience, ensuring their well-being across all aspects of life.

Source 1: [The Bioecological Model of Human Development](#)

Citation: Bronfenbrenner, U., & Morris, P. A. (2006). The bioecological model of human development. In R. M. Lerner & W. Damon (Eds.), *Handbook of child psychology: Theoretical models of human development* (6th ed., pp. 793–828). John Wiley & Sons, Inc.
<https://doi.org/10.1002/9780470147658.chpsy0114>

Summary: This article outlines the “bioecological” model of human development, a system for exploring the complex interplays between environmental factors and individual development across a person’s lifespan. The model involves four components—process, person, context, and time—and the dynamic interactions among them. The model posits that an individual’s biology as well as their abilities, knowledge, and skills interact with their environments, all across their lifespan. This continuous, dynamic interaction constitutes the processes whereby human development is nurtured.

As such, an individual’s own attributes, and the attributes of their learning environments, both play equally important roles in the development of that individual. Further, the attributes of the learning environment matter not only in how they are objectively designed—i.e., the features of their physical space, the other humans within them, the content and skills waiting to be learned—but also in how they are subjectively experienced, meaning that each individual will have unique interactions with the environment.

Ultimately, then, learning environments—and the adults in them—should attend not only to students’ common goals for learning, but also the highly individualized ways that learning will occur.

Source 2: [We Feel, Therefore We Learn: The Relevance of Affective and Social Neuroscience to Education](#)

Citation: Immordino–Yang, M. H., & Damasio, A. (2007). We feel, therefore we learn: The relevance of affective and social neuroscience to education. *Mind, Brain, and Education*, 1(1), 3–10.
<https://doi.org/10.1111/j.1751-228X.2007.00004.x>

Summary: This article explores the connection between emotion, social functioning, and cognitive activities such as learning, attention, and decision-making. The authors find that cognitive functions are deeply integrated with emotional processes, suggesting that emotional states are not just influential but foundational to student learning.

Insights from neuroscience show that damage to emotional processing centers in the brain can severely impair social behaviors and decision-making, underlining the role of emotion in effective cognitive function. Furthermore, emotional processes are crucial for transferring learned skills and knowledge from structured educational settings to real-world applications, acting as an “emotional rudder” that guides decision-making and action.

In discussing the importance of self-awareness, the authors underscore that emotional awareness is a cornerstone of effectively managing and understanding one's emotional states, which in turn supports effective learning and decision-making. This self-awareness is essential for students to effectively navigate and utilize their emotional and cognitive resources in academic and social environments.

Source 3: Developing Identification with Humanity and Social Well-Being Through Social Identification with Peer Groups in Adolescence

Citation: Albarello, F., Crocetti, E., & Rubini, M. (2021). Developing identification with humanity and social well-being through social identification with peer groups in adolescence. *Journal of Youth and Adolescence*, 50, 1157–1172. <https://doi.org/10.1007/s10964-020-01214-0>

Summary: This study investigates the relationship between adolescents' identification with close peer groups, such as classmates and friends, and with broader, more inclusive groups, such as humanity. It explores the effects of these identifications on adolescents' social well-being and feeling of being integrated and functional within their communities.

Identifying both with peer groups and with humanity were positively related to social well-being, suggesting that a strong sense of belonging and inclusiveness is associated with better social integration and functioning.

Adolescence is a critical period for developing a sense of identity that is connected to local social interactions. Schools and community programs that foster positive peer relationships might enhance not only interpersonal but also global social cohesion. The results support the importance of fostering environments that enhance positive social identifications, as these are linked to greater well-being and societal integration.

Source 4: Within-Person Changes in Mindfulness and Self-Compassion Predict Enhanced Emotional Well-Being in Healthy, but Stressed Adolescents

Citation: Galla B. M. (2016). Within-person changes in mindfulness and self-compassion predict enhanced emotional well-being in healthy, but stressed adolescents. *Journal of Adolescence*, 49, 204–217. <https://doi.org/10.1016/j.adolescence.2016.03.016>

Summary: This study investigates how meditation training can improve emotional well-being in adolescents by developing mindfulness and self-compassion. Specific improvements included reductions in stress, depression, and rumination, alongside increases in positive affect and life satisfaction.

The authors highlight that self-compassion, which involves a sympathetic awareness of one's own situation and a commitment to improving it, predicts improvements in emotional well-being more consistently than mindfulness alone. This suggests that self-compassion might play a critical role in transforming self-awareness into proactive steps towards personal well-being.

A core component of self-compassion is self-awareness, which enables adolescents to recognize and understand their emotions and thoughts without over-identification. This awareness, when combined with a compassionate attitude towards oneself, helps in mitigating negative emotional states and promotes a healthier handling of stress and challenges. The findings suggest that meditation training that emphasizes both mindfulness and self-compassion can be an effective strategy for supporting adolescent emotional health, particularly in stress management and emotional regulation.

Source 5: [Can Happiness Be Taught?](#)

Citation: Seligman, M. E. P. (2004). Can happiness be taught? *Daedalus*, 133(2), 80–87.
<http://www.jstor.org/stable/20027916>

Summary: Positive education is described as integrating the teaching of traditional academic skills with the skills necessary for students' happiness and well-being, which can improve both students' academic performance and their individual well-being. The author describes various interventions designed to promote resilience, happiness, and engagement among students. The interventions aim to teach students skills such as realistic optimism, creative brainstorming, decision making, relaxation, and coping with problems, which are all part of enhancing students' overall well-being. These interventions can significantly reduce symptoms of depression and improve social skills and academic engagement.

The article calls for more widespread adoption of positive education principles, suggesting that such an approach could lead to a "new prosperity" where educational systems not only prepare students for economic success but also for a fulfilling life.

Conclusion

Well-being is supported by cognitive, social, psychological, environmental, and physical elements that play distinct but interconnected roles in human development. Academic literature robustly supports the idea that self-awareness regarding the status and fulfillment of one's emotional, physical, and social needs is crucial for supporting academic growth and achievement.

Recognizing Conflict

Introduction

Students manage conflicts by thoughtfully addressing tension and constructively navigating differences. They become aware of their own and others' physical, mental, and emotional reactions during a conflict, applying strategies to stay calm and remain productive. Students also explore the deeper forces driving conflicts, enabling them to address root causes rather than just surface-level disagreements.

Source 1: Social Cognitive Development During Adolescence

Citation: Choudhury, S., Blakemore, S-J., & Charman, T. (2006). Social cognitive development during adolescence. *Social Cognitive and Affective Neuroscience*, 1(3), 165–174.
<https://doi.org/10.1093/scan/nsl024>

Summary: There are numerous specific parts of the human brain that undergo significant development during adolescence that enhance the ability to differentiate between one's own and others' perspectives. This development improves the efficiency and strategy of perspective-taking, transitioning from egocentric (self-focused) to allocentric (other-focused) viewpoints.

The authors of this study argue that adolescence is a pivotal time for developing the ability to recognize and appreciate the perspectives of others, which is essential for recognizing moments of conflict with the ideas of others and emotionally regulating one's response in order to avoid escalation. Adolescents who excel in perspective-taking are better equipped for leadership, collaboration, conflict resolution, and other social roles.

Positive school climates, characterized by feelings of connectedness, peer cohesion, and teacher support, have been associated with better perspective-taking in some studies. However, longitudinal studies suggest that school climate alone may not be sufficient without specific, intentional instructional interventions focused on the skill.

Source 2: The Development of Conflict Resolution Skills: Preschool to Adulthood

Citation: Sandy, S. V. (2014). The development of conflict resolution skills: Preschool to adulthood. In P. T. Coleman, M. Deutsch, & E. C. Marcus (Eds.), *The handbook of conflict resolution: Theory and practice* (3rd ed., pp. 430–463). Jossey-Bass/Wiley.

Summary: Conflict can create valuable learning opportunities in educational settings. This chapter points out that adolescents are developing advanced levels of social and emotional competence that are essential for handling interpersonal conflicts. These skills include empathy, perspective-taking, and the regulation of emotions, which help in understanding and resolving conflicts with peers, family, and authority figures.

The social dynamics of adolescence, including the increased influence of peer groups, are discussed as both a challenge and an opportunity for practicing conflict resolution. Adolescents often face conflicts that stem from peer pressure, bullying, and the establishment of social hierarchies. Learning to navigate these conflicts is an important part of social development.

The author links poor academic performance and school dropout rates to deficiencies in social-emotional skills, arguing that social competence and appropriate behavior are strong predictors of academic success, surpassing traditional measures like intellectual ability or sensory deficits.

Schools are identified as key arenas for adolescents to learn and practice conflict resolution skills. The article suggests that curricula should include structured opportunities for students to engage in role-plays, discussions, and other activities that help them practice and refine these skills in a supportive environment.

Source 3: Adolescents' Emotion Regulation in Daily Life: Links to Depressive Symptoms and Problem Behavior

Citation: Silk, J. S., Steinberg, L., & Morris, A. S. (2003). Adolescents' emotion regulation in daily life: links to depressive symptoms and problem behavior. *Child Development*, 74(6), 1869–1880.
<https://doi.org/10.1046/j.1467-8624.2003.00643.x>

Summary: The ability to regulate emotions effectively is linked to lower reports of problem behaviors such as aggressiveness and antisocial behavior.

This study differentiates between types of strategies for emotional regulation. “Engagement strategies,” which involve dealing with the emotion directly, were generally associated with better emotional regulation and outcomes. In contrast, “involuntary” (e.g., rumination) and “disengagement” (e.g., denial and avoidance) strategies were correlated with poorer regulation and more negative outcomes. This finding supports the idea that recognizing and intentionally managing emotional responses enables better conflict resolution.

The authors underscore the importance of teaching adolescents to both identify their emotional reactions and practice using effective strategies for reducing potential negative effects of those emotional reactions. The authors emphasize the critical role of emotion regulation in adolescent mental health and behavior, suggesting that interventions focusing on improving these skills may benefit adolescents significantly.

Source 4: Teaching Self-Regulation Skills to Students with Behavior Problems: Essential Instructional Components

Citation: Smith, S. W., Cumming, M. M., Merrill, K. L., Pitts, D. L., & Daunic, A. P. (2015). Teaching self-regulation skills to students with behavior problems: Essential instructional components. *Beyond Behavior*, 24(3), 4–13. <https://doi.org/10.1177/107429561502400302>

Summary: The authors of this article argue that self-regulation instruction should be an integral part of school curriculum, citing neuroscientific evidence that self-regulation relies on internal cognitive mechanisms, or executive functions, that serve as an “air-traffic control system” for the brain. These executive functions are still in development during adolescence and can be strengthened through structured practice in classroom settings.

The article highlights three key instructional areas critical for developing effective self-regulatory processes in adolescents, particularly those with behavioral challenges. “Emotion regulation” refers to the ability to manage internal emotional responses through strategies such as situation modification, attention shifting, cognitive change, and response modulation, which reduce behavioral issues and improve interpersonal interactions. “Social problem-solving” is a cognitive-behavioral process involving identifying problems in group settings, generating solutions, evaluating these solutions, and enacting the best solution. Teaching this process helps students handle interpersonal conflicts more effectively. “Goal setting” involves identifying personal goals based on personal values, planning how to achieve these goals, overcoming external and internal barriers, and monitoring progress.

Students benefit from explicit teaching of self-regulation strategies followed by practical exercises, such as role-plays, which help them apply what they've learned in real-life scenarios. Although school professionals have a limited amount of time and resources, self-regulation instruction can be integrated feasibly throughout the school day, and the reinforcement of these skills throughout the day supports generalization. The authors encourage collaboration among school personnel to ensure consistent reinforcement of self-regulation skills across different settings and throughout the school day.

Source 5: Collaboration, Intragroup Conflict, and Social Skills in Project-Based Learning

Citation: Lee, D., Huh, Y., & Reigeluth, C. M. (2015). Collaboration, intragroup conflict, and social skills in project-based learning. *Instructional Science*, 43(5), 561–590. <http://www.jstor.org/stable/43575307>

Summary: This case study explores how social skills such as open communication, empathy, and the ability to understand and accommodate diverse viewpoints decrease the impact of intragroup conflicts in collaborative project-based learning (PBL) in high school settings. In particular, the authors examined task-related conflicts, process-related conflicts, and relationship-related conflicts. While task-related conflicts often catalyze collaboration, process- and relationship-related conflicts are detrimental to collaboration. Social skills at the group level are more influential than one-on-one social skills in promoting effective collaboration and reducing conflicts, and groups with high collective social skills experience less conflict and more productive collaboration.

The authors suggest strategies for leveraging task-related conflict toward the end of productive group behavior, such as creating student learning groups with a diverse (conflicting) set of interests to catalyze collaboration. They also recommend specific strategies to undermine process-related conflict during project-based learning experiences, including the development of clear procedures and scaffolding for successful completion of the projects; interruption by teachers of social loafing behaviors; and letting students work in the same groups over multiple projects. Finally, the authors note the importance of supporting social emotional learning in general to manage relationship-related conflicts.

Conclusion

Adolescence is a period in which the brain is primed to develop substantial emotional self-awareness. One dynamic of this development that is important for lifelong learning is the ability to recognize the tension of conflict, as it is through this recognition that one is able to effectively consider alternative ideas and viewpoints that may support learning. This type of emotional regulation also supports adolescent mental health and behavior, suggesting that interventions focusing on improving these skills can significantly benefit adolescents.

Understanding Self

Introduction

Students reflect on their strengths and difficulties, using this awareness to confidently learn new material, navigate new tasks, and overcome obstacles. By recognizing how their skills and mindsets have shaped their journey, students develop a clear vision for their post high school future, making them better equipped to succeed personally and professionally. This self-awareness fosters resilience and growth, laying a strong foundation for lifelong learning and achievement.

Source 1: Evolution of Self-Awareness and the Cultural Emergence of Academic and Non-academic Self-Concepts

Citation: Geary, D., & Xu, K.M. (2022). Evolution of self-awareness and the cultural emergence of academic and non-academic self-concepts. *Educational Psychology Review*, 34, 2323–2349. <https://doi.org/10.1007/s10648-022-09669-2>

Summary: Academic self-concepts, which include perceptions of one's capabilities in various educational areas, have a strong impact on academic performance. Individuals with a positive perception of their academic abilities tend to engage more effectively in learning activities, which leads to better grades and higher achievement in educational settings. The article suggests that students with a strong academic self-concept are more motivated and engaged in their studies. They are more likely to take on challenging tasks, persist in the face of difficulties, and have a proactive approach to learning.

A positive academic self-concept also opens up better educational and career opportunities. Students with a strong belief in their academic capabilities are more likely to pursue higher education and secure jobs that require advanced skills, contributing to better economic outcomes over their lifetimes.

The development of academic self-concepts is significantly influenced by the cultural environment, which shapes the educational systems and the importance placed on academic success. Understanding the cultural basis of academic self-concepts is crucial for educators and policymakers. The authors argue that educators should consider how cultural factors influence the development of students' self-concepts and, subsequently, their academic engagement and performance.

Source 2: Competence and Motivation during Adolescence

Citation: Yeager, D. S., Lee, H. Y., & Dahl, R. E. (2017). Competence and motivation during adolescence. In A. J. Elliot, C. S. Dweck, & D. S. Yeager (Eds.), *Handbook of competence and motivation: Theory and application* (2nd ed., pp. 431–448). The Guilford Press.

Summary: Adolescents are commonly portrayed and perceived as disengaged and uninterested in school settings. However, emerging neuroscientific research suggests that adolescents possess unique cognitive flexibilities and decision-making capabilities that can potentially enhance learning, particularly in environments that engage their specific developmental sensitivities.

A key factor influencing adolescents' motivation to learn is their perceived level of their own competence. This chapter argues that when educational experiences allow adolescents to feel competent, particularly based on their own internally held standards and related to their authentic interests and lives, it can lead to enhanced motivation and engagement.

The authors call for educational practices and interventions that respect and utilize the natural developmental progressions of adolescents. They emphasize creating learning environments that offer meaningful challenges and support autonomy, relatedness, and competence.

Source 3: Motivation Interventions in Education: A Meta-Analytic Review

Citation: Lazowski, R. A., & Hulleman, C. S. (2016). Motivation interventions in education: A meta-analytic review. *Review of Educational Research*, 86(2), 602–640.
<https://doi.org/10.3102/0034654315617832>

Summary: Being responsible, accountable, and prepared to pursue and achieve future goals requires the mobilization of multiple motivational processes. This review argues that adolescence is a pivotal time for intentional cultivation of such processes, which can nurture the self-awareness and self-efficacy that serve as foundations for effective motivation.

For example, interventions that help students recognize and align their learning activities with their personal values and goals can enhance their self-awareness and perceived control over their learning, which in turn can increase intrinsic motivation. Further, interventions that seek to nurture students' beliefs in their capabilities to execute tasks successfully help students set achievable goals and provide feedback that helps them understand their progress and areas for improvement. This, in turn, directly supports the development of self-awareness of their skills and capabilities.

The interventions discussed in this review are effective in shaping student engagement and attitudes toward learning, highlighting the complex relationship between self-awareness and motivation. These interventions are grounded in robust educational and psychological theories and have been shown to improve not only specific task performance but also general academic engagement and success.

Source 4: A Social-Cognitive Approach to Motivation and Personality

Citation: Dweck, C. S., & Leggett, E. L. (1988). A social-cognitive approach to motivation and personality. *Psychological Review*, 95(2), 256–273. <https://doi.org/10.1037/0033-295X.95.2.256>

Summary: The authors propose that individuals operate based on implicit theories—either viewing intelligence as a fixed trait (entity theory) or as a malleable quality (incremental theory). These theories significantly affect individuals' goals: those with an entity theory focus on performance goals (aiming to prove their competence), while those with an incremental theory focus on learning goals (aiming to improve their competence).

The article distinguishes between two behavioral patterns in response to challenges: the maladaptive "helpless" response characterized by avoidance of challenge and decline in performance when faced with obstacles, and the adaptive "mastery-oriented" response, where individuals seek challenging tasks, examine their past efforts (whether successful or unsuccessful) in order to improve, and maintain or increase their effort when faced with difficulty.

The authors underscore the importance of implicit beliefs and goal orientations in shaping how individuals approach challenges and opportunities for growth, advocating for a focus on learning goals to foster resilience and lifelong development.

Source 5: Cognitive Ability, Cognitive Self-Awareness, and School Performance: From Childhood to Adolescence

Citation: Demetriou, A., Kazi, S., Makris, N., & Spanoudis, G. (2020). Cognitive ability, cognitive self-awareness, and school performance: From childhood to adolescence. *Intelligence*, 79, Article 101432. <https://doi.org/10.1016/j.intell.2020.101432>

Summary: This study explores the complex interactions between cognitive abilities, self-awareness in the form of self-evaluation and self-concept, and their impacts on academic performance from childhood through adolescence. Students with a positive cognitive self-concept are more likely to show higher engagement in schoolwork. This positive self-view fosters a more resilient attitude towards learning, where students are more likely to persevere through difficulties and setbacks. It also facilitates the practice of self-evaluation, which is focused on identifying one's own performance and capabilities on specific cognitive tasks.

The authors argue that self-evaluation is crucial because it affects how students approach learning and problem-solving. Students who can accurately judge their performance tend to adjust their learning strategies more effectively, which in turn can enhance their academic performance. This ability to self-evaluate becomes increasingly important as students progress through school, where learning often becomes more self-directed.

The research findings highlight that self-evaluation adds significant predictive power to understanding academic performance, especially in older students. This indicates that as children grow and their educational environments become more complex, the ability to accurately understand and assess their own cognitive processes plays a critical role in navigating academic challenges.

Understanding these dynamics can help educators tailor interventions and teaching strategies to individual students' profiles, potentially enhancing educational outcomes by focusing on developing both cognitive skills and cognitive self-awareness.

Conclusion

Self-efficacy and self-concept are crucial components of motivation. When students are aware of their skills, strengths, postsecondary goals, and challenge areas, they are more likely to feel capable of achieving personally meaningful positive outcomes. When they feel capable of achieving positive outcomes, they are more likely to feel motivation and perseverance when pursuing their academic goals.

Receiving Feedback

Introduction

Whether it's for assignments, work, or relationships, feedback is crucial for learning and growth. To benefit from feedback, we need to listen, reflect, and take action. This process allows us to grow in ways we can't achieve on our own. By being open to feedback, we reduce biases and gain valuable insights from others.

Source 1: The Power of Feedback

Citation: Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research*, 77(1), 81–112. <https://doi.org/10.3102/O03465430298487>

Summary: The authors of this meta-analysis explore the impact of feedback on student learning, finding that effective feedback plays a crucial role in student learning by helping bridge the gap between what students know and what they are aiming to learn.

Effective feedback can motivate students, clarify their understanding, and improve their learning outcomes by addressing specific aspects of their performance. When feedback is targeted well, it can increase students' effort, attention, and engagement, helping them refine their approach to tasks and improve their performance. A productive culture of feedback can take hold in classroom climates that encourage peer and self-assessment and foster open discussions about learning goals.

The paper identifies different levels at which feedback can be directed—task-level, process-level, self-regulation, and personal feedback—and highlights that feedback is most beneficial when focused on tasks and self-regulation rather than personal praise. Teachers are encouraged to provide feedback that guides students in developing strategies for self-regulation and self-assessment, helping them become more independent and reflective learners.

Source 2: Formative Assessment Techniques to Support Student Motivation and Achievement

Citation: Cauley, K. M., & McMillan, J. H. (2010). Formative assessment techniques to support student motivation and achievement. *The Clearing House: A Journal of Educational Strategies, Issues and Ideas*, 83(1), 1–6. <https://doi.org/10.1080/O0098650903267784>

Summary: Formative assessment is the practice of using assessment to understand where students are in their learning in order to better support their learning. In contrast, traditional assessment is typically used to measure and judge total attainment of knowledge at a curricular endpoint.

The application of formative assessment techniques leads to higher levels of student engagement and better achievement by engaging students with meaningful feedback that helps them improve their performance. Teachers can use the insights gained from formative assessments to make instructional decisions that cater more closely to the needs of their students.

Instructional strategies that support formative assessment include providing clear learning targets, offering feedback about progress, and encouraging mastery-focused goals. Focusing on mastery and understanding rather than performance or competing against peers promotes intrinsic motivation.

Source 3: The Impact of Peer Assessment on Academic Performance: A Meta-analysis of Control Group Studies

Citation: Double, K.S., McGrane, J.A., & Hopfenbeck, T.N. (2020). The impact of peer assessment on academic performance: A meta-analysis of control group studies. *Educational Psychology Review* 32, 481–509. <https://doi.org/10.1007/s10648-019-09510-3>

Summary: The authors of this meta-analysis examine 54 experimental and quasi-experimental studies that evaluated the effect of peer assessment on academic performance in primary, secondary, or tertiary students across subjects and domains.

The analysis finds that peer assessment is more effective for encouraging learning than receiving no assessment or teacher-driven assessment, and has an impact similar to self-assessment. The positive effects of peer assessment are consistent across primary, secondary, and tertiary education levels. Effectiveness of this process remains stable regardless of the mode of delivery (online or offline), anonymity, and whether the feedback was scaffolded or freeform. Repeated peer assessment yielded better outcomes compared to a single instance.

Source 4: Formative Assessment: Assessment Is for Self-Regulated Learning

Citation: Clark, I. (2012). Formative assessment: Assessment is for self-regulated learning. *Educational Psychology Review*, 24(2), 205–249. <https://doi.org/10.1007/s10648-011-9191-6>

Summary: Formative assessment supports the creation of a learning environment where feedback is perceived as a constructive part of the learning process rather than a judgment of student capability. This approach can diminish the anxiety that typically accompanies evaluations and encourages students to view feedback more positively and as a tool for improvement. Through this mechanism, formative assessment practices encourage students to seek feedback more frequently. Because these practices integrate feedback as a regular and ongoing part of learning activities, students become accustomed to receiving and utilizing feedback without the usual stress associated with high-stakes testing.

The implementation of formative assessment is linked to the development of lifelong learning skills, as it prepares students to manage their own learning and adapt to various educational and professional challenges throughout their lives.

Source 5: Using Formative Assessment and Metacognition to Improve Student Achievement

Citation: Hudesman, J., Crosby, S., Flugman, B., Issac, S., Everson, H.T., & Clay, D.B. (2013). Using formative assessment and metacognition to improve student achievement. *Journal of Developmental Education*, 37(1).

Summary: This study examines the impact of a model designed to improve students' academic performance in mathematics by developing their metacognitive skills through formative assessment. The model includes teaching students how to use self-regulation strategies effectively during their learning process. The self-regulated learning component of this model involves continuous feedback cycles broken down into planning, practice, and evaluation phases. This approach helps students to continually adjust their learning strategies based on feedback, fostering a deeper understanding of the learning process and improving their ability to tackle academic challenges. The process not only provides content-related feedback but also focuses on students' learning strategies, helping them develop skills to assess their own learning and make adjustments.

Students engaging in courses based on this model have consistently achieved higher pass rates in both the course itself and on standardized tests like the ACT mathematics portion. The model has also shown to be beneficial in subsequent college-level mathematics courses.

Conclusion

Feedback is a powerful tool for promoting academic achievement and metacognition; but, depending on when and how it is delivered, feedback can lead to students feeling anxious and/or not seeking more feedback. Formative feedback, which is delivered during the learning process, holds significant promise in this regard.

Pursuing Goals

Introduction

Students effectively manage resources, create actionable plans, and adapt their strategies when faced with challenges. By honing these abilities, students learn to navigate obstacles and set a clear path toward their future success, both in school and beyond. They can take ownership of their goals, ensuring they are well-prepared for the demands of postsecondary life and their professional careers.

Source 1: What has fMRI Told Us About the Development of Cognitive Control Through Adolescence?

Citation: Luna, B., Padmanabhan, A., & O'Hearn, K. (2010). What has fMRI told us about the development of cognitive control through adolescence? *Brain and Cognition*, 72(1), 101–113.
<https://doi.org/10.1016/j.bandc.2009.08.005>

Summary: Cognitive control, which allows for goal-directed behavior and decision-making, continues to develop throughout adolescence. This review examines neuroscience studies focusing on two fundamental aspects of cognitive control: response inhibition (the ability to suppress inappropriate actions) and working memory (the ability to maintain and manipulate information over short periods).

Studies show improvement in adolescents' ability to inhibit responses, which is crucial for flexible and appropriate behavior. Similar to response inhibition, working memory shows developmental progression during adolescence. These aspects of cognitive control are crucial foundations for the development of other crucial capacities, such as time management, action planning, and pursuing goals.

Source 2: Metacognition, Achievement Goals, Study Strategies and Academic Achievement: Pathways to Achievement

Citation: Vrugt, A., & Oort, F.J. (2008). Metacognition, achievement goals, study strategies and academic achievement: Pathways to achievement. *Metacognition Learning*, 3, 123–146.
<https://doi.org/10.1007/s11409-008-9022-4>

Summary: Metacognitive strategies, or practices that facilitate awareness and regulation of one's own learning processes, can positively influence students' achievement goals and their utilization of appropriate study strategies (e.g., deep cognitive strategies for understanding, surface strategies for rote learning, and resource management for time and environmental factors).

A key metacognitive strategy outlined in this study is metacognitive regulation, which refers to the regulation of thinking and learning through planning, monitoring, and evaluating one's learning processes. It includes planning (selecting appropriate strategies and allocating resources before beginning a task), monitoring (keeping track of one's understanding and progress during the task), and evaluation (reviewing and assessing one's learning after completing a task to understand what was effective and what needs improvement).

The study emphasizes how these skills interact with achievement goals to influence the use of different study strategies, ultimately impacting academic performance, and underscores the importance of a holistic approach that considers cognitive, metacognitive, and motivational factors in fostering academic success. While this study was undertaken with college students, the cognitive findings translate well across different age spans.

Source 3: The Relation Between Self-Regulated Learning and Academic Achievement Across Childhood and Adolescence: A Meta-Analysis

Citation: Dent, A. L., & Koenka, A. C. (2016). The relation between self-regulated learning and academic achievement across childhood and adolescence: A meta-analysis. *Educational Psychology Review*, 28(3), 425–474. <https://doi.org/10.1007/s10648-015-9320-8>

Summary: As adolescents grow, their cognitive capabilities mature, making them better able to employ and benefit from learning strategies including setting realistic academic goals, planning and executing tasks effectively, and monitoring their own understanding and progress. These skills help adolescents navigate more complex academic tasks and manage the greater autonomy and responsibility typical of high school education.

This meta-analysis finds that learning strategies such as these become significantly more impactful as students progress into high school. This age group shows a stronger association between the use of these strategies and academic performance compared to younger students. The complexities of high school tasks and the maturity of high school students allow them to utilize these strategies more effectively.

Overall, the analysis underscores the critical role of self-regulated learning in enhancing academic achievement among high school students, emphasizing that as students' cognitive capacities expand, so too does the effectiveness of these learning strategies.

Source 4: College Knowledge: What it Really Takes for Students to Succeed and What We Can Do to Get Them Ready.

Citation: Conley, D. T. (2005). *College knowledge: What it really takes for students to succeed and what we can do to get them ready* (pp. 49–100). Jossey-Bass.

Summary: Cognitive and metacognitive skills such as self-regulated learning, strategic learning, and reflective thinking are key to preparing high school students for college success. These skills enable students to monitor and manage their own learning processes effectively, and as such are crucial for pursuing and achieving post-secondary educational goals.

The author suggests that students must be able to assess their understanding and skills critically, plan their learning strategies accordingly, and adjust their approaches based on outcomes. These capacities enable students to navigate the academic challenges of college more successfully, as they are better prepared to handle the complexities of college-level coursework and the independence required in higher education settings.

Educational practices and curricular designs should help students develop these metacognitive skills throughout their high school years, ensuring they are not only knowledgeable in academic content but also adept at understanding how they learn best, setting goals, and making informed decisions to reach those goals.

Source 5: Goal Setting and Student Achievement: A Longitudinal Study

Citation: Moeller, A.J., Theiler, J.M., & Wu, C. (2012). Goal setting and student achievement: A longitudinal study. *The Modern Language Journal*, 96, (153–169).
<https://doi.org/10.1111/j.1540-4781.2011.01231.x>

Summary: This study examines how goal setting—the process of establishing clear, achievable objectives— influences student achievement over time.

The study finds that structured goal setting is positively correlated with improved proficiency scores. Students who actively engage in setting and reflecting on personal learning goals show greater achievement over time. The findings emphasize the importance of self-regulation and learner autonomy in educational success.

Goal setting as part of a structured learning process helps students internalize objectives and actively engage in their learning. The study supports the inclusion of goal-setting processes in curriculum design, highlighting its role in enhancing student motivation and achievement.

Conclusion

Adolescence is a key phase in the early development of certain habits and mindsets that enable individuals to effectively pursue their goals; these include setting realistic academic goals, planning and executing tasks effectively, and monitoring one's own understanding and progress. Learning experiences that provide a structured approach to practicing such ways of thinking are effective in helping students develop these capacities.

Self-Regulation

Introduction

Students effectively manage their emotions and reactions in challenging situations. Through self-monitoring, they apply mindful strategies to stay focused and process their emotions. By practicing self-reflection, they identify emotional patterns, anticipate their responses, and prepare to handle situations with composure. This competency is crucial for helping students build resilience and emotional intelligence, setting them up for success in both personal and professional life.

Source 1: Maturation of Brain Microstructure and Metabolism Associates with Increased Capacity for Self-Regulation during the Transition from Childhood to Adolescence

Citation: Baron Nelson, M., O'Neil, S. H., Wisnowski, J. L., Hart, D., Sawardekar, S., Rauh, V., Perera, F., Andrews, H. F., Hoepner, L. A., Garcia, W., Algermissen, M., Bansal, R., & Peterson, B. S. (2019). Maturation of brain microstructure and metabolism associates with increased capacity for self-regulation during the transition from childhood to adolescence. *The Journal of Neuroscience*, 39(42), 8362–8375. <https://doi.org/10.1523/JNEUROSCI.2422-18.2019>

Summary: Significant brain development occurs as children transition into adolescence, particularly in regions associated with self-regulation—the ability to control emotions, thoughts, and actions. These include the frontal cortex, which is important for decision-making and impulse control, and the striatum and thalamus, both of which are involved in processing emotions and rewards.

The maturation of these brain areas enhances adolescents' ability to manage their behavior and emotions more effectively. Improved self-regulation is linked to better performance in school, more successful social interactions, and greater emotional stability. Enhancing self-regulation through targeted educational and developmental practices can have long-term benefits, including improved academic success and better mental health outcomes.

By aligning educational and developmental interventions with the natural growth patterns of the brain, schools and teachers can better support adolescents' journey towards becoming well-adjusted adults.

Source 2: Self-Control Development in Adolescence Predicts Love and Work in Adulthood

Citation: Allemand, M., Job, V., & Mroczek, D. K. (2019). Self-control development in adolescence predicts love and work in adulthood. *Journal of Personality and Social Psychology*, 117(3), 621–634. <https://doi.org/10.1037/pspp0000229>

Summary: This study finds that improvements in adolescent self-control are associated with better adult relationships, characterized by higher satisfaction and lower conflict. Development in self-control also predicts better work-life outcomes, including greater job satisfaction and engagement.

Self-control, defined as the ability to prioritize long-term goals over immediate desires, is essential for achieving personal and professional success. The study emphasizes that changes in self-control during adolescence, not just the initial levels, have significant implications for adult life.

The authors underscore the importance of adolescence as a critical period for developing self-control, which has significant implications for an individual's future ability to form and maintain satisfying relationships and succeed in their careers. The authors suggest that efforts to enhance self-control during adolescence could have long-term benefits for individuals' social and professional success.

Source 3: Longitudinal Intra-Individual and Inter-Individual Relations Between Cognitive and Emotional Self-Regulation Across Adolescence

Citation: Memmott-Elison, M. K., & Moilanen, K. L. (2021). Longitudinal intra-individual and inter-individual relations between cognitive and emotional self-regulation across adolescence. *Journal of Youth and Adolescence*, 50(10), 1970–1981. <https://doi.org/10.1007/s10964-021-01488-y>

Summary: This study examines how cognitive self-regulation (e.g., planning and decision-making) and emotional self-regulation (e.g., controlling negative emotions) interact and develop during the adolescent years.

The study finds statistically significant effects in which cognitive self-regulation influences emotional self-regulation over time, suggesting that the ability to plan and make decisions can help adolescents manage their emotions better. Interventions aimed at enhancing adolescents' cognitive self-regulation skills could have secondary benefits for emotional self-regulation, which is important for students' long-term adjustment and overall healthy adolescent development.

This study contributes to our understanding of self-regulation in adolescents by delineating the directional influences between its cognitive and emotional components. It emphasizes the potential of targeting cognitive self-regulation to improve emotional outcomes, which could inform both educational practices and mental health interventions.

Source 4: Adolescents' Emotion Regulation in Daily Life: Links to Depressive Symptoms and Problem Behavior

Citation: Silk, J. S., Steinberg, L., & Morris, A. S. (2003). Adolescents' emotion regulation in daily life: Links to depressive symptoms and problem behavior. *Child Development*, 74(6), 1869–1880. <https://doi.org/10.1046/j.1467-8624.2003.00643.x>

Summary: The strategies related to managing emotions are relevant for conflict situations, as stronger emotion regulation skills can lead to better interpersonal interactions and less reactive behaviors during conflicts.

The study found that adolescents' responses to negative emotions included disengagement techniques, such as avoidance or suppression, and involuntary engagement, such as rumination. Disengagement and involuntary engagement are linked to higher instances of depressive symptoms and increased problem behaviors. The research highlights the importance of teaching adolescents effective techniques of emotion regulation in order to help them manage their emotional responses more constructively.

Enhancing adolescents' emotion regulation capabilities could significantly reduce psychological distress and behavioral issues. The study underscores the importance of supportive interventions that focus on developing healthy emotional and behavioral responses.

Source 5: [A Synergistic Mindsets Intervention Protects Adolescents from Stress](#)

Citation: Yeager, D. S., Bryan, C. J., Gross, J. J., Murray, J. S., Krettek Cobb, D., H F Santos, P., Graveling, H., Johnson, M., & Jamieson, J. P. (2022). A synergistic mindsets intervention protects adolescents from stress. *Nature*, 607(7919), 512–520. <https://doi.org/10.1038/s41586-022-04907-7>

Summary: Adolescents face high levels of social-evaluative stress, impacting their mental health and causing them to avoid challenging situations that are necessary for learning and skill development.

This study tests the idea of a "synergistic mindsets" intervention, combining focuses on growth mindsets (the belief that abilities can be developed) and stress-can-be-enhancing mindsets (the belief that physiological stress responses can be beneficial).

When students reframe their perceptions of intellectual challenges and physiological stress responses as opportunities for growth and enhanced performance, it improves their stress-related cognitions, cardiovascular reactivity, and cortisol levels, indicating better physiological and psychological responses to stress. That is, a synergistic mindset supports students in regulating their biopsychological states and by extension their behaviors.

Participants in this study showed improved well-being, greater academic success, and reduced anxiety symptoms, particularly during the COVID-19 pandemic lockdowns. The study found that addressing both growth and stress mindsets together was crucial—separately targeting these mindsets did not yield the same benefits. This shift in mindset allowed students to engage more effectively with academic challenges and manage stress more healthily.

Conclusion

Adolescence is a key period in which people develop emotional self-regulation capacities. These capacities support academic achievement, mental and physical well-being, and healthy social relationships. A key aspect of effective emotional self-regulation is the ability to maintain positive mindsets (such as the belief that one can develop new and stronger skills, or that physiological stress responses can be beneficial) in the face of challenges or setbacks.

Self-Motivation

Introduction

Students take control of their learning by creating conditions that boost their engagement and success. By aligning their work with personal interests and goals, they make learning relevant and meaningful. They learn to recognize and recreate the learning conditions that bring them joy, building a lifelong passion for learning.

Source 1: Interest Development and Learning

Citation: Renninger, K. A., & Hidi, S. E. (2019). Interest development and learning. In *The cambridge handbook of motivation and learning* (pp. 265–290). Cambridge University Press.
<https://doi.org/10.1017/9781316823279.013>

Summary: The authors of this review explore how interest develops (from initial engagement triggered by novel or engaging content, called triggered situational interest) to a deep and enduring personal commitment (well-developed individual interest). Each phase involves different levels of self-regulation, knowledge, and value associated with the content. The authors argue that everyone has the potential to develop interest in any content area, regardless of their starting level of interest or background, due to the physiological basis of interest.

The authors suggest that interest can be sparked by content that is surprising, has personal relevance, or captures imagination. Even without initial interest, structured tasks or social interactions can initiate interest development. As interest develops, it needs supportive conditions like relevant tasks, meaningful activities, and encouragement to progress to higher phases. Interest not only increases engagement but also improves persistence, even when tasks become challenging.

The authors encourage educators to use strategies that align with the different phases of interest development, recognizing that students' engagement can be enhanced by linking new content to their existing interests or by introducing novel and relevant applications of the content. When integrating interests into curriculum, educators should consider the variability in students' developmental phases of interest, which can help in accommodating diverse learning needs and in maximizing educational outcomes.

Source 2: The Effect of Motive–Goal Congruence on Adolescents' Academic Goal Engagement and Disengagement

Citation: Yau, P. S., Cho, Y., Kay, J., & Heckhausen, J. (2022). The effect of motive–goal congruence on adolescents' academic goal engagement and disengagement. *Motivation and Emotion*, 46(4), 447–460. <https://doi.org/10.1007/s11031-022-09946-1>

Summary: This study differentiates between implicit motives (deep-seated desires not always conscious) and explicit goals (conscious targets like achieving high grades), showing that both components independently and in interaction (when congruent) significantly influence students' engagement with academic goals.

Adolescence is highlighted as a critical period for aligning implicit motives with explicit goals. During this stage, individuals are forming their identity and academic aspirations, making it an optimal time for interventions that can help align their intrinsic motivations with their explicit academic goals, thereby enhancing engagement and reducing disengagement.

The findings suggest that educational interventions should focus on enhancing the congruence between students' implicit motives and their explicit academic goals. For example, helping students set realistic yet challenging goals that resonate with their deeper motivations can lead to improved engagement. Schools might use formative assessments and feedback mechanisms to continuously align students' learning activities with their motives and goals, promoting sustained engagement and preventing disengagement.

Source 3: States of Curiosity Modulate Hippocampus-Dependent Learning via the Dopaminergic Circuit

Citation: Gruber, M.J., Gelman, B.D., & Ranganath, C. (2014). States of curiosity modulate hippocampus-dependent learning via the dopaminergic circuit. *Neuron*, 84(2), 486–496.
<https://doi.org/10.1016/j.neuron.2014.08.060>

Summary: Using functional magnetic resonance imaging (fMRI), the authors of this study determined that people are better at learning new information if they are curious about the topic.

Further, memory of information that is not inherently interesting, but is introduced during a period of heightened curiosity, is also easier to learn. In particular, arousal in the mid-brain and nucleus accumbens increases during the experience of curiosity. Retention of information presented during the period of this neurological arousal is enhanced both in immediate and delayed tests of recall.

These findings suggest a link between the neurological structures implicated in extrinsic motivation (e.g., rewards) and those implicated in intrinsic motivation (e.g., curiosity). The authors suggest that these findings provide strong motivation for educators and learning designers to build learning experiences that honor students' inherent areas of interest and curiosity, as such experiences will have greater effectiveness in terms of students' retention of the material.

Source 4: The Role of the Learning Environment in Adolescents' Motivational Development

Citation: Raufelder, D., & Kulakow, S. (2021). The role of the learning environment in adolescents' motivational development. *Motivation and Emotion*, 45(3), 299–311.
<https://doi.org/10.1007/s11031-021-09879-1>

Summary: This study compares a conventional teacher-directed learning (TDL) environment and a student-directed learning (SDL) environment. The study involved 1,153 students from grades 7 to 10, with an average age of 13.97 years. Changes in general self-efficacy and intrinsic value (aspects of motivation) were measured.

The study finds significant differences in the motivational trajectories of students in the two learning environments. The SDL environment appeared to better support students' motivational needs, countering the typical decline seen in conventional TDL settings.

The authors concluded that the learning environment plays a critical role in shaping adolescent motivational development, and that tailoring learning environments to better meet the developmental needs of adolescents could mitigate commonly seen declines in motivation. Strategies include offering more autonomy and opportunities for self-directed learning as students progress through school.

Source 5: [The Importance of Motivation as a Predictor of School Achievement](#)

Citation: Steinmayr, R., & Spinath, B. (2009). The importance of motivation as a predictor of school achievement. *Learning and Individual Differences*, 19(1), 80–90.
<https://doi.org/10.1016/j.lindif.2008.05.004>

Summary: There are a number of motivational factors that predict academic success among high school students beyond the influence of intelligence. This study examines how a variety of motivational constructs impact learning, including domain-specific values (the relevance, usefulness, and enjoyment a student ascribes to specific subjects or tasks), self-perceptions of ability (self-efficacy), goals (both learning- and performance-oriented), and achievement motives (desire for success or fear of failure).

Different motivational constructs significantly contribute to predicting school achievement. Particularly, domain-specific values and self-perceptions of ability added the most to the prediction. The study suggests that while intelligence is a strong predictor of academic performance, motivation adds a crucial, independent layer of prediction.

The authors underscore the significant role of personal relevance and motivation in educational contexts, highlighting the importance of fostering motivational aspects in classroom settings to enhance academic performance, and suggesting that educational interventions should focus not only on enhancing cognitive abilities, but also on fostering positive motivational attitudes such as authentic student interest.

Conclusion

Most people aren't highly motivated by things that don't interest or affect them, and adolescents tend to experience this feeling particularly acutely. When students are personally invested in what they're learning, it not only increases engagement but also improves persistence, even when tasks become challenging. Learning environments that allow students to practice self-direction and that feature supportive social opportunities are most likely to motivate adolescents.

Wayfinding

Introduction

Wayfinders are self-reliant, resilient, and can adapt to life's ups and downs. They know how to set personal goals, identify key resources, make decisions based on real-time feedback, and uncover hidden opportunities in the most unexpected places.

Source 1: The Relation Between Self-Regulated Learning and Academic Achievement Across Childhood and Adolescence: A Meta-Analysis

Citation: Dent, A. L., & Koenka, A. C. (2016). The relation between self-regulated learning and academic achievement across childhood and adolescence: A meta-analysis. *Educational Psychology Review*, 28(3), 425–474. <https://doi.org/10.1007/s10648-015-9320-8>

Summary: This meta-analysis finds that self-regulated learning strategies become significantly more impactful as students progress into high school. This age group shows a stronger association between the use of self-regulated learning strategies and academic performance compared to younger students. The complexities of high school tasks and the maturity of high school students allow them to utilize these strategies more effectively.

The authors highlight that as adolescents grow, their cognitive capabilities mature, making them better able to employ and benefit from self-regulated learning strategies. This includes setting realistic academic goals, planning and executing tasks effectively, and monitoring their own understanding and progress. For high school students, metacognitive processes like planning, monitoring, and evaluating are particularly important. These skills help them navigate more complex academic tasks and manage the greater autonomy and responsibility typical of high school education.

The authors underscore the critical role of self-regulated learning in enhancing academic achievement among high school students, emphasizing that as students' cognitive capacities expand, so too does the effectiveness of these learning strategies.

Source 2: Self-Determination as a Source of Self-Esteem in Adolescence

Citation: Owens, T. J., Mortimer, J. T., & Finch, M. D. (1996). Self-determination as a source of self-esteem in adolescence. *Social Forces*, 74(4), 1377–1404. <https://doi.org/10.2307/2580355>

Summary: This study finds that self-determination in family, school, and part-time job settings significantly influences adolescents' self-esteem; adolescents who feel a sense of control and autonomy in these areas tend to have higher self-esteem.

The article explores how self-determination within family decisions, academic engagement, and workplace tasks correlates with positive self-perceptions. The authors argue that experiences that allow personal discretion and involve engaging tasks (found to foster positive outcomes in adults) are also critical in adolescence. Such experiences are pivotal in shaping an adolescent's self-concept and esteem.

The findings suggest that skills and attitudes developed through autonomous experiences in one domain can influence perceptions and behaviors in others, highlighting the interconnectedness of different life areas in adolescent development.

Source 3: Youth Development Programs: Supporting Self-Motivation in Project-Based Learning

Citation: Larson, R. W., McGovern, G., Orson, C., Renninger, K. A., & Hidi, S. E. (2019). Youth development programs: Supporting self-motivation in project-based learning. In *The Cambridge Handbook of Motivation and Learning* (pp. 111–138). Cambridge University Press.
<https://doi.org/10.1017/9781316823279.007>

Summary: This chapter explores how self-motivation and wayfinding skills can be developed through project-based learning, discussing how project-based afterschool youth development programs, including those focused on subjects such as arts, leadership, and STEM, serve as significant contexts for learning.

Projects introduce students to challenges that occasionally overwhelm and disrupt their efforts, but the structured support from teachers and peers helps them learn from these experiences and develop their ability to recognize and navigate such challenges through strategies like iterative thinking.

Project-based learning boosts motivation significantly, often surpassing the levels of motivation experienced in traditional school settings. The chapter presents findings that motivation not only starts high but also tends to increase over time in these learning experiences.

Source 4: Fostering High-School Students' Self-Regulated Learning Online and Across Academic Domains

Citation: Greene, J. A., Bolick, C. M., Caprino, A. M., Deekens, V. M., McVea, M., Yu, S., & Jackson, W. P. (2015). Fostering high-school students' self-regulated learning online and across academic domains. *The High School Journal*, 99(1), 88–106. <https://doi.org/10.1353/hsj.2015.0019>

Summary: Self-regulated learning skills are critical for managing and utilizing online information effectively. The authors of this article discuss key aspects of self-regulated learning, such as planning, goal-setting, and strategy development; monitoring and adjusting their strategies based on feedback and self-assessment, and reflecting on what worked and what didn't, which helps improve future learning strategies.

The article highlights the pivotal role of teachers in scaffolding and supporting the development of self-regulated learning skills through direct instruction, modeling effective strategies, and providing feedback.

Source 5: [Interest Development and Learning](#)

Citation: Renninger, K. A. & Hidi, S. E. (2019). Interest development and learning. In *The Cambridge Handbook of Motivation and Learning* (pp. 265–290). Cambridge University Press.

<https://doi.org/10.1017/9781316823279.013>

Summary: The authors of this chapter explore how interest develops from initial engagement to a deep and enduring personal commitment. Each phase involves different levels of self-regulation, knowledge, and value associated with the subject.

Self-guided exploration and choice are crucial aspects of interest development, especially to encourage learners' transition from situational interest (e.g., "That was an interesting fact") to individual interest (e.g., "I am going to try to learn more about that"). Teachers can support this process and help students make more informed decisions about how to engage with content and what learning paths to pursue.

As learners' interest deepens, they increasingly take initiative in their learning journey, seeking out information and activities related to their interests. Educational environments that allow for choice and provide opportunities for deeper engagement with content facilitate this transition. These settings help learners feel more competent and autonomous, which are key drivers of intrinsic motivation.

Conclusion

Cognitive capabilities mature in adolescence, making adolescents better able to employ and benefit from self-regulated learning skills, such as planning, monitoring, and/or adapting one's learning strategies. Self-directed learning opportunities and project-based learning environments are effective for developing students' self-regulated learning skills, positive self-views, and strong self-efficacy.

Self-Advocacy

Introduction

Students identify and connect with key allies who can support their success. By learning to communicate persuasively and seek mentorship, students build a strong network around their personal and professional goals. This skill not only helps them navigate challenges but also transforms opportunities, ensuring they have the support and guidance needed to craft and achieve their vision of success.

Source 1: Older and Wiser: Mentoring Relationships in Childhood and Adolescence.

Citation: Rhodes, J. E. (1994). Older and wiser: Mentoring relationships in childhood and adolescence. *The Journal of Primary Prevention*, 14(3), 187–196. <https://doi.org/10.1007/BF01324592>

Summary: This review examines the impact of mentoring on resilience in high-risk youth and synthesizes literature across various fields to provide a comprehensive understanding of mentor relationships.

The author underscores the role of mentors in contributing to resilience among youth, particularly those in high-risk environments, drawing on literature on social support, resilience, and mentoring. Mentoring relationships can have a protective influence, providing critical social support that can shield youth from developmental risks and promote healthy outcomes. Mentors are also valuable additions to the social networks of youth, providing guidance, emotional support, and acting as role models.

The article suggests that mentoring programs, which do not require extensive resources, could be an effective, culturally sensitive approach to intervention and prevention with youth. The findings confirm the importance of mentor relationships in shaping and protecting youth, making a strong case for the integration of mentoring programs within broader educational and developmental frameworks.

Source 2: Stand by Me: The Risks and Rewards of Mentoring Today's Youth.

Citation: Rhodes, J. E. (2002). *Stand by me: The risks and rewards of mentoring today's youth*. Harvard University Press. <https://doi.org/10.2307/j.ctvjz8142>

Summary: Mentoring can offer significant emotional, behavioral, and academic benefits to youth, providing them with role models and increasing their resilience and opportunities for success.

This book also discusses potential downsides, noting that ineffective mentoring relationships can harm vulnerable youth, stressing the importance of careful mentor selection and training. The author critiques the lack of depth in current research, the small effect sizes found, and the often overly positive portrayal of mentoring effects in the media.

The author advocates for a more measured approach to mentoring, suggesting enhancements in how mentors are trained and matched with mentees and to improved research mechanisms to better inform policy and practice.

While acknowledging the potential of mentoring to significantly benefit youth, the author urges a balanced view that recognizes the complexities and challenges of effectively implementing mentoring programs.

Source 3: [It's Who You Know That Matters: Identifying Which Type of Informal Mentor Is Most Likely to Promote Economic Mobility for Vulnerable Youth](#)

Citation: Gowdy, G., & Spencer, R. (2021). It's who you know that matters: Identifying which type of informal mentor is most likely to promote economic mobility for vulnerable youth. *The Journal of Primary Prevention*, 42(3), 217–238. <https://doi.org/10.1007/s10935-021-00630-7>

Summary: This study explores the impact of different types of mentoring on the economic mobility of youth from low-income backgrounds. The research focuses on how informal mentoring relationships, distinct from structured programs, support upward economic mobility for youth facing significant socioeconomic disadvantages.

The findings of the study indicate that “capital mentoring,” which involves mentors who provide significant connections to external resources and networks, are most heavily associated with greater economic mobility. These mentors provide access to social networks and resources beyond the youth's immediate environment, which is crucial for economic advancement.

The authors underscore the importance of both social capital and social support provided by mentors, which contribute to better educational and vocational outcomes, hence facilitating upward mobility. The findings suggest that interventions aimed at promoting economic mobility among vulnerable youth should focus on connecting them with mentors who can provide substantial social capital and access to broader networks.

Source 4: [Does Natural Mentoring Matter? A Multilevel Meta-analysis on the Association Between Natural Mentoring and Youth Outcomes](#)

Citation: Van Dam, L., Smit, D., Wildschut, B., Branje, S. J. T., Rhodes, J. E., Assink, M., & Stams, G. J. J. M. (2018). Does natural mentoring matter? A multilevel meta-analysis on the association between natural mentoring and youth outcomes. *American Journal of Community Psychology*, 62(1–2), 203–220. <https://doi.org/10.1002/ajcp.12248>

Summary: This meta-analysis provides a comprehensive view of the impact of “natural mentoring” (mentorship that occurs through naturally forming relationships between youth and non-parental adults who are part of their existing social networks) on academic and vocational functioning, social-emotional development, physical health, and psychosocial problems.

The authors examine data from 30 studies from 1992 to 2018; they observe that the presence of a natural mentor was significantly associated with positive youth outcomes across all domains. A more substantial effect size was observed when considering the quality of the mentoring relationship, emphasizing the importance of the depth, duration, and frequency of engagement within the relationship over mere presence.

The largest effect sizes were found in social-emotional development and academic and vocational functioning, indicating these areas benefit most from high-quality mentoring relationships. The benefits of natural mentoring relationships were not moderated by the youth's risk status, suggesting that such relationships are generally beneficial regardless of the youth's background or challenges.

Source 5: How Economic Disadvantage Affects the Availability and Nature of Mentoring Relationships During the Transition to Adulthood

Citation: Raposa, E. B., Erickson, L. D., Hagler, M., & Rhodes, J. E. (2018). How economic disadvantage affects the availability and nature of mentoring relationships during the transition to adulthood. *American Journal of Community Psychology*, 61(1-2), 191–203. <https://doi.org/10.1002/ajcp.12228>

Summary: This study examines how economic constraints influence the accessibility and quality of natural mentoring relationships among adolescents, finding that adolescents from economically disadvantaged backgrounds have significantly less access to mentoring from nonparental, nonfamilial adults. They are more likely to receive mentoring from family and friends within their immediate social networks.

When lower-income youth do have mentors, these relationships often focus more on immediate, practical support rather than on long-term educational or career guidance. These mentors are less likely to serve as role models or provide connections that facilitate upward social mobility. The study highlights that socio-economic status substantially shapes the social supports available to youth, potentially perpetuating cycles of poverty.

The findings suggest that interventions should not only increase the quantity of mentoring available but also focus on enhancing the quality of mentoring relationships to include more nonfamilial mentors who can provide career and educational guidance.

Conclusion

Self-directed learners know when to ask for help, and this often entails asking for support or guidance from an adult. Formal and informal mentorship relationships can significantly contribute to adolescents' learning, social-emotional development, and academic and career opportunities.